



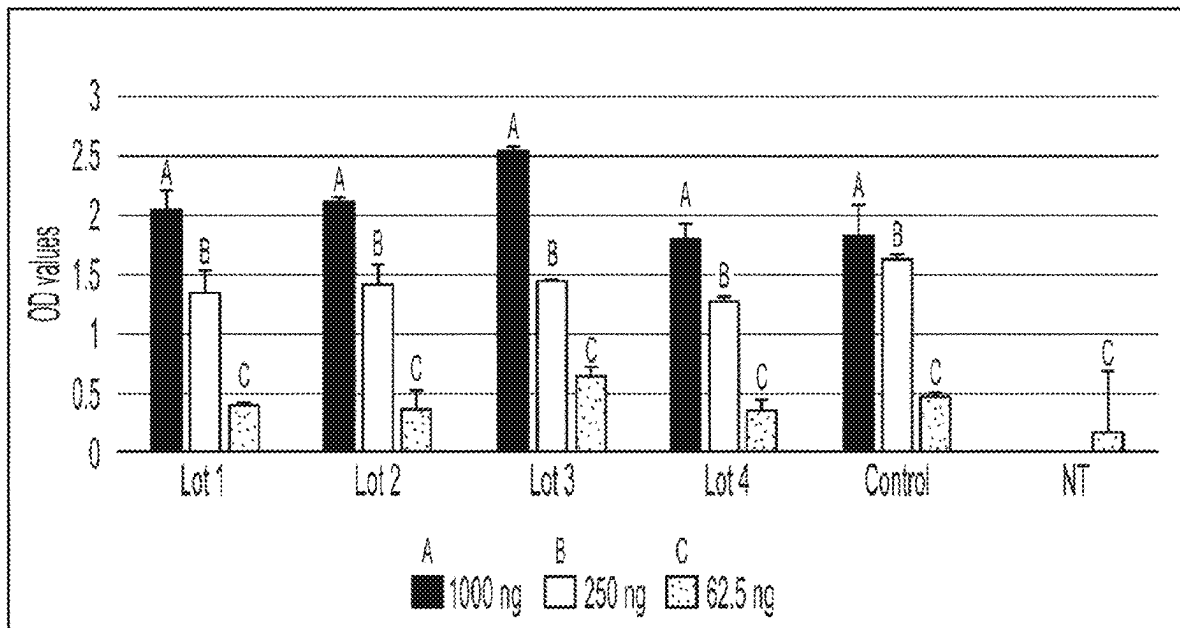
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(19) **United States**(12) **Patent Application Publication**
Smith(10) **Pub. No.: US 2025/0281418 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **LIPID NANOPARTICLE FORMULATIONS
AND METHODS OF SYNTHESIS THEREOF****Related U.S. Application Data**(60) Provisional application No. 63/232,110, filed on Aug.
11, 2021.(71) Applicant: **ModernaTX, Inc.**, Cambridge, MA
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§ 371 (c)(1),

(2) Date: **Feb. 9, 2024**(57) **ABSTRACT**

Stabilized formulations of lipids and nucleic acids, including lipid nanoparticle formulations which encapsulate nucleic acids are disclosed herein. Methods of using the formulations are also provided.



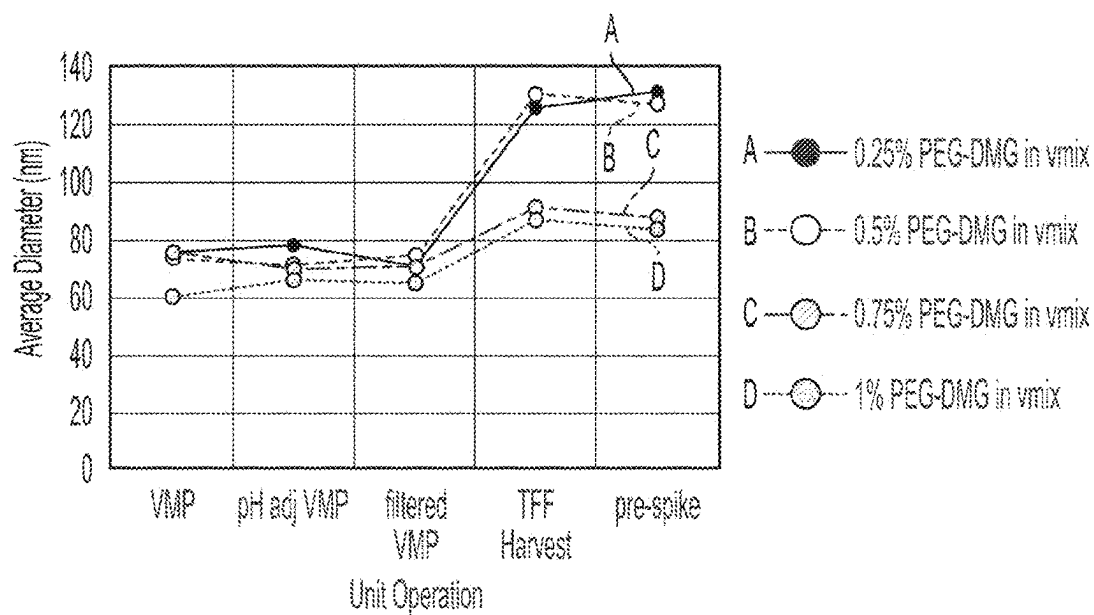


FIG. 1

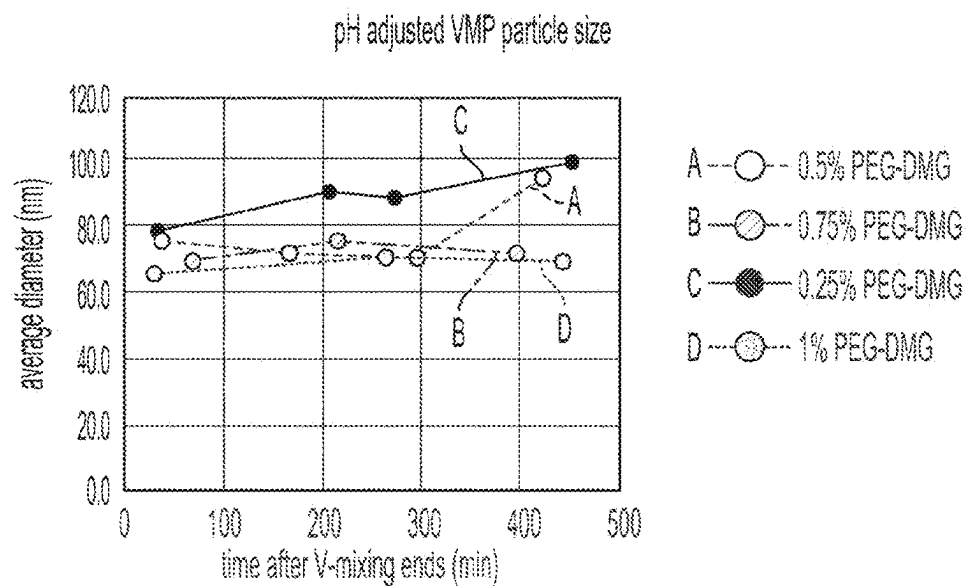


FIG. 2

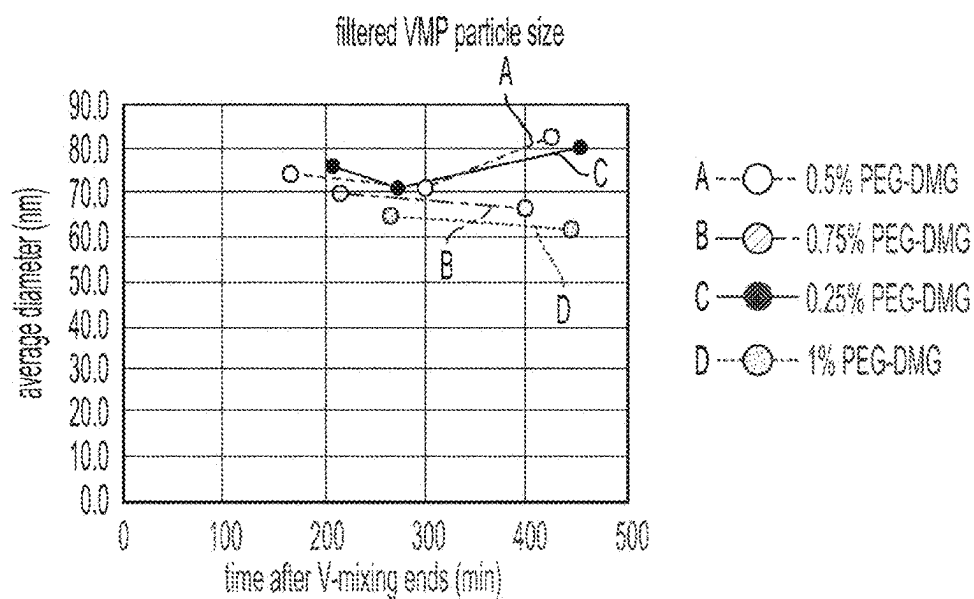


FIG. 3

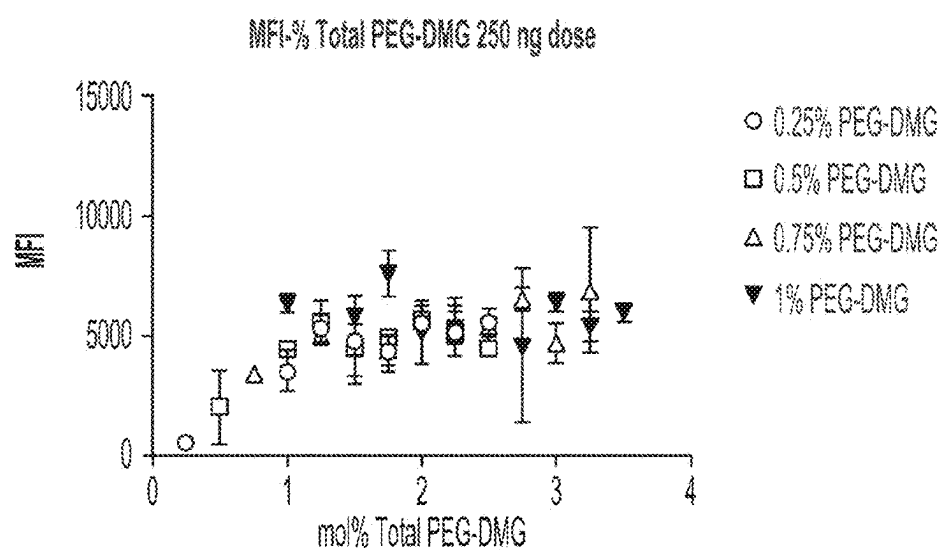


FIG. 4A

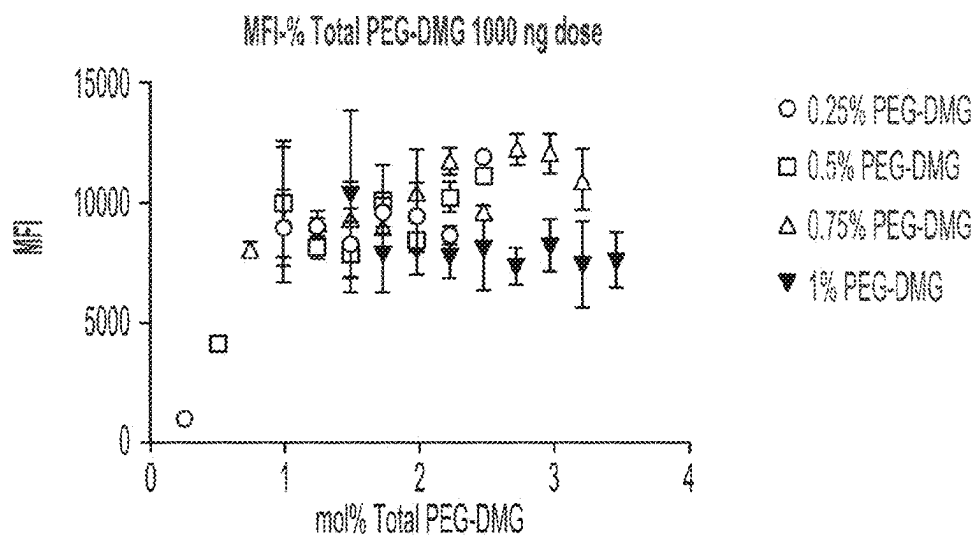


FIG. 4B

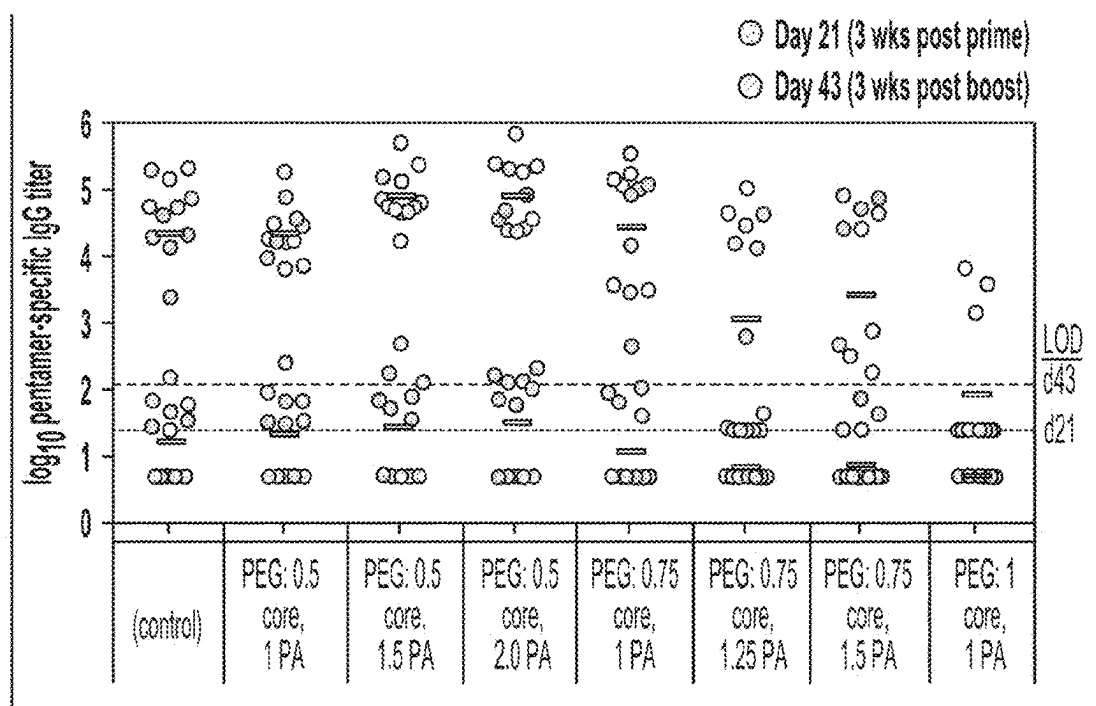


FIG. 5

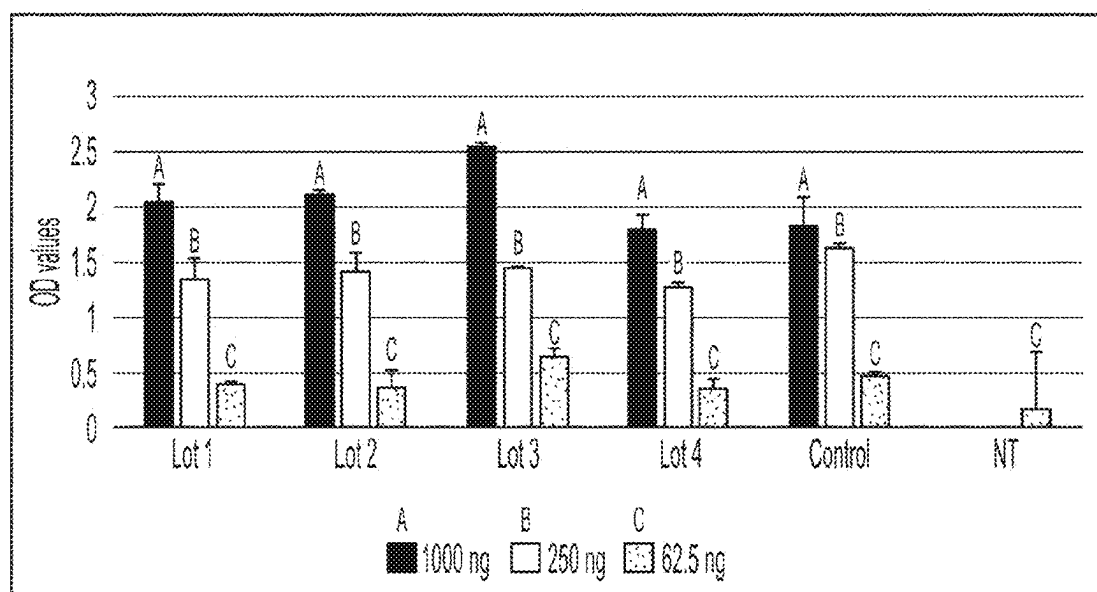


FIG. 6

LIPID NANOPARTICLE FORMULATIONS AND METHODS OF SYNTHESIS THEREOF

RELATED APPLICATIONS

[0001] This application is a national stage filing under 35 U.S.C. § 371 of international PCT Application PCT/US2022/039921, filed Aug. 10, 2022, which claims priority under 35 U.S.C. § 119 (c) to U.S. Provisional Patent Application No. 63/232,110, filed Aug. 11, 2021, each of which is hereby incorporated by reference in its entirety.

BACKGROUND

[0002] The use of messenger RNA as a pharmaceutical agent is of great interest for a variety of applications, including in therapeutics, vaccines, and diagnostics. Effective in vivo delivery of mRNA formulations represents a continuing challenge, as many such formulations are inherently unstable, activate an immune response, are susceptible to degradation by nucleases, or fail to reach their target organs or cells within the body due to issues with biodistribution. Each of these challenges results in loss of translational potency and therefore hinders efficacy of conventional mRNA pharmaceutical agents.

[0003] Various non-viral delivery systems, including nanoparticle formulations, present attractive opportunities to overcome many challenges associated with mRNA delivery. In particular, lipid nanoparticles (LNPs) have drawn particular attention in recent years as various LNP formulations have shown promise in a variety of pharmaceutical applications.

[0004] However, lipids have been shown to degrade nucleic acids including mRNA, and lipid nanoparticle formulations undergo rapid loss of purity when stored as refrigerated liquids. It is also evident that the stability of mRNA is poorer when encapsulated within LNPs than when stored unencapsulated.

SUMMARY

[0005] Provided in this disclosure, among other things, are lipid nanoparticles (LNPs) having improved biophysical properties, in vitro expression, in vivo immunogenicity, and long term stability, surprisingly, even at refrigerated or warmer temperatures.

[0006] The disclosure, in some aspects, provides a method of producing a lipid nanoparticle composition, the method comprising: a) mixing a lipid solution comprising 0.25 mol % to 0.75 mol % of a first PEG-lipid, an ionizable lipid, a phospholipid, and a structural lipid with an aqueous buffer to thereby form a precursor lipid nanoparticle; and b) adding a lipid nanoparticle modifier comprising 1.5 mol % to 2.5 mol % of a second PEG-lipid to the precursor lipid nanoparticle thereby forming a modified lipid nanoparticle.

[0007] In some embodiments, the method further comprises mixing a nucleic acid (e.g., mRNA) with the lipid solution and the aqueous buffer to thereby form the precursor lipid nanoparticle, which is a precursor nucleic acid lipid nanoparticle. In some embodiments, the method further comprises mixing a nucleic acid with the precursor lipid nanoparticle to thereby form a nucleic acid lipid nanoparticle.

[0008] In some embodiments, the lipid nanoparticle modifier is added in two separate steps and the total amount of the second PEG-lipid added is 1.5 mol % to 2.5 mol %. In some

embodiments, the lipid nanoparticle modifier is added in three, four, or five separate steps and the total amount of the second PEG-lipid added is 1.5 mol % to 2.5 mol %.

[0009] In some embodiments, the lipid solution comprises 0.5 mol % of the first PEG-lipid. In some embodiments, the lipid nanoparticle modifier comprises 2.0 mol % of the second PEG-lipid.

[0010] In some embodiments, the nucleic acid lipid nanoparticle comprises a total of 2.5 mol % of the first PEG-lipid and the second PEG-lipid. In some embodiments, the nucleic acid lipid nanoparticle comprises a total of 3 mol % of the first PEG-lipid and the second PEG-lipid.

[0011] In some embodiments, the first PEG-lipid is PEG-DMG or 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108, 111,114,117,120,123,126,129,132-tetratetracontaoxatetratetracontahexyl stearate. In some embodiments, the second PEG-lipid is PEG-DMG or 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetracontaoxatetratetracontahexyl stearate.

[0012] In some embodiments, the ionizable lipid is compound I (heptadecan-9-yl 8-((2-hydroxyethyl)(6-oxo-6-(undecyloxy)hexyl)amino)octanoate), compound II (heptadecan-9-yl 8-((2-hydroxyethyl)(8-(nonyloxy)-8-oxooctyl)amino)octanoate), compound III (heptadecan-9-yl 8-((3-((2-(methylamino)-3,4-dioxocyclobut-1-en-1-yl)amino)propyl)(8-oxo-8-(undecan-3-yloxy)octyl)amino)octanoate), or compound IV (3-butylheptyl 8-((8-(heptadecan-9-yloxy)-8-oxooctyl)(2-hydroxyethyl)amino)octanoate).

[0013] In some embodiments, the phospholipid is distearoylphosphatidylcholine (DSPC). In some embodiments, the structural lipid is a sterol. In some embodiments, the sterol is cholesterol.

[0014] In some embodiments, the lipid solution comprises ethanol.

[0015] In some embodiments, the mixing in (a) comprises turbulent mixing ("T-mix"), vortex mixing ("V-mix"), microfluidic mixing, or a combination thereof.

[0016] In some embodiments, the nucleic acid comprises messenger ribonucleic acid (mRNA).

[0017] In some embodiment, the disclosure provides a composition comprising any one of the lipid nanoparticles described herein.

[0018] In some aspects, the disclosure provides a composition comprising a lipid nanoparticle (LNP) and a nucleic acid, wherein the LNP comprises 2-4 mol % PEG-modified lipid, 5-25 mol % neutral lipid, 25-55 mol % sterol, and 20-60 mol % ionizable amino lipid.

[0019] In some embodiments, the composition is stable for at least six months at a temperature of about 5° C. In some embodiments, the composition is stable for at least six months at a temperature of about 2° C. to about 6° C. In some embodiments, the composition is stable for at least six months at room temperature.

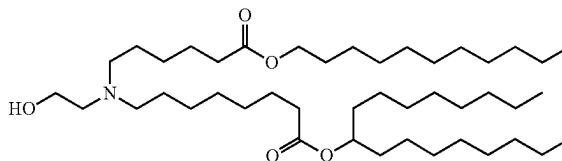
[0020] In some embodiments, the LNP comprises about 2-3 mol % PEG-modified lipid, 5-25 mol % neutral lipid, 25-55 mol % sterol, and 20-60% ionizable amino lipid. In some embodiments, the LNP comprises about 2.5 mol % PEG-modified lipid, 5-25 mol % neutral lipid, 25-55 mol % sterol, and 20-60% ionizable amino lipid. In some embodiments, the LNP comprises about 2.5 mol % PEG-modified lipid, 11 mol % neutral lipid, 38.5 mol % sterol, and 48 mol

% ionizable amino lipid. In some embodiments, the LNP comprises about 3 mol % PEG-modified lipid, 5-25 mol % neutral lipid, 25-55 mol % sterol, 20-60% ionizable amino lipid, or any combination thereof. In some embodiments, the LNP comprises about 3 mol % PEG-modified lipid, 11 mol % neutral lipid, 39 mol % sterol, and 47 mol % ionizable amino lipid.

[0021] In some embodiments, the PEG-modified lipid is 1,2 dimyristoyl-sn-glycerol, methoxypolyethyleneglycol (PEG-DMG). In some embodiments, the PEG-modified lipid is 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetradecacontaoxatetracontahexyl stearate. In some embodiments, the neutral lipid is 1,2 distearoyl-sn-glycero-3-phosphocholine (DSPC). In some embodiments, the sterol is cholesterol.

[0022] In some embodiments, the ionizable amino lipid has the structure of Compound I:

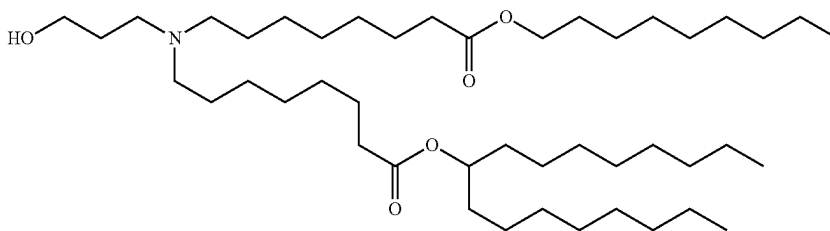
(Compound I)



(heptadecan-9-yl 8-((2-hydroxyethyl)(6-oxo-6-(undecyloxy)hexyl)amino) octanoate). In some embodiments, the LNP comprises about 2.5 mol % PEG-DMG, 11 mol % DSPC, 38.5 mol % cholesterol, and 48 mol % Compound I.

[0023] In some embodiments, the ionizable amino lipid has the structure of Compound II:

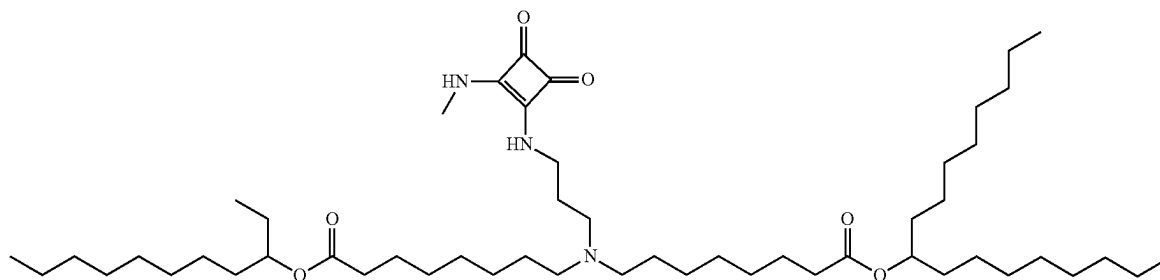
(Compound II)



(heptadecan-9-yl 8-((2-hydroxyethyl)(8-(nonyloxy)-8-oxooctyl)amino) octanoate). In some embodiments, the LNP comprises about 3 mol % 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetradecacontaoxatetracontahexyl stearate 11 mol % DSPC, 39 mol % cholesterol, and 47 mol % Compound II.

[0024] In some embodiments, the ionizable amino lipid has the structure of Compound III:

(Compound III)

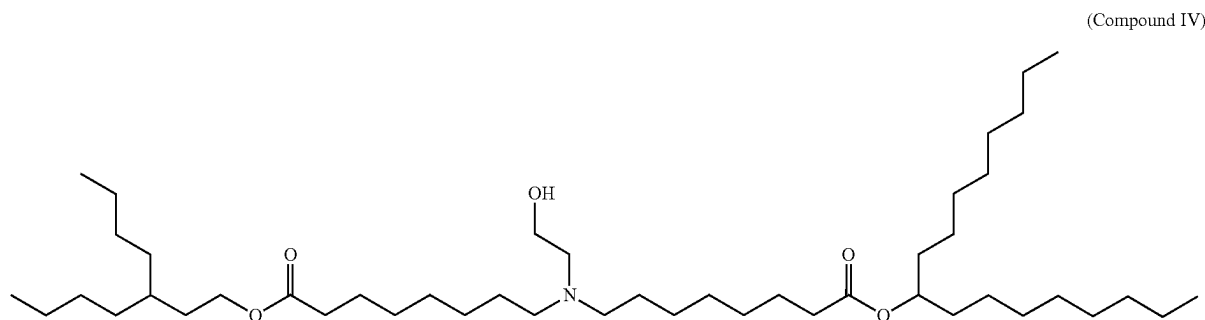


(heptadecan-9-yl 8-((3-((2-(methylamino)-3,4-dioxoclobut-1-en-1-yl)amino)propyl)(8-oxo-8-(undecan-3-yloxy)octyl)amino)octanoate). In some embodiments, the LNP comprises about 3 mol % 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetracontaoxatetratriacontahexyl stearate, 11 mol % DSPC, 39 mol % cholesterol, and 47 mol % Compound III.

[0025] In some embodiments, the ionizable amino lipid has the structure of Compound IV:

hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetracontaoxatetratriacontahexyl stearate is in the core of the LNP.

[0028] The disclosure, in some aspects, provides a composition comprising a lipid nanoparticle (LNP) and a nucleic acid, wherein the LNP comprises about 3 mol % 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetracon-



(3-butylheptyl 8-((8-(heptadecan-9-yloxy)-8-oxooctyl)(2-hydroxyethyl)amino)octanoate). In some embodiments, the LNP comprises about 3 mol % 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetracontaoxatetratriacontahexyl stearate, 11 mol % DSPC, 39 mol % cholesterol, and 47 mol % Compound IV.

[0026] The disclosure, in some aspects, provides a composition comprising a lipid nanoparticle (LNP) and a nucleic acid, wherein the LNP comprises about 2.5 mol % PEG-DMG, 11 mol % DSPC, 38.5 mol % cholesterol, and 48 mol % Compound I, and wherein about 0.25 to about 0.5 mol % of the PEG-DMA is in the core of the LNP. 10 The disclosure, in some aspects, provides a composition comprising a lipid nanoparticle (LNP) and a nucleic acid, wherein the LNP comprises about 3 mol % 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetracontaoxatetratriacontahexyl stearate, 11 mol % DSPC, 39 mol % cholesterol, and 47 mol % Compound II, and wherein about 0.25 to about 0.5 mol % of the 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetracontaoxatetratriacontahexyl stearate is in the core of the LNP.

[0027] The disclosure, in some aspects, provides a composition comprising a lipid nanoparticle (LNP) and a nucleic acid, wherein the LNP comprises about 3 mol % 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetracontaoxatetratriacontahexyl stearate, 11 mol % DSPC, 39 mol % cholesterol, and 47 mol % Compound III, and wherein about 0.25 to about 0.5 mol % of the 134-

taoxatetratriacontahexyl stearate, 11 mol % DSPC, 39 mol % cholesterol, and 47 mol % Compound IV, and wherein about 0.25 to about 0.5 mol % of the 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetracontaoxatetratriacontahexyl stearate is in the core of the LNP.

[0029] Other advantages and novel features will become apparent from the following detailed description of various non-limiting embodiments when considered in conjunction with the accompanying figures. In cases where the present specification and a document incorporated by reference include conflicting and/or inconsistent disclosure, the present specification shall control. If two or more documents incorporated by reference include conflicting and/or inconsistent disclosure with respect to each other, then the document having the later effective date shall control.

BRIEF DESCRIPTION OF THE DRAWINGS

[0030] Non-limiting embodiments are provided by way of example with reference to the accompanying figures, which are schematic and are not intended to be drawn to scale. In the figures, each identical or nearly identical component illustrated is typically represented by a single numeral. For purposes of clarity, not every component is labeled in every figure, nor is every component of each embodiment shown where illustration is not necessary to allow those of ordinary skill in the art to understand the subject matter. In the figures:

[0031] FIG. 1 is a graph showing the average diameter of process intermediates during the manufacture of an exemplary construct comprising mRNA and the lipid nanoparticle formulations described herein. The amount of PEG-DMG present during mixing was varied for each lot. Material collected during the v-mixed product (VMP), pH adj VMP, and filtered VMP were run neat. Material collected during

TFF Harvest and pre-spike (1 day after TFF Harvest) steps was diluted in PBS to 0.05 mg/ml prior to running on a Wyatt Plate reader in triplicate.

[0032] FIG. 2 is a graph showing the pH adjusted VMP particle size plotted against the time after V-mixing. Aliquots were run neat on a Wyatt DLS Plate reader. Aliquot was stored at room temperature.

[0033] FIG. 3 is a graph showing the filtered VMP particle size plotted against the time after V-mixing. Aliquots were run neat on a Wyatt DLS Plate reader. Aliquot was stored at room temperature.

[0034] FIGS. 4A-4B are graphs showing pentamer protein expression levels as a function of total mol % PEG-DMG. The expression was determined using flow cytometry. The mean fluorescent intensity is plotted against mol % Total PEG-DMG. Cells were given a A) 250 ng mRNA dose (FIG. 4A) or 1000 ng mRNA dose (FIG. 4B). Error bars represent standard error of the mean (SEM).

[0035] FIG. 5 is a graph showing that the amount of PEG-DMG present during V-mixing has a significant impact on the pentamer specific titer levels in Balb/c mice. Mice were given 1 ug doses on days 1 and 22. Amount in core refers to amount of PEG-DMG present during V-mixing. PA refers to lipid mol % of PEG-DMG added post addition. Serum was harvested on Day 21 and Day 43.

[0036] FIG. 6 is a graph showing in vitro mRNA expression levels resulting from mRNA formulated in lipid nanoparticles (LNPs) comprising different percentages of components (ionizable amino lipid:neutral lipid:sterol:PEG lipid).

DETAILED DESCRIPTION

[0037] The present disclosure is based, at least in part, on the discovery that the concentration of PEG-stabilized lipids in lipid nanoparticles (LNPs) can impact an LNP's biophysical properties, in vitro expression, in vivo immunogenicity, and long term stability. As described herein, the LNPs can comprise ionizable amino (cationic) lipid, neutral lipid, sterol, and PEG lipid components. Without wishing to be bound by theory, it is thought that the addition of steric stabilizers, such as polyethylene glycol (PEG) at different stages of the production process, increase the long term stability of LNPs, while preserving mRNA functionality, in the vial and in serum at temperatures above freezing. As described herein, the timing of the addition of PEG lipid components during the manufacturing process can impact the LNP's properties. In particular, it was found that the addition of PEG lipids during the nanoprecipitation step (following pH adjust) and/or during excipient addition enhances the stability of the LNP while preserving the biological activity of the RNA/LNP.

Synthesis of Lipid Nanoparticles

[0038] The present disclosure provides methods of producing a nucleic acid LNP composition having enhanced stability, even at temperatures well above freezing. The stabilized LNP comprises, in some embodiments 2.5 mol % PEG-lipid or more, such as from 2.5 mol % to 3.0 mol % PEG-lipid. The PEG lipid components can be added during the nanoprecipitation step and/or during excipient addition, which can maintain the potency of the LNP (and nucleic acid) while enhancing its stability for six months or more.

[0039] Some embodiments comprise mixing a lipid solution comprising a first PEG lipid to produce a precursor lipid nanoparticle. In some embodiments, the method comprises mixing a lipid solution comprising an ionizable lipid and a first PEG-lipid, or a precursor lipid nanoparticle with a first PEG-lipid, with a solution comprising a nucleic acid (e.g., mRNA) thereby forming a precursor nucleic acid lipid nanoparticle and then adding a lipid nanoparticle modifier comprising an additional, e.g., second, PEG-lipid (which can be the same or different from the first PEG-lipid) to the precursor nucleic acid lipid nanoparticle thereby forming a modified nucleic acid lipid nanoparticle. Some embodiments further comprise adding a third PEG-lipid (which can be the same or different from the first and/or second PEG-lipid), thereby forming a further modified nucleic acid lipid nanoparticle. The modified and/or further modified nucleic acid lipid nanoparticle can then be processed, forming the nucleic acid lipid nanoparticle composition. Some embodiments comprise further PEG addition steps, e.g., fourth, fifth, sixth, etc., PEG addition steps. In some embodiments, a PEG-lipid is added in stages. For example, a second PEG-lipid can be added in two, three, four, five, or more stages. Moreover, one or more stages can be added at a different point in the production process, e.g., before or after different processing steps. In some embodiments, one or more stages can be performed before or after the same processing step.

[0040] In some embodiments, the precursor nucleic acid lipid nanoparticle (which may or may not contain an initial amount of a lipid nanoparticle modifier) is prepared prior to adding a lipid nanoparticle modifier. As used herein, such embodiments in which a modifier is added after formation of a precursor lipid nanoparticle may be referred to as "post addition" methods or steps. Some embodiments comprise using two or more post addition steps.

[0041] In some embodiments, the lipid solution does not include a PEG lipid. In some embodiments, the lipid solution comprises a first PEG lipid. In some embodiments, the lipid solution comprises 0.25 mol % or more of the first PEG lipid, such as 0.5 mol %, 0.75 mol %, 1 mol %, or more of the first PEG lipid.

[0042] In certain embodiments, the first PEG lipid is PEG-DMG. In some embodiments, the lipid solution comprises 0.25 mol % of PEG-DMG. In certain embodiments, the lipid solution comprises 0.50 mol % of PEG-DMG. In some embodiments, the lipid solution comprises 0.75 mol % of PEG-DMG. In certain embodiments, the lipid solution comprises 1 mol % of PEG-DMG.

[0043] In certain embodiments, the first PEG lipid is 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetracontaoxatetrahectyl stearate. In some embodiments, the lipid solution comprises 0.25 mol % of 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetracontaoxatetrahectyl stearate. In certain embodiments, the lipid solution comprises 0.50 mol % of 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetracontaoxatetrahectyl stearate. In some embodiments, the lipid solution comprises 0.75 mol % of 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,

69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetraconta-oxatetriacontahectyl stearate. In certain embodiments, the lipid solution comprises 1 mol % of 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetraconta-oxatetriacontahectyl stearate.

[0044] In some embodiments, the precursor nucleic acid lipid nanoparticle does not include a PEG lipid. In some embodiments, the precursor nucleic acid lipid nanoparticle comprises a first PEG lipid. In certain embodiments, the first PEG lipid is PEG-DMG. In some embodiments, the precursor nucleic acid lipid nanoparticle comprises 0.25 mol % of PEG-DMG. In certain embodiments, the precursor nucleic acid lipid nanoparticle comprises 0.50 mol % of PEG-DMG. In some embodiments, the precursor nucleic acid lipid nanoparticle comprises 0.75 mol % of PEG-DMG. In certain embodiments, the precursor nucleic acid lipid nanoparticle comprises 1 mol % of PEG-DMG.

[0045] In certain embodiments, the first PEG lipid is 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetraconta-oxatetriacontahectyl stearate. In some embodiments, the lipid solution comprises 0.25 mol % of 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetraconta-oxatetriacontahectyl stearate. In certain embodiments, the lipid solution comprises 0.50 mol % of 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetraconta-oxatetriacontahectyl stearate. In certain embodiments, the lipid solution comprises 0.75 mol % of 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetraconta-oxatetriacontahectyl stearate.

[0046] In some embodiments, the second PEG lipid is PEG-DMG or 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetraconta-oxatetriacontahectyl stearate.

[0047] In some embodiments, the first PEG lipid, the second PEG lipid, and optionally, other additional PEG lipids are the same PEG lipid. In some embodiments, the first PEG lipid and the additional PEG lipid are both PEG-DMG. In some embodiments, the first PEG lipid and the additional PEG lipid are both 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetraconta-oxatetriacontahectyl stearate. In some embodiments, the first PEG lipid and the second PEG lipid are not the same (e.g., the first PEG lipid is PEG-DMG and the second PEG lipid is 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,

117,120,123,126,129,132-tetratetraconta-oxatetriacontahectyl stearate, or the first PEG lipid is 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetraconta-oxatetriacontahectyl stearate and the second PEG lipid is PEG-DMG).

[0048] In some embodiments, the molar ratio of the first PEG lipid to the second PEG lipid is in a range of about 1:100 to about 2:1, for example, about 1:50 to about 2:1, about 1:25 to about 2:1, about 1:10 to about 2:1. In some embodiments, the molar ratio of the first PEG lipid to the second PEG lipid is in a range of about 1:25 to about 2:1, for example, about 1:10 to about 2:1. In some embodiments, the molar ratio of the first PEG lipid to the second PEG lipid is in a range of about 1:10 to about 2:1. In some embodiments, the molar ratio of the first PEG lipid to the second PEG lipid is in a range of about 1:100 to about 1:1, for example, about 1:50 to about 1:1, about 1:25 to about 1:1, about 1:10 to about 1:1. In some embodiments, the molar ratio of the first PEG lipid to the second PEG lipid is in a range of about 1:10 to about 1:1. In some embodiments, the molar ratio of the first PEG lipid to the second PEG lipid is in a range of about 1:5 to about 1:1. In some embodiments, the molar ratio of the first PEG lipid to the second PEG lipid is in a range of about 1:3 to about 1:1.

[0049] In some embodiments, the molar ratio of the first PEG lipid to the additional (e.g., a combination of second, third, fourth, etc., if used) PEG lipid is in a range of about 1:100 to about 2:1, for example, about 1:50 to about 2:1, about 1:25 to about 2:1, about 1:10 to about 2:1. In some embodiments, the molar ratio of the first PEG lipid to the additional PEG lipid is in a range of about 1:25 to about 2:1, for example, about 1:10 to about 2:1. In some embodiments, the molar ratio of the first PEG lipid to the additional PEG lipid is in a range of about 1:10 to about 2:1. In some embodiments, the molar ratio of the first PEG lipid to the additional PEG lipid is in a range of about 1:100 to about 1:1, for example, about 1:50 to about 1:1, about 1:25 to about 1:1, about 1:10 to about 1:1. In some embodiments, the molar ratio of the first PEG lipid to the additional PEG lipid is in a range of about 1:10 to about 1:1. In some embodiments, the molar ratio of the first PEG lipid to the additional PEG lipid is in a range of about 1:5 to about 1:1. In some embodiments, the molar ratio of the first PEG lipid to the additional PEG lipid is in a range of about 1:3 to about 1:1.

[0050] In some embodiments, the first PEG lipid is added to the core of the LNP and the additional PEG lipid is added to the surface of the LNP. Accordingly, in some embodiments, the molar ratio of the core PEG lipid to the surface PEG lipid is in a range of about 1:100 to about 2:1, for example, about 1:50 to about 2:1, about 1:25 to about 2:1, about 1:10 to about 2:1. In some embodiments, the molar ratio of the core PEG lipid to the surface PEG lipid is in a range of about 1:25 to about 2:1, for example, about 1:10 to about 2:1. In some embodiments, the molar ratio of the core PEG lipid to the surface PEG lipid is in a range of about 1:10 to about 2:1. In some embodiments, the molar ratio of the core PEG lipid to the surface PEG lipid is in a range of about 1:100 to about 1:1, for example, about 1:50 to about 1:1, about 1:25 to about 1:1, about 1:10 to about 1:1. In some embodiments, the molar ratio of the core PEG lipid to the surface PEG lipid is in a range of about 1:10 to about 1:1.

In some embodiments, the molar ratio of the core PEG lipid to the surface PEG lipid is in a range of about 1:5 to about 1:1. In some embodiments, the molar ratio of the core PEG lipid to the surface PEG lipid is in a range of about 1:3 to about 1:1.

[0051] The lipid mixture can be solubilized in a water miscible organic solvent, for example, absolute ethanol. In some embodiments, the organic solvent is used in the form in which it is commercially available. In one exemplary embodiment, the mixture of lipids is a mixture of an ionizable amino lipid and a first PEG lipid are co-solubilized in the organic solvent. In some embodiments, the lipid mixture consists essentially of an ionizable amino lipid and a PEG lipid, and optionally a phospholipid (neutral lipid) and/or a structural lipid (e.g., a sterol). The total concentration of lipid is preferably less than 25 mg/ml, preferably less than 5 mg/ml. The lipid mixture may be filtered through a membrane, e.g. a 0.45 or 0.2 μ m filter.

[0052] As described herein, the lipid mixture may be combined with a nucleic acid solution, preferably in the form of a buffered aqueous solution. The buffered aqueous solution may be a solution in which the buffer has a pH less than the pKa of a protonated lipid in the lipid mixture. Examples of suitable buffers include, but are not limited to, citrate, phosphate, and acetate. In some embodiments, the buffer is acetate buffer. In some embodiments, the buffers are in the concentration range of 1-1000 mM of the anion, depending on the chemistry of the nucleic acid being encapsulated, and optimization of buffer concentration may be significant to achieving high loading levels. It may be suitable to add a cryoprotectant, and/or a non-ionic solute, which can balance the osmotic potential across the particle membrane, e.g., when the particles are dialyzed to remove ethanol, increase the pH, or mixed with a pharmaceutically acceptable carrier or diluent. The amount of nucleic acid in buffer is preferably from about 0.01 to 1.0 mg/mL, preferably 0.08 to 0.8 mg/mL.

[0053] At the time of addition of the lipid solution (e.g., ethanol), the temperature of the aqueous nucleic acid solution is 25 to 45° C., for example, 30 to 40° C. In some embodiments, briefly heating the aqueous nucleic acid solution at elevated temperature may be useful, e.g., 1-2 minutes at 65° C. The lipid solution may be added to the aqueous solution either by spraying on the air-water interface, in a narrow stream, or through a liquid-liquid interface between lipid solution delivered through a tube that is submerged in the aqueous nucleic acid solution.

[0054] The organic lipid solution may be added by gravity or by a pump delivering the organic lipid solution to the aqueous nucleic acid solution at a controlled rate, preferably a constant rate. In some embodiments, the delivery of the organic lipid is continuous (e.g., by a pump operating under continuous flow). The delivery of the organic lipid solution can be completed in 1 minute to 6 hours, in 1 minute to 100 minutes, or in 1 to 25 minutes. The organic lipid solution may be added through a single spray or stream, through a tube or outlet, or through a multi-outlet system. While the lipid organic solution is added into the nucleic acid aqueous solution, the resulting solution may be mixed by stirring, shaking, or recirculation. As used herein, “mixing” preferably comprises turbulent mixing (“T-mix”), vortex mixing (“V-mix”), microfluidic mixing, or a combination thereof. In some embodiments, “mixing” comprises vortex mixing. The addition/mixing step results in a final concentration that is

10 to 45% ethanol, for example 11 to 30% ethanol, or 12.5 to 25% ethanol. Preferably, formation involves either turbulent or microfluidic mixing of solutions to induce precipitation lipids in organic phase with nucleic acid in aqueous phase, or extrusion of an already phase-separated mixture of nucleic acid and lipids through membranes to create LNPs.

[0055] In one step of the process a lipid solution comprising a first PEG lipid can be mixed with a solution comprising a nucleic acid thereby forming a precursor nucleic acid lipid nanoparticle. In some embodiments, the nucleic acid is provided. In some embodiments, the nucleic acid is any one of the nucleic acids provided herein.

[0056] In some embodiments, precursor lipid nanoparticles are provided. As used herein, a “precursor lipid nanoparticle” refers to a lipid nanoparticle that is further modified by additional structural components, e.g., lipids and/or nucleic acids to produce a subsequent precursor lipid nanoparticle or final lipid nanoparticle. In some embodiments, a precursor lipid nanoparticle may be formed and/or exist during one or more steps in the particle formulation process. In some embodiments, multiple precursor lipid nanoparticles (e.g., a first, second, third, etc. precursor lipid nanoparticle) are formed during preparation of a final lipid nanoparticle. In some embodiments, in which a lipid nanoparticle comprises a PEG molecule, the precursor lipid nanoparticle may comprise a relatively low percentage of PEG molecules (e.g., at least about 0.01 mol % and less than or equal to about 1.0 mol %, at least about 0.05 mol %, at least about 0.1 mol %, at least about 0.2 mol %, at least about 0.3 mol %, at least about 0.4 mol %, at least about 0.5 mol %, at least about 0.6 mol %, at least about 0.7 mol %, or 0.8 mol %).

[0057] “Precursor nucleic acid lipid nanoparticle” refers to a lipid nanoparticle that comprises a nucleic acid that is further modified by adding additional structural components, e.g., lipids and/or nucleic acids. In some embodiments, all of the nucleic acid in the precursor nucleic acid lipid nanoparticle is associated with the ionizable lipid. In some embodiments, between about 80% and about 100%, between about 85% and about 100%, or between about 90% and about 100% of the nucleic acid in the precursor nucleic acid lipid nanoparticle is associated with the ionizable lipid, preferably about 95% to about 100%, preferably about 98% to about 100%, preferably about 99% to about 100%.

[0058] In some embodiments, in which a lipid nanoparticle comprises a PEG molecule, the precursor lipid nanoparticle may have more nucleic acid associated with the ionizable lipid than the PEG molecule. For instance, at least about 50%, at least about 60%, at least about 70%, at least about 80%, or at least about 90% of the nucleic acid in the precursor lipid nanoparticle is associated with the ionizable lipid. In some such cases, less than about 50%, less than about 40%, less than about 30%, less than about 20%, or less than about 10% of the nucleic acid in the precursor lipid nanoparticle is associated with the PEG molecule (e.g., PEG lipid). In some embodiments, a ratio of nucleic acid associated with the ionizable lipid to nucleic acid associated with the PEG lipid in the precursor lipid nanoparticles is at least about 2:1. In some embodiments, a composition comprising precursor lipid nanoparticles may comprise one or more organic solvents (e.g., ethanol). In some embodiments, the nucleic acid lipid nanoparticle composition may be enriched in precursor lipid nanoparticles. For instance, at least about

50% of the lipid nanoparticles in the nucleic acid lipid nanoparticle composition may be precursor lipid nanoparticles.

[0059] In some embodiments, the precursor nucleic acid lipid nanoparticle comprises about 30-60 mol % ionizable lipid; about 0-30 mol % phospholipid (neutral lipid); about 15-50 mol % structural lipid; and about 0.01-10 mol % first PEG lipid. In some embodiments, the precursor nucleic acid lipid nanoparticle comprises about 30-60 mol % ionizable lipid; about 0-30 mol % phospholipid; about 15-50 mol % structural lipid; and about 0.01-1 mol % first PEG lipid. In some embodiments, the precursor nucleic acid lipid nanoparticle comprises about 40-60 mol % ionizable lipid; about 5-15 mol % phospholipid; about 35-45 mol % structural lipid; and about 0.01-10 mol % first PEG lipid. In some embodiments, the precursor nucleic acid lipid nanoparticle comprises about 40-60 mol % ionizable lipid; about 5-15 mol % phospholipid; about 35-45 mol % structural lipid; and about 0.01-1 mol % first PEG lipid. In some embodiments, the precursor nucleic acid lipid nanoparticle comprises about 40-60 mol % ionizable lipid; about 5-15 mol % phospholipid; about 35-45 mol % structural lipid; and about 0.25-1 mol % of the first PEG lipid. In some embodiments, the precursor nucleic acid lipid nanoparticle comprises about 40-60 mol % ionizable lipid; about 5-15 mol % phospholipid; about 35-45 mol % structural lipid; and about 0.5-1 mol % first PEG lipid. In some embodiments, the precursor nucleic acid lipid nanoparticle comprises about 30-60 mol % ionizable lipid; about 0-30 mol % phospholipid; about 15-50 mol % structural lipid; and about 0.01-0.5 mol % first PEG lipid. In some embodiments, the precursor nucleic acid lipid nanoparticle comprises about 30-60 mol % ionizable lipid; about 0-30 mol % phospholipid; about 15-50 mol % structural lipid; and about 0.25-1.0 mol % first PEG lipid. In some embodiments, the precursor nucleic acid lipid nanoparticle comprises about 30-60 mol % ionizable lipid; about 0-30 mol % phospholipid; about 15-50 mol % structural lipid; and about 0.5-1.0 mol % first PEG lipid.

[0060] Some embodiments comprise processing a precursor LNP. Some embodiments comprise adding a PEG lipid before, during, and/or after one or more processing steps. In some embodiments, the processing may involve treating to remove an organic solvent (i.e., ethanol), by dialysis or filtration, preferably by diafiltration. As used herein, "processing" includes steps to purify, pH adjust, buffer exchange, and/or concentrate LNPs. In some embodiments, the processing comprises a filtration such as a sterile filtration. In some embodiments, the processing comprises a tangential flow filtration (TFF). While the ethanol is removed, the aqueous solution can be converted to a one buffered at a neutral pH, for example, pH 6.5 to 7.8, pH 6.8 to pH 7.5, pH 7.0 to pH 7.5, or pH 7.0 to pH 7.2, for example using a phosphate or HEPES buffer. The resulting aqueous solution is preferably sterilized before storage or use, such as, for example by filtration through a 0.22 μ m filter.

[0061] In some embodiments, the processing may comprise a freezing and/or lyophilizing. Lyophilizing steps may be carried out in a suitable glass receptacle, preferably a 1 ml to 10 ml (e.g., 3 ml), cylindrical glass vial. The glass vial should withstand extreme changes in temperatures of less

than -40° C. and greater than room temperature in short periods of time, and be cut in a uniform shape. The composition comprising the nucleic acid lipid nanoparticle can be added to the vial, preferably in a volume ranging from about 0.1 ml to about 5 ml, from 0.2 ml to about 3 ml, from 0.3 ml to about 1 ml, or from about 0.4 ml to about 0.8 ml (e.g., about 0.5 ml), and preferably with about 9 mg/ml lipid. The step of lyophilizing may comprise freezing the composition at a temperature of greater than about -40° C., or e.g. less than about -30° C., forming a frozen composition; and drying the frozen composition to form the lyophilized composition. The freezing step preferably results in a linear decrease in temperature to the final over about 100 to 180 minutes (e.g., about 130 minutes), preferably at 0.1 to 1° C./minute (e.g., about 0.5° C./minute) from 20 to -40° C. More preferably, sucrose at 5-15% (e.g., 8-12%) may be used, and the drying step is at about 50-150 mTorr, first at a low temperature of about -15 to about -35° C., and thereafter at a higher temperature of room temperature to about 25° C., and is completed in three to seven days. In another embodiment of the present disclosure the drying step is at about 50-100 mTorr, first at a low temperature of about -40° C. to about -20° C., and then at the higher temperature.

[0062] In some embodiments, the method may further comprise packing the nucleic acid lipid nanoparticle composition. As used herein, "storage" or "packing" may refer to storing drug product in its final state or in-process storage of LNPs before they are placed into final packaging. Modes of storage and/or packing include, but are not limited to refrigeration in sterile bags, refrigerated or frozen formulations in vials, lyophilized formulations in vials and syringes, etc.

[0063] In some embodiments, the concentration of the non-ionic surfactant in the nucleic acid LNP formulation ranges from about 0.00001% w/v to about 1% w/v, e.g., from about 0.00005% w/v to about 0.5% w/v, or from about 0.0001% w/v to about 0.1% w/v.

[0064] In some embodiments, the concentration of the non-ionic surfactant in the nucleic acid LNP formulation ranges from about 0.000001 wt % to about 1 wt %, e.g., from about 0.000002 wt % to about 0.8 wt %, or from about 0.000005 wt % to about 0.5 wt %.

[0065] In some embodiments, the concentration of the PEG lipid in the stabilized LNP formulation ranges from about 0.01% by molar to about 50% by molar, e.g., from about 0.05% by molar to about 20% by molar, from about 0.07% by molar to about 10% by molar, from about 0.1% by molar to about 8% by molar, from about 0.2% by molar to about 5% by molar, or from about 0.25% by molar to about 3% by molar. In some embodiments, the concentration of the PEG lipid in the stabilized LNP formulation is about 2.5% or 3% by molar.

[0066] In some embodiments, the distribution of one or more components in the lipid nanoparticle may be dictated, at least in part, by the process by which the components are assembled. For instance, in some embodiments, the distribution (e.g., accessibility, arrangement) of nucleic acid (e.g., mRNA) within the lipid nanoparticle may be controlled, at least in part, by the formulation process. For example, the formulation process may comprise one or more steps that allow the distribution of mRNA to be tailored, as described in more detail below. For example, the formulation process may use a relatively low weight percentage of certain

components (e.g., PEG lipid) during the particle formation step (e.g., nanoprecipitation reaction) and/or add certain lipid nanoparticle components after particle formation.

[0067] In some embodiments, a lipid nanoparticle and/or composition, described herein, may have a beneficial amount of nucleic acid (e.g., mRNA) that is at least partially (e.g., fully) encapsulated. For instance, in some embodiments, at least about 30%, at least about 35%, at least about 40%, at least about 45%, at least about 50%, at least about 55%, at least about 60%, at least about 65%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, or at least about 95% of the total amount of nucleic acid in the lipid nanoparticle and/or a composition is at least partially (e.g., fully) encapsulated. In some embodiments, less than or equal to about 70% or less than or equal to about 90% of the total amount of nucleic acid in the lipid nanoparticle and/or a composition is at least partially (e.g., fully) encapsulated. In some embodiments, the percentage of at least partially (e.g., fully) encapsulated nucleic acid may be determined by an *in vitro* assay (e.g., IEX) as described herein.

[0068] In some embodiments, the resulting LNPs undergo further processing, such as one or more steps to purify, pH adjust, buffer exchange, and/or concentrate LNPs. In some embodiments, the step of processing the LNP solution comprises filtering the LNP solution.

[0069] In some embodiments, the filtration removes an organic solvent (e.g., ethanol) from the LNP solution. In some embodiments, the processing comprises a filtration such as a sterile filtration. In some embodiments, the processing comprises a tangential flow filtration (TFF). In some embodiments, upon removal of the organic solvent (e.g., ethanol), the LNP solution is converted to a solution buffered at a neutral pH, pH 6.5 to 7.8, pH 6.8 to pH 7.5, pH 7.0 to pH 7.5, or pH 7.0 to pH 7.2 (e.g., a phosphate or HEPES buffer). In some embodiments, the resulting LNP solution is preferably sterilized before storage or use, e.g., by filtration (e.g., through a 0.22 μ m filter).

[0070] In some embodiments, the step of processing the LNP solution further comprises packing the LNP solution. Modes of storage and/or packing include, but are not limited to refrigeration in sterile bags, refrigerated or frozen formulations in vials, lyophilized formulations in vials and syringes, etc.

[0071] In some embodiments, the step of packing the LNP solution comprises one or more of the following steps: adding a cryoprotectant to the LNP solution; and lyophilizing the LNP solution, thereby forming a lyophilized LNP composition; storing the LNP solution or the lyophilized LNP composition; and adding a reconstituting solution to the LNP solution or the lyophilized LNP composition, thereby forming the LNP formulation.

[0072] In some embodiments, the cryoprotectant is added to the LNP solution prior to the lyophilization. In some embodiments, the cryoprotectant comprises one or more cryoprotective agents, and each of the one or more cryoprotective agents is independently a polyol (e.g., a diol or a triol such as propylene glycol (i.e., 1,2-propanediol), 1,3-propanediol, glycerol, (+/-)-2-methyl-2,4-pentanediol, 1,6-hexanediol, 1,2-butanediol, 2,3-butanediol, ethylene glycol, or diethylene glycol), a nondetergent sulfobetaine (e.g., NDSB-201 (3-(1-pyridino)-1-propane sulfonate), an osmolyte (e.g., L-proline or trimethylamine N-oxide dihydrate), a polymer (e.g., polyethylene glycol 200 (PEG 200), PEG

400, PEG 600, PEG 1000, PEG 3350, PEG 4000, PEG 8000, PEG 10000, PEG 20000, polyethylene glycol monomethyl ether 550 (mPEG 550), mPEG 600, mPEG 2000, mPEG 3350, mPEG 4000, mPEG 5000, polyvinylpyrrolidone (e.g., polyvinylpyrrolidone K 15), pentacrythritol propoxylate, or polypropylene glycol P 400), an organic solvent (e.g., dimethyl sulfoxide (DMSO) or ethanol), a sugar (e.g., D-(+)-sucrose, D-sorbitol, trehalose, D-(+)-maltose monohydrate, meso-erythritol, xylitol, myo-inositol, D-(+)-raffinose pentahydrate, D-(+)-trehalose dihydrate, or D-(+)-glucose monohydrate), or a salt (e.g., lithium acetate, lithium chloride, lithium formate, lithium nitrate, lithium sulfate, magnesium acetate, sodium chloride, sodium formate, sodium malonate, sodium nitrate, sodium sulfate, or any hydrate thereof), or any combination thereof. In some embodiments, the cryoprotectant comprises sucrose.

[0073] In some embodiments, the lyophilization carried out in a suitable glass receptacle (e.g., a 2, 3, 5, or 10 ml cylindrical glass vial). The glass receptacle is able to withstand extreme changes in temperatures between lower than -40° C. and higher than room temperature in short periods of time, and/or be cut in a uniform shape. In some embodiments, the step of lyophilizing comprises freezing the LNP solution at a temperature lower than about -40° C., thereby forming a frozen LNP solution; and drying the frozen LNP solution to form the lyophilized LNP composition. The freezing step, in some embodiments, results in a linear decrease in temperature to the final over about 100 to 180 minutes (e.g., about 130 minutes) or at 0.1 to 1° C./minute (e.g., about 0.5° C./minute) from 20 to -40° C. In some embodiments, sucrose at 5-15% (e.g., 8-12%) may be used, and the drying step is performed at a vacuum ranging from about 50 mTorr to about 150 mTorr, for example, first at a low temperature lower than -10° C. (e.g., from about -35° C. to about -15° C.), lower than -20° C., lower than -30° C., or lower than -40° C., and then at a higher temperature ranging from room temperature to about 25° C. In some embodiments, the drying step is completed in three to seven days. In some embodiments, the drying step is performed at a vacuum ranging from about 50 mTorr to about 100 mTorr, for example, first at a low temperature below about 0° C., below about -10° C., below about -20° C., or below about -30° C. (e.g., -35° C.), and then at a higher temperature.

[0074] In some embodiments, the LNP solution or the lyophilized LNP composition is stored at a temperature of about -40° C., about -35° C., about -30° C., about -25° C., about -20° C., about -15° C., about -10° C., about -5° C., about 0° C., about 5° C., about 10° C., about 15° C., about 20° C., or about 25° C. prior to adding the reconstituting solution.

[0075] In some embodiments, the LNP solution or the lyophilized LNP composition is stored at a temperature of ranging from about -40° C. to about 0° C., from about -35° C. to about -5° C., from about -30° C. to about -10° C., from about -25° C. to about -15° C., from about -22° C. to about -18° C., or from about -21° C. to about -19° C. prior to adding the reconstituting solution.

[0076] In some embodiments, the LNP solution or the lyophilized LNP composition is stored at a temperature of about -20° C. prior to adding the reconstituting solution.

[0077] In some embodiments, the LNP solution or the lyophilized LNP composition is stored at a temperature of ranging from about -15° C. to about 25° C., from about -10° C. to about 20° C., from about -5° C. to about 15° C., from

about 0° C. to about 10° C., from about 1° C. to about 9° C., or from about 2° C. to about 8° C. prior to adding the reconstituting solution.

[0078] In some embodiments, the LNP solution or the lyophilized LNP composition is stored for about 30 minutes, about 1 hour, about 2 hours, about 4 hours, about 6 hours, about 12 hours, about 1 day, about 2 days, about 3 days, about 4 days, about 5 days, about 6 days, about 1 week, about 2 weeks, about 3 weeks, about 4 weeks, about 1 month, about 2 months, about 3 months, about 4 months, about 6 months, about 9 months, about 1 year, about 2 years, about 3 years, about 4 years, about 5 years, about 6 years, about 7 years about 8 years, about 9 years, or about 10 years prior to adding the reconstituting solution.

[0079] In some embodiments, the LNP solution or the lyophilized LNP composition is stored for a time period ranging from about 1 month to about 10 years, from about 3 months to about 8 years, from about 6 months to about 6 years, from about 9 months to about 4 years, from about 1 year to about 3 years, or from about 1.5 years to about 2.5 years prior to adding the reconstituting solution.

[0080] In some embodiments, the LNP solution or the lyophilized LNP composition is stored for about 2 years prior to adding the reconstituting solution.

[0081] In some embodiments, the accessibility of the nucleic acid in a LNP composition comprising LNPs may be determined by one or more assays (e.g., in vitro assay). In general, any suitable in vitro assay may be used. Suitable assays are able to distinguish between different encapsulation states of the nucleic acid and/or association states of the nucleic acid with components of the lipid nanoparticle. For example, the accessibility of a nucleic acid may be determined by an ion-exchange chromatography (IEX) assay. In some embodiments, the in vitro assay may be used to generate a quantitative value of the amount of accessible or inaccessible nucleic acids (e.g., mRNA) in the lipid nanoparticles or composition. For example, an ion-exchange chromatography (IEX) assay may be used to generate a quantitative value of the amount of accessible or inaccessible mRNA in a composition comprising lipid nanoparticles. In general, the amount of inaccessible or accessible nucleic acids may be determined for the total composition and/or a fraction of the composition (e.g., fraction comprising certain lipid nanoparticles).

[0082] In some embodiments, the accessibility of the nucleic acid within the lipid nanoparticle may correlate to one or more biological properties of the lipid nanoparticle. In certain embodiments, the accessibility of the nucleic acid within the lipid nanoparticle may correlate with protein expression levels and/or the efficacy of intracellular nucleic acid delivery. For instance, in some embodiments, a relatively high percentage of inaccessible nucleic acid, and accordingly a relatively low percentage of accessible nucleic acid, may produce high levels of protein expression (e.g., in vitro, in vivo). In such cases, a composition having a low percentage of accessible mRNA may have a higher level of mRNA expression than a comparative composition having a higher percentage of accessible mRNA.

[0083] In some aspects, the present disclosure provides a method of characterizing a LNP composition (e.g., the LNP composition prepared by a method of the present disclosure) using a chromatography assay.

[0084] In some embodiments, a quantitative value of an amount of the nucleic acid (e.g., mRNA) encapsulated in the LNP composition is measured using the chromatography assay.

[0085] In some embodiments, the chromatography assay is an ion-exchange (IEX) chromatography assay.

[0086] In some embodiments, at least about 50%, at least about 55%, at least about 60%, at least about 65%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 95%, or at least about 95% of the LNPs in the LNP composition have mRNA encapsulated therein, as determined by the ion-exchange chromatography (IEX) assay.

[0087] An ion exchange (IEX) chromatography method determines encapsulation efficiency for mRNAs encapsulated in ionizable-lipid-based LNPs. IEX chromatography can be used to separate bound versus free mRNA. The IEX screening method separates free mRNA from LNPs when there is a gradient change from low to high salt concentration. The LNPs elute in the void (peak 1) and mRNA elutes when gradient changes from low to high salt concentration (peak 2, termed “accessible mRNA”).

[0088] Without wishing to being bound in theory, it is believed that within a population of LNPs (e.g., LNPs encapsulating mRNA), mRNA can exist in a variety of different encapsulation states, including, for example, fully encapsulated, surface-associated, loosely encapsulated (or other physical states). To exemplify the utility of the IEX method of the invention, a LNP sample population can be subjected to an art-recognized separation technique, for example, size-exclusion chromatography (SEC). This fractionates particles based on size. Fractions can be subjected, for example, to a biological assay, e.g., in vitro protein expression assay. Fractions can likewise be subjected to determination of encapsulation efficiency according to the IEX methods of the invention.

Lipid Nanoparticle Formulations

[0089] A lipid nanoparticle (LNP) refers to a nanoscale construct (e.g., a nanoparticle, typically less than 100 nm in diameter) comprising lipid molecules, preferably arranged in a substantially spherical (e.g., spheroid) geometry, sometimes encapsulating one or more additional molecular species. In some embodiments, the LNP contains a bleb region, e.g., as described in Brader et al., *Biophysical Journal* 120:1-5 (2021). A LNP may comprise one or more types of lipids, including but not limited to amino lipids (e.g., ionizable amino lipids), neutral lipids, neutral lipids, charged lipids, PEG-modified lipids, phospholipids, structural lipids and sterols. In some embodiments, a LNP may further comprise one or more cargo molecules, including but not limited to nucleic acids (e.g., mRNA, plasmid DNA, DNA or RNA oligonucleotides, siRNA, shRNA, snRNA, snoRNA, lncRNA, etc.), small molecules, proteins, and peptides. A LNP may have a unilamellar structure (i.e., having a single lipid layer or lipid bilayer surrounding a central region) or a multilamellar structure (i.e., having more than one lipid layer or lipid bilayer surrounding a central region). In some embodiments, a lipid nanoparticle may be a liposome. A liposome is a nanoparticle comprising lipids arranged into one or more concentric lipid bilayers around a central region. The central region of a liposome may comprise an aqueous solution, suspension, or other aqueous composition.

[0090] In some embodiments, nucleic acids are formulated as lipid nanoparticle (LNP) compositions. Lipid nanoparticles typically comprise amino lipid, phospholipid, structural lipid and PEG lipid components along with the nucleic acid cargo of interest. The lipid nanoparticles provided herein can be generated using components, compositions, and methods as are generally known in the art, see for example PCT/US2016/052352; PCT/US2016/068300; PCT/US2017/037551; PCT/US2015/027400; PCT/US2016/047406; PCT/US2016/000129; PCT/US2016/014280; PCT/US2017/038426; PCT/US2014/027077; PCT/US2014/055394; PCT/US2016/052117; PCT/US2012/069610; PCT/US2017/027492; PCT/US2016/059575; PCT/US2016/069491; PCT/US2016/069493; and PCT/US2014/066242, all of which are incorporated by reference herein in their entirety. Each optional component of the lipid nanoparticles described herein is described below.

Polyethylene Glycol (PEG)-Lipids

[0091] Effective *in vivo* delivery of nucleic acids represents a continuing medical challenge. Exogenous nucleic acids (i.e., originating from outside of a cell or organism) are readily degraded in the body, e.g., by the immune system. Accordingly, effective delivery of nucleic acids to cells often requires the use of a particulate carrier (e.g., lipid nanoparticles). The particulate carrier should be formulated to have minimal particle aggregation, be relatively stable prior to intracellular delivery, effectively deliver nucleic acids intracellularly, and illicit no or minimal immune response. To achieve minimal particle aggregation and pre-delivery stability, many conventional particulate carriers have relied on the presence and/or concentration of certain components (e.g., PEG-lipid). However, it has been discovered that certain components may decrease the stability of encapsulated nucleic acids (e.g., mRNA molecules). The reduced stability may limit the broad applicability of the particulate carriers. As such, there remains a need for methods by which to improve the stability of nucleic acid (e.g., mRNA) encapsulated within lipid nanoparticles.

[0092] As described herein, the lipid nanoparticles (LNPs) may comprise one or more polyethylene glycol (PEG) lipids. As used herein, the term “PEG-lipid” or “PEG-modified lipid” refers to polyethylene glycol (PEG)-modified lipids. Non-limiting examples of PEG-lipids include PEG-modified phosphatidylethanolamine and phosphatidic acid, PEG-ceramide conjugates (e.g., PEG-CerC14 or PEG-CerC20), PEG-modified dialkylamines and PEG-modified 1,2-diacyloxypropan-3-amines. Such lipids are also referred to as PEGylated lipids. For example, a PEG lipid can be PEG-c-DMOG, PEG-DMG, PEG-DLPE, PEG-DMPE, PEG-DPPC, or a PEG-DSPE lipid.

[0093] In some embodiments, the PEG-lipid includes, but is not limited to 1,2-dimyristoyl-sn-glycerol methoxypoly-

ethylene glycol (PEG-DMG), 1,2-distearoyl-sn-glycero-3-phosphoethanolamine-N-[amino (polyethylene glycol)] (PEG-DSPE), PEG-disteryl glycerol (PEG-DSG), PEG-dipalmitoyl, PEG-dioleoyl, PEG-distearoyl, PEG-diacylglycamide (PEG-DAG), PEG-dipalmitoyl phosphatidylethanolamine (PEG-DPPE), or PEG-1,2-dimyristyloxypropyl-3-amine (PEG-c-DMA).

[0094] In some embodiments, the PEG-lipid is selected from the group consisting of a PEG-modified phosphatidylethanolamine, a PEG-modified phosphatidic acid, a PEG-modified ceramide, a PEG-modified dialkylamine, a PEG-modified diacylglycerol, a PEG-modified dialkylglycerol, and mixtures thereof. In some embodiments, the PEG-modified lipid is PEG-DMG, PEG-c-DOMG (also referred to as PEG-DOMG), PEG-DSG and/or PEG-DPG.

[0095] In some embodiments, the lipid moiety of the PEG-lipids includes those having lengths of from about C₁₄ to about C₂₂, preferably from about C₁₄ to about C₁₆. In some embodiments, a PEG moiety, for example an mPEG-NH₂, has a size of about 1000, 2000, 5000, 10,000, 15,000 or 20,000 daltons. In some embodiments, the PEG-lipid is PEG_n-DMG.

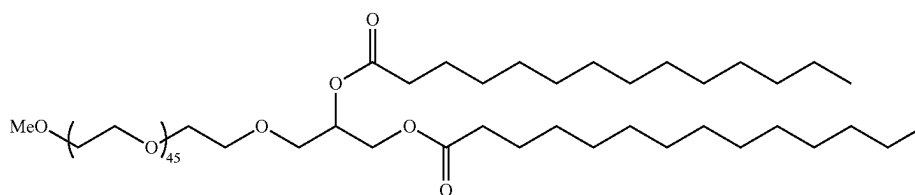
[0096] In some embodiments, the lipid nanoparticles provided herein can comprise a PEG lipid which is a non-diffusible PEG. Non-limiting examples of non-diffusible PEGs include PEG-DSG and PEG-DSPE.

[0097] PEG-lipids are known in the art, such as those described in U.S. Pat. No. 8,158,601 and International Publ. No. WO 2015/130584 A2, which are incorporated herein by reference in their entirety.

[0098] In general, some of the other lipid components (e.g., PEG lipids) of various formulae provided herein may be synthesized as described International Patent Application No. PCT/US2016/000129, filed Dec. 10, 2016, entitled “Compositions and Methods for Delivery of Therapeutic Agents,” which is incorporated by reference in its entirety.

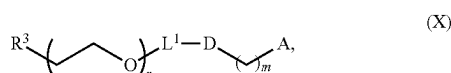
[0099] The lipid component of a lipid nanoparticle composition may include one or more molecules comprising polyethylene glycol, such as PEG or PEG-modified lipids. Such species may be alternately referred to as PEGylated lipids. A PEG lipid is a lipid modified with polyethylene glycol. A PEG lipid may be selected from the non-limiting group including PEG-modified phosphatidylethanolamines, PEG-modified phosphatidic acids, PEG-modified ceramides, PEG-modified dialkylamines, PEG-modified diacylglycerols, PEG-modified dialkylglycerols, and mixtures thereof. For example, a PEG lipid may be PEG-c-DOMG, PEG-DMG, PEG-DLPE, PEG-DMPE, PEG-DPPC, or a PEG-DSPE lipid.

[0100] In some embodiments the PEG-modified lipids are a modified form of PEG DMG. PEG-DMG has the following structure:



[0101] In some embodiments, PEG lipids can be PEGylated lipids described in International Publication No. WO2012099755, the contents of which is herein incorporated by reference in its entirety. Any exemplary PEG lipids may be modified to comprise a hydroxyl group on the PEG chain. In certain embodiments, the PEG lipid is a PEG-OH lipid. As generally defined herein, a “PEG-OH lipid” (also referred to herein as “hydroxy-PEGylated lipid”) is a PEGylated lipid having one or more hydroxyl (—OH) groups on the lipid. In certain embodiments, the PEG-OH lipid includes one or more hydroxyl groups on the PEG chain. In certain embodiments, a PEG-OH or hydroxy-PEGylated lipid comprises an —OH group at the terminus of the PEG chain. Each possibility represents a separate embodiment.

[0102] In certain embodiments, a PEG lipid is a compound of Formula (X):



[0103] or salts thereof, wherein:

[0104] R^3 is ---OR^O ;

[0105] R^O is hydrogen, optionally substituted alkyl, or an oxygen protecting group;

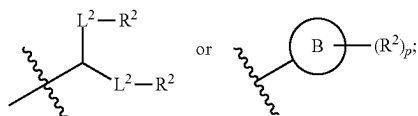
[0106] r is an integer between 1 and 100, inclusive;

[0107] L^1 is optionally substituted C_{1-10} alkylene, wherein at least one methylene of the optionally substituted C_{1-10} alkylene is independently replaced with optionally substituted carbocyclylene, optionally substituted heterocyclylene, optionally substituted arylene, optionally substituted heteroarylene, O, $\text{N}(\text{R}^N)$, S, $\text{C}(\text{O})$, $\text{C}(\text{O})\text{N}(\text{R}^N)$, $\text{NR}^N\text{C}(\text{O})$, $\text{C}(\text{O})\text{O}$, $\text{OC}(\text{O})$, $\text{OC}(\text{O})\text{O}$, $\text{OC}(\text{O})\text{N}(\text{R}^N)$, $\text{NR}^N\text{C}(\text{O})\text{O}$, or $\text{NR}^N\text{C}(\text{O})\text{N}(\text{R}^N)$;

[0108] D is a moiety obtained by click chemistry or a moiety cleavable under physiological conditions;

[0109] m is 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10;

[0110] A is of the formula:



[0111] each instance of L^2 is independently a bond or optionally substituted C_{1-6} alkylene, wherein one methylene unit of the optionally substituted C_{1-6} alkylene is optionally replaced with O, $\text{N}(\text{R}^N)$, S, $\text{C}(\text{O})$, $\text{C}(\text{O})\text{N}(\text{R}^N)$, $\text{NR}^N\text{C}(\text{O})$, $\text{C}(\text{O})\text{O}$, $\text{OC}(\text{O})$, $\text{OC}(\text{O})\text{O}$, $\text{OC}(\text{O})\text{N}(\text{R}^N)$, $\text{---NR}^N\text{C}(\text{O})\text{O}$, or $\text{NR}^N\text{C}(\text{O})\text{N}(\text{R}^N)$;

[0112] each instance of R^2 is independently optionally substituted C_{1-30} alkyl, optionally substituted C_{1-30} alkenyl, or optionally substituted C_{1-30} alkynyl; optionally wherein one or more methylene units of R^2 are independently replaced with optionally substituted carbocyclylene, optionally substituted heterocyclylene, optionally substituted arylene, optionally substituted heteroarylene, $\text{N}(\text{R}^N)$, O, S, $\text{C}(\text{O})$, $\text{C}(\text{O})\text{N}(\text{R}^N)$, $\text{NR}^N\text{C}(\text{O})$, $\text{NR}^N\text{C}(\text{O})\text{N}(\text{R}^N)$, $\text{C}(\text{O})\text{O}$, $\text{OC}(\text{O})$, $\text{---OC}(\text{O})\text{O}$, $\text{OC}(\text{O})\text{N}(\text{R}^N)$, $\text{NR}^N\text{C}(\text{O})\text{O}$, $\text{C}(\text{O})\text{S}$, $\text{SC}(\text{O})$, $\text{C}(\text{=NR}^N)$, $\text{C}(\text{=NR}^N)\text{N}(\text{R}^N)$, $\text{NR}^N\text{C}(\text{=NR}^N)$, $\text{NR}^N\text{C}(\text{=NR}^N)\text{N}$

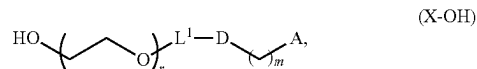
(R^N) , $\text{C}(\text{S})$, $\text{C}(\text{S})\text{N}(\text{R}^N)$, $\text{NR}^N\text{C}(\text{S})$, $\text{NR}^N\text{C}(\text{S})\text{N}(\text{R}^N)$, $\text{S}(\text{O})$, $\text{OS}(\text{O})$, $\text{S}(\text{O})\text{O}$, $\text{---OS}(\text{O})\text{O}$, $\text{OS}(\text{O})_2$, $\text{S}(\text{O})_2\text{O}$, $\text{OS}(\text{O})_2\text{O}$, $\text{N}(\text{R}^N)\text{S}(\text{O})$, $\text{S}(\text{O})\text{N}(\text{R}^N)$, $\text{N}(\text{R}^N)\text{S}(\text{O})\text{N}(\text{R}^N)$, $\text{OS}(\text{O})\text{N}(\text{R}^N)$, $\text{N}(\text{R}^N)\text{S}(\text{O})\text{O}$, $\text{S}(\text{O})_2$, $\text{N}(\text{R}^N)\text{S}(\text{O})_2$, $\text{S}(\text{O})_2\text{N}(\text{R}^N)$, $\text{N}(\text{R}^N)\text{S}(\text{O})_2\text{N}(\text{R}^N)$, $\text{OS}(\text{O})_2\text{N}(\text{R}^N)$, or $\text{---N}(\text{R}^N)\text{S}(\text{O})_2\text{O}$;

[0113] each instance of R^N is independently hydrogen, optionally substituted alkyl, or a nitrogen protecting group;

[0114] Ring B is optionally substituted carbocyclyl, optionally substituted heterocyclyl, optionally substituted aryl, or optionally substituted heteroaryl; and

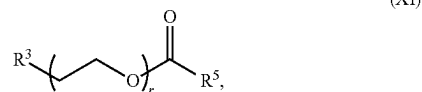
[0115] p is 1 or 2.

[0116] In certain embodiments, the compound of Formula (X) is a PEG-OH lipid (i.e., R^3 is ---OR^O , and R^O is hydrogen). In certain embodiments, the compound of Formula (X) is of Formula (X-OH):



[0117] or a salt thereof.

[0118] In certain embodiments, a PEG lipid is a PEGylated fatty acid. In certain embodiments, a PEG lipid is a compound of Formula (XI). Provided herein are compounds of Formula (XI):



[0119] or a salts thereof, wherein:

[0120] R^3 is ---OR^O ;

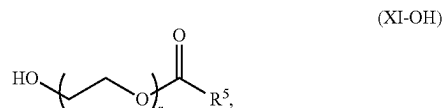
[0121] R^O is hydrogen, optionally substituted alkyl or an oxygen protecting group;

[0122] r is an integer between 1 and 100, inclusive;

[0123] R^5 is optionally substituted C_{10-40} alkyl, optionally substituted C_{10-40} alkenyl, or optionally substituted C_{10-40} alkynyl; and optionally one or more methylene groups of R^5 are replaced with optionally substituted carbocyclylene, optionally substituted heterocyclylene, optionally substituted arylene, optionally substituted heteroarylene, $\text{N}(\text{R}^N)$, O, S, $\text{C}(\text{O})$, $\text{---C}(\text{O})\text{N}(\text{R}^N)$, $\text{NR}^N\text{C}(\text{O})$, $\text{NR}^N\text{C}(\text{O})\text{N}(\text{R}^N)$, $\text{C}(\text{O})\text{O}$, $\text{OC}(\text{O})$, $\text{OC}(\text{O})\text{O}$, $\text{OC}(\text{O})\text{N}(\text{R}^N)$, $\text{NR}^N\text{C}(\text{O})\text{O}$, $\text{C}(\text{O})\text{S}$, $\text{SC}(\text{O})$, $\text{C}(\text{=NR}^N)$, $\text{C}(\text{=NR}^N)\text{N}(\text{R}^N)$, $\text{NR}^N\text{C}(\text{=NR}^N)$, $\text{NR}^N\text{C}(\text{=NR}^N)\text{N}(\text{R}^N)$, $\text{C}(\text{S})$, $\text{---C}(\text{S})\text{N}(\text{R}^N)$, $\text{NR}^N\text{C}(\text{S})$, $\text{NR}^N\text{C}(\text{S})\text{N}(\text{R}^N)$, $\text{S}(\text{O})$, $\text{OS}(\text{O})$, $\text{S}(\text{O})\text{O}$, $\text{OS}(\text{O})\text{O}$, $\text{OS}(\text{O})_2$, $\text{S}(\text{O})_2\text{O}$, $\text{---OS}(\text{O})_2\text{O}$, $\text{N}(\text{R}^N)\text{S}(\text{O})$, $\text{S}(\text{O})\text{N}(\text{R}^N)$, $\text{N}(\text{R}^N)\text{S}(\text{O})\text{N}(\text{R}^N)$, $\text{OS}(\text{O})\text{N}(\text{R}^N)$, $\text{N}(\text{R}^N)\text{S}(\text{O})\text{O}$, $\text{S}(\text{O})_2$, $\text{---N}(\text{R}^N)\text{S}(\text{O})_2$, $\text{S}(\text{O})_2\text{N}(\text{R}^N)$, $\text{N}(\text{R}^N)\text{S}(\text{O})_2\text{N}(\text{R}^N)$, $\text{OS}(\text{O})_2\text{N}(\text{R}^N)$, or $\text{N}(\text{R}^N)\text{S}(\text{O})_2\text{O}$; and

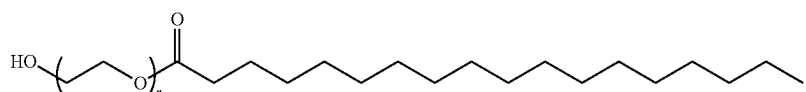
[0124] each instance of R^N is independently hydrogen, optionally substituted alkyl, or a nitrogen protecting group.

[0125] In certain embodiments, the compound of Formula (XI) is of Formula (XI-OH):



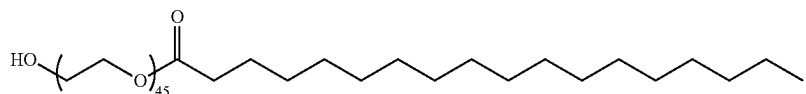
[0126] or a salt thereof. In some embodiments, r is 40-50.

[0127] In yet other embodiments the compound of Formula (XI) is:

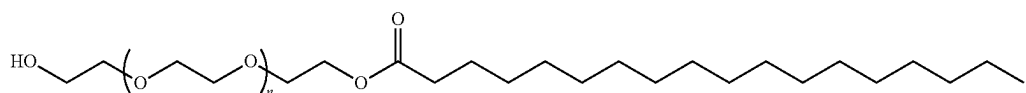


[0128] or a salt thereof.

[0129] In one embodiment, the compound of Formula (XI) is



[0130] In some embodiments, the PEG-lipid comprises 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48, 51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102, 105,108,111,114,117,120,123,126,129,132-tetratetracontaoxatetatriacontahexyl stearate:



[0131] In some embodiments, the PEG-lipids may be one or more of the PEG lipids described in U.S. Application No. U.S. Ser. No. 15/674,872.

Ionizable Amino Lipids

[0132] In some embodiments, a LNP provided herein may include one or more ionizable molecules (e.g., amino lipids or ionizable lipids). The ionizable molecule may comprise a charged group and may have a certain pKa. In certain embodiments, the pKa of the ionizable molecule may be greater than or equal to about 6, greater than or equal to about 6.2, greater than or equal to about 6.5, greater than or equal to about 6.8, greater than or equal to about 7, greater than or equal to about 7.2, greater than or equal to about 7.5, greater than or equal to about 7.8, greater than or equal to about 8. In some embodiments, the pKa of the ionizable molecule may be less than or equal to about 10, less than or equal to about 9.8, less than or equal to about 9.5, less than or equal to about 9.2, less than or equal to about 9.0, less than or equal to about 8.8, or less than or equal to about 8.5.

Combinations of the above referenced ranges are also possible (e.g., greater than or equal to 6 and less than or equal to about 8.5). Other ranges are also possible. In embodiments in which more than one type of ionizable molecule are present in a particle, each type of ionizable molecule may independently have a pKa in one or more of the ranges described above.

[0133] In general, an ionizable molecule comprises one or more charged groups. In some embodiments, an ionizable molecule may be positively charged or negatively charged. For instance, an ionizable molecule may be positively charged. For example, an ionizable molecule may comprise an amine group. As used herein, the term “ionizable molecule” has its ordinary meaning in the art and may refer to a molecule or matrix comprising one or more charged moiety. As used herein, a “charged moiety” is a chemical moiety that carries a formal electronic charge, e.g., monovalent (+1, or -1), divalent (+2, or -2), trivalent (+3, or -3), etc. The charged moiety may be anionic (i.e., negatively charged) or cationic (i.e., positively charged). Examples of

positively-charged moieties include amine groups (e.g., primary, secondary, and/or tertiary amines), ammonium groups, pyridinium groups, guanidine groups, and imidazolium groups. In a particular embodiment, the charged moieties comprise amine groups. Examples of negatively-charged groups or precursors thereof, include carboxylate groups, sulfonate groups, sulfate groups, phosphonate groups, phosphate groups, hydroxyl groups, and the like. The charge of the charged moiety may vary, in some cases, with the environmental conditions, for example, changes in pH may alter the charge of the moiety, and/or cause the moiety to become charged or uncharged. In general, the charge density of the molecule and/or matrix may be selected as desired.

[0134] In some cases, an ionizable molecule (e.g., an amino lipid or ionizable lipid) may include one or more precursor moieties that can be converted to charged moieties. For instance, the ionizable molecule may include a neutral moiety that can be hydrolyzed to form a charged moiety, such as those provided above. As a non-limiting specific example, the molecule or matrix may include an amide, which can be hydrolyzed to form an amine, respectively. Those of ordinary skill in the art will be able to determine whether a given chemical moiety carries a formal electronic charge (for example, by inspection, pH titration, ionic conductivity measurements, etc.), and/or whether a given chemical moiety can be reacted (e.g., hydrolyzed) to form a chemical moiety that carries a formal electronic charge.

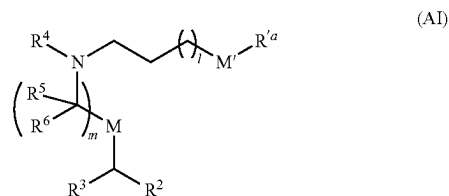
[0135] The ionizable molecule (e.g., amino lipid or ionizable lipid) may have any suitable molecular weight. In certain embodiments, the molecular weight of an ionizable molecule is less than or equal to about 2,500 g/mol, less than or equal to about 2,000 g/mol, less than or equal to about 1,500 g/mol, less than or equal to about 1,250 g/mol, less than or equal to about 1,000 g/mol, less than or equal to about 900 g/mol, less than or equal to about 800 g/mol, less than or equal to about 700 g/mol, less than or equal to about 600 g/mol, less than or equal to about 500 g/mol, less than or equal to about 400 g/mol, less than or equal to about 300 g/mol, less than or equal to about 200 g/mol, or less than or equal to about 100 g/mol. In some instances, the molecular weight of an ionizable molecule is greater than or equal to about 100 g/mol, greater than or equal to about 200 g/mol, greater than or equal to about 300 g/mol, greater than or equal to about 400 g/mol, greater than or equal to about 500 g/mol, greater than or equal to about 600 g/mol, greater than or equal to about 700 g/mol, greater than or equal to about 1000 g/mol, greater than or equal to about 1,250 g/mol, greater than or equal to about 1,500 g/mol, greater than or equal to about 1,750 g/mol, greater than or equal to about 2,000 g/mol, or greater than or equal to about 2,250 g/mol. Combinations of the above ranges (e.g., at least about 200 g/mol and less than or equal to about 2,500 g/mol) are also possible. In embodiments in which more than one type of ionizable molecules are present in a particle, each type of ionizable molecule may independently have a molecular weight in one or more of the ranges described above.

[0136] It should be understood that the terms “charged” or “charged moiety” does not refer to a “partial negative charge” or “partial positive charge” on a molecule. The terms “partial negative charge” and “partial positive charge” are given their ordinary meaning in the art. A “partial negative charge” may result when a functional group com-

prises a bond that becomes polarized such that electron density is pulled toward one atom of the bond, creating a partial negative charge on the atom. Those of ordinary skill in the art will, in general, recognize bonds that can become polarized in this way.

[0137] In some embodiments, the lipid nanoparticle comprises at least one ionizable amino lipid, at least one neutral lipid, at least one sterol, and/or at least one polyethylene glycol (PEG)-modified lipid.

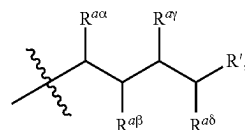
[0138] In some embodiments, the ionizable amino lipid is a compound of Formula (AI):



or its N-oxide, or a salt or isomer thereof,

[0139] wherein R'^a is $R'^{branched}$, wherein

[0140] $R'^{branched}$ is:

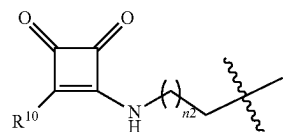


wherein $\text{---}\zeta\text{---}$ denotes a point of attachment;

[0141] wherein $R^{\alpha\alpha}$, $R^{\alpha\beta}$, $R^{\alpha\gamma}$, and $R^{\alpha\delta}$ are each independently selected from the group consisting of H, C_{2-12} alkyl, and C_{2-12} alkenyl;

[0142] R^2 and R^3 are each independently selected from the group consisting of C_{1-14} alkyl and C_{2-14} alkenyl;

[0143] R^4 is selected from the group consisting of $\text{---}(\text{CH}_2)_n\text{OH}$, wherein n is selected from the group consisting of 1, 2, 3, 4, and 5, and



[0144] wherein $\text{---}\zeta\text{---}$ denotes a point of attachment; wherein

[0145] R^{10} is N(R)_2 ; each R is independently selected from the group consisting of C_{1-6} alkyl,

[0146] C_{2-3} alkenyl, and H; and n_2 is selected from the group consisting of 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10;

[0147] each R^5 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;

[0148] each R^6 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;

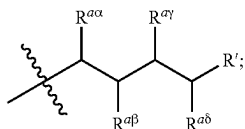
[0149] M and M' are each independently selected from the group consisting of ---C(O)O--- and ---OC(O)--- ;

[0150] R' is a C₁₋₁₂ alkyl or C₂₋₁₂ alkenyl;

[0151] l is selected from the group consisting of 1, 2, 3, 4, and 5; and

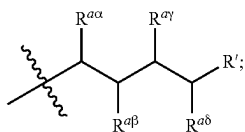
[0152] m is selected from the group consisting of 5, 6, 7, 8, 9, 10, 11, 12, and 13.

[0153] In some embodiments of the compounds of Formula (AI), R^{1a} is R'^{branched}; R'^{branched} is



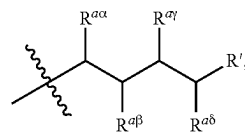
—— denotes a point of attachment; R^{αα}, R^{αβ}, R^{αγ}, and R^{αδ} are each H; R² and R³ are each C₁₋₁₄ alkyl; R⁴ is —(CH₂)_nOH; n is 2; each R⁵ is H; each R⁶ is H; M and M' are each —C(O)O—; R' is a C₁₋₁₂ alkyl; l is 5; and m is 7.

[0154] In some embodiments of the compounds of Formula (AI), R^{1a} is R'^{branched}; R'^{branched} is

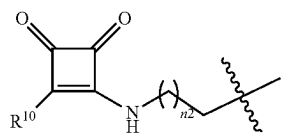


—— denotes a point of attachment; R^{αα}, R^{αβ}, R^{αγ}, and R^{αδ} are each H; R² and R³ are each C₁₋₁₄ alkyl; R⁴ is —(CH₂)_nOH; n is 2; each R⁵ is H; each R⁶ is H; M and M' are each —C(O)O—; R' is a C₁₋₁₂ alkyl; l is 3; and m is 7.

[0155] In some embodiments of the compounds of Formula (AI), R^{1a} is R'^{branched}; R'^{branched} is

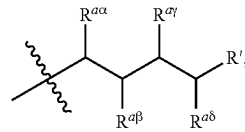


—— denotes a point of attachment; R^{αα} is C₂₋₁₂ alkyl; R^{αβ}, R^{αγ}, and R^{αδ} are each H; R² and R³ are each C₁₋₁₄ alkyl; R⁴ is



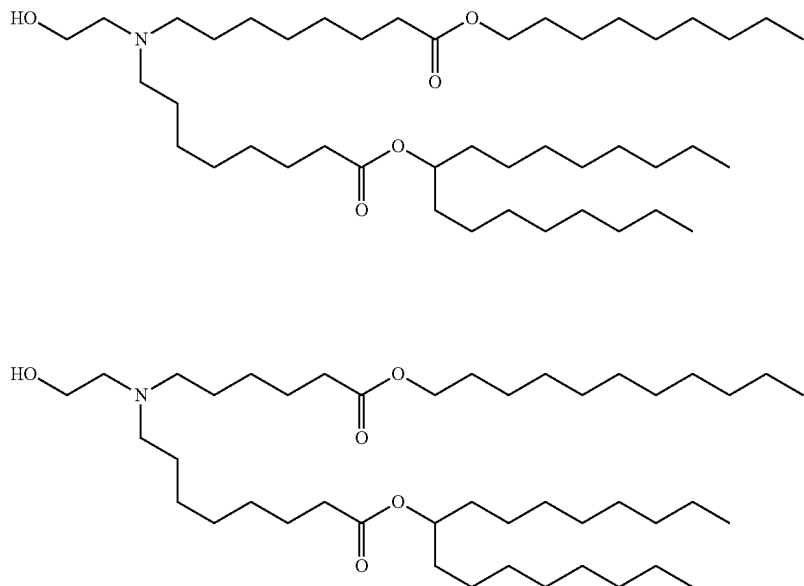
R¹⁰ NH(C₁₋₆ alkyl); n₂ is 2; R⁵ is H; each R⁶ is H; M and M' are each —C(O)O—; R' is a C₁₋₁₂ alkyl; l is 5; and m is 7.

[0156] In some embodiments of the compounds of Formula (I), R^{1a} is R'^{branched}; R'^{branched} is

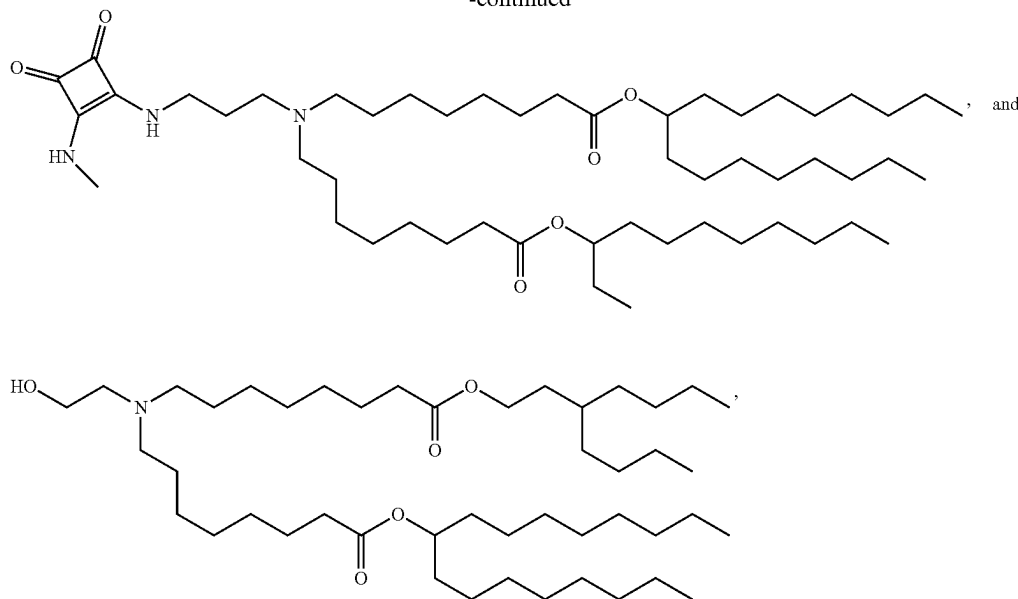


—— denotes a point of attachment; R^{αα}, R^{αβ}, and R^{αδ} are each H; R^{αγ} is C₂₋₁₂ alkyl; R² and R³ are each C₁₋₁₄ alkyl; R⁴ is —(CH₂)_nOH; n is 2; each R⁵ is H; each R⁶ is H; M and M' are each —C(O)O—; R' is a C₁₋₁₂ alkyl; l is 5; and m is 7.

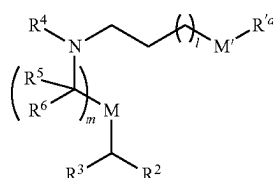
[0157] In some embodiments, the compound of Formula (I) is selected from:



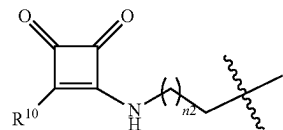
-continued



[0158] In some embodiments, the ionizable amino lipid is a compound of Formula (A1a):



(A1a)



[0164] wherein denotes a point of attachment; wherein

[0165] R^{10} is $N(R)_2$; each R is independently selected from the group consisting of C_{1-6} alkyl, C_{2-3} alkenyl, and H; and n_2 is selected from the group consisting of 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10;

[0166] each R^5 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;

[0167] each R^6 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;

[0168] M and M' are each independently selected from the group consisting of $-C(O)O-$ and $-OC(O)-$;

[0169] R' is a C_{1-12} alkyl or C_{2-12} alkenyl;

[0170] 1 is selected from the group consisting of 1, 2, 3, 4, and 5; and

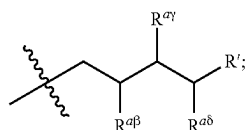
[0171] m is selected from the group consisting of 5, 6, 7, 8, 9, 10, 11, 12, and 13.

[0172] In some embodiments, the ionizable amino lipid is a compound of Formula (A1b):

or its N-oxide, or a salt or isomer thereof,

[0159] wherein R^{1a} is $R^{branched}$; wherein

[0160] $R^{branched}$ is:

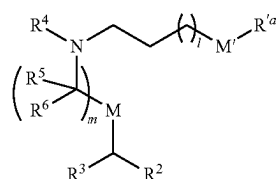


wherein denotes a point of attachment;

[0161] wherein $R^{aβ}$, $R^{aγ}$, and $R^{aδ}$ are each independently selected from the group consisting of H, C_{2-12} alkyl, and C_{2-12} alkenyl;

[0162] R^2 and R^3 are each independently selected from the group consisting of C_{1-14} alkyl and C_{2-14} alkenyl;

[0163] R^4 is selected from the group consisting of $-(CH_2)_nOH$ wherein n is selected from the group consisting of 1, 2, 3, 4, and 5, and

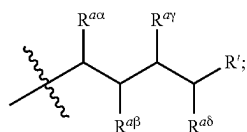


(A1b)

or its N-oxide, or a salt or isomer thereof,

[0173] wherein R^{1a} is $R^{branched}$, wherein

[0174] $R^{branched}$ is:



wherein $\text{---}\zeta\text{---}$ denotes a point of attachment;

[0175] wherein R^{α} , R^{β} , R^{γ} , and R^{δ} are each independently selected from the group consisting of H, C_{2-12} alkyl, and C_{2-12} alkenyl;

[0176] R^2 and R^3 are each independently selected from the group consisting of C_{1-14} alkyl and C_{2-14} alkenyl;

[0177] R^4 is $-(CH_2)_nOH$, wherein n is selected from the group consisting of 1, 2, 3, 4, and 5;

[0178] each R^5 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;

[0179] each R^6 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;

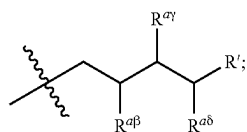
[0180] M and M' are each independently selected from the group consisting of $-C(O)O-$ and $-OC(O)-$;

[0181] R' is a C_{1-12} alkyl or C_{2-12} alkenyl;

[0182] l is selected from the group consisting of 1, 2, 3, 4, and 5; and

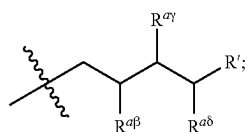
[0183] m is selected from the group consisting of 5, 6, 7, 8, 9, 10, 11, 12, and 13.

[0184] In some embodiments of Formula (AI) or (AII), R^{1a} is $R^{branched}$; $R^{branched}$ is



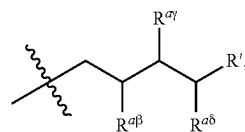
$\text{---}\zeta\text{---}$ denotes a point of attachment; R^{α} , R^{β} , R^{γ} , and R^{δ} are each H; R^2 and R^3 are each C_{1-14} alkyl; R^4 is $-(CH_2)_nOH$; n is 2; each R^5 is H; each R^6 is H; M and M' are each $-C(O)O-$; R' is a C_{1-12} alkyl; l is 5; and m is 7.

[0185] In some embodiments of Formula (AI) or (AII), R^{1a} is $R^{branched}$; $R^{branched}$ is



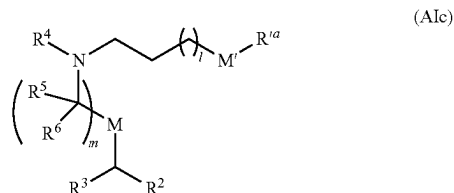
$\text{---}\zeta\text{---}$ denotes a point of attachment; R^{α} , R^{β} , R^{γ} , and R^{δ} are each H; R^2 and R^3 are each C_{1-14} alkyl; R^4 is $-(CH_2)_nOH$; n is 2; each R^5 is H; each R^6 is H; M and M' are each $-C(O)O-$; R' is a C_{1-12} alkyl; l is 3; and m is 7.

[0186] In some embodiments of Formula (AI) or (AII), R^{1a} is $R^{branched}$; $R^{branched}$ is



[0187] $\text{---}\zeta\text{---}$ denotes a point of attachment; R^{α} and R^{δ} are each H; R^{γ} is C_{2-12} alkyl; R^2 and R^3 are each C_{1-14} alkyl; R^4 is $-(CH_2)_nOH$; n is 2; each R^5 is H; each R^6 is H; M and M' are each $-C(O)O-$; R' is a C_{1-12} alkyl; l is 5; and m is 7.

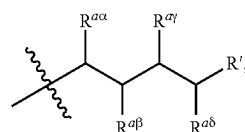
[0188] In some embodiments, the ionizable amino lipid is a compound of Formula (AIIc):



or its N-oxide, or a salt or isomer thereof,

[0189] wherein R^{1a} is $R^{branched}$, wherein

[0190] $R^{branched}$ is:

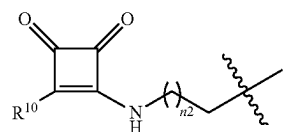


wherein $\text{---}\zeta\text{---}$ denotes a point of attachment;

[0191] wherein R^{α} , R^{β} , R^{γ} , and R^{δ} are each independently selected from the group consisting of H, C_{2-12} alkyl, and C_{2-12} alkenyl;

[0192] R^2 and R^3 are each independently selected from the group consisting of C_{1-14} alkyl and C_{2-14} alkenyl;

[0193] R^4 is



[0194] wherein $\text{---}\zeta\text{---}$ denotes a point of attachment; wherein R^{10} is $N(R)_2$; each R is independently selected from the group consisting of C_{1-6} alkyl, C_{2-3} alkenyl, and H; n2 is selected from the group consisting of 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10;

[0195] each R^5 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;

[0196] each R^6 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;

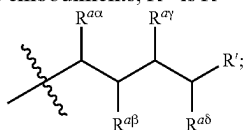
[0197] M and M' are each independently selected from the group consisting of —C(O)O— and —OC(O)— ;

[0198] R' is a C_{1-12} alkyl or C_{2-12} alkenyl;

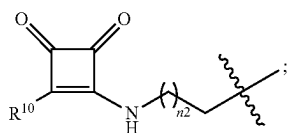
[0199] l is selected from the group consisting of 1, 2, 3, 4, and 5; and

[0200] m is selected from the group consisting of 5, 6, 7, 8, 9, 10, 11, 12, and 13.

[0201] In some embodiments, R^{1a} is $\text{R}^{\text{branched}}$; $\text{R}^{\text{branched}}$ is

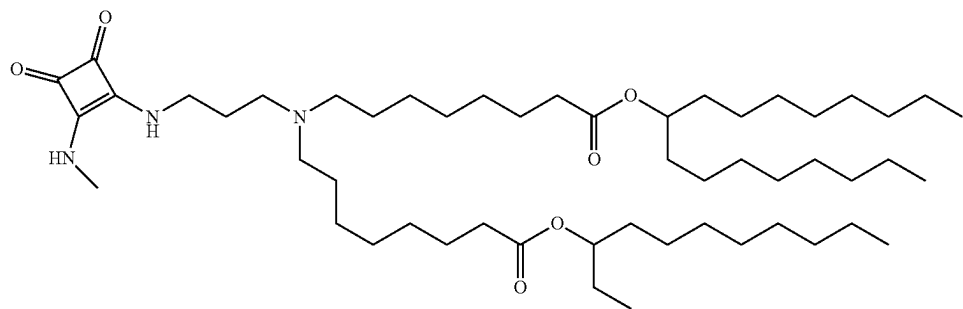


$\text{—}\zeta\text{—}$ denotes a point of attachment; $\text{R}^{\alpha\beta}$, $\text{R}^{\alpha\gamma}$, and $\text{R}^{\alpha\delta}$ are each H; $\text{R}^{\alpha\alpha}$ is C_{2-12} alkyl; R^2 and R^3 are each C_{1-14} alkyl; R^4 is

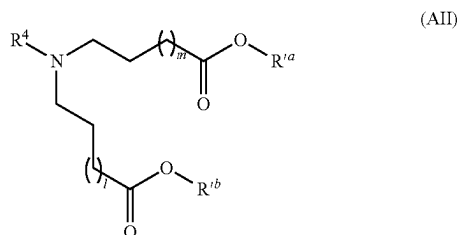


$\text{—}\zeta\text{—}$ denotes a point of attachment; R^{10} is $\text{NH}(\text{C}_{1-6}$ alkyl); n_2 is 2; each R^5 is H; each R^6 is H; M and M' are each —C(O)O— ; R' is a C_{1-12} alkyl; l is 5; and m is 7.

[0202] In some embodiments, the compound of Formula (A1c) is:



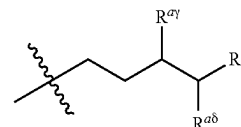
[0203] In some embodiments, the ionizable amino lipid is a compound of Formula (AII):



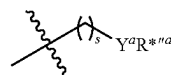
or its N-oxide, or a salt or isomer thereof,

[0204] wherein R^{1a} is $\text{R}^{\text{branched}}$ or R^{cyclic} , wherein

[0205] $\text{R}^{\text{branched}}$ is:

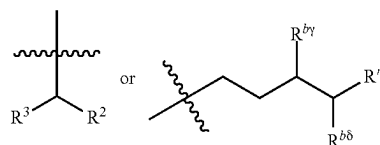


and R^{cyclic} is:



and

[0206] R^b is:



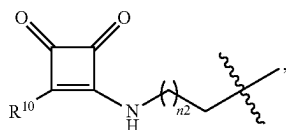
[0207] wherein $\text{—}\zeta\text{—}$ denotes a point of attachment;

[0208] $\text{R}^{\alpha\gamma}$ and $\text{R}^{\alpha\delta}$ are each independently selected from the group consisting of H, C_{1-12} alkyl, and C_{2-12} alkenyl, wherein at least one of $\text{R}^{\alpha\gamma}$ and $\text{R}^{\alpha\delta}$ is selected from the group consisting of C_{1-12} alkyl and C_{2-12} alkenyl;

[0209] $\text{R}^{\text{b}\gamma}$ and $\text{R}^{\text{b}\delta}$ are each independently selected from the group consisting of H, C_{1-12} alkyl, and C_{2-12} alkenyl, wherein at least one of $\text{R}^{\text{b}\gamma}$ and $\text{R}^{\text{b}\delta}$ is selected from the group consisting of C_{1-12} alkyl and C_{2-12} alkenyl;

[0210] R^2 and R^3 are each independently selected from the group consisting of C_{1-14} alkyl and C_{2-14} alkenyl;

[0211] R^4 is selected from the group consisting of $\text{—(CH}_2\text{)}_n\text{OH}$ wherein n is selected from the group consisting of 1, 2, 3, 4, and 5, and



[0212] wherein $\text{---}\zeta\text{---}$ denotes a point of attachment; wherein R^{10} is $N(R)_2$; each R is independently selected from the group consisting of C_{1-6} alkyl, C_{2-3} alkenyl, and H; and n_2 is selected from the group consisting of 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10;

[0213] each R' independently is a C_{1-12} alkyl or C_{2-12} alkenyl;

[0214] Y^a is a C_{3-6} carbocycle;

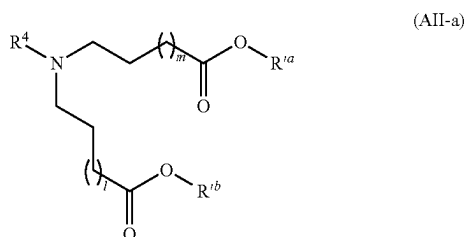
[0215] R^{*m_a} is selected from the group consisting of C_{1-15} alkyl and C_{2-15} alkenyl; and

[0216] s is 2 or 3;

[0217] m is selected from 1, 2, 3, 4, 5, 6, 7, 8, and 9;

[0218] l is selected from 1, 2, 3, 4, 5, 6, 7, 8, and 9.

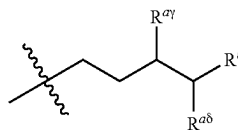
[0219] In some embodiments, the ionizable amino lipid is a compound of Formula (AII-a):



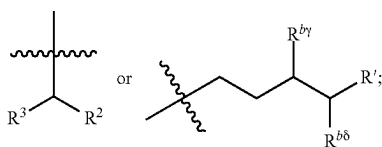
or its N-oxide, or a salt or isomer thereof,

[0220] wherein R^{1a} is $R^{branched}$ or R^{cyclic} , wherein

[0221] $R^{branched}$ is:



and R^{1b} is:



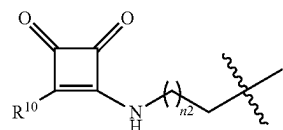
[0222] wherein $\text{---}\zeta\text{---}$ denotes a point of attachment;

[0223] $R^{aγ}$ and $R^{aδ}$ are each independently selected from the group consisting of H, C_{1-12} alkyl, and C_{2-12} alkenyl, wherein at least one of $R^{aγ}$ and $R^{aδ}$ is selected from the group consisting of C_{1-12} alkyl and C_{2-12} alkenyl;

[0224] $R^{bγ}$ and $R^{bδ}$ are each independently selected from the group consisting of H, C_{1-12} alkyl, and C_{2-12} alkenyl, wherein at least one of $R^{bγ}$ and $R^{bδ}$ is selected from the group consisting of C_{1-12} alkyl and C_{2-12} alkenyl;

[0225] R^2 and R^3 are each independently selected from the group consisting of C_{1-14} alkyl and C_{2-14} alkenyl;

[0226] R^4 is selected from the group consisting of $-(CH_2)_nOH$ wherein n is selected from the group consisting of 1, 2, 3, 4, and 5, and



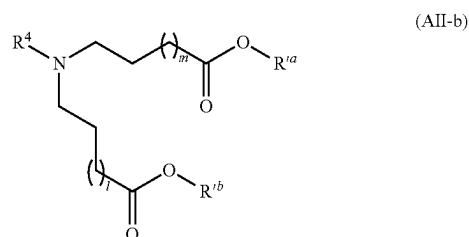
[0227] wherein $\text{---}\zeta\text{---}$ denotes a point of attachment; wherein R^{10} is $N(R)_2$; each R is 5 independently selected from the group consisting of C_{1-6} alkyl, C_{2-3} alkenyl, and H; and n_2 is selected from the group consisting of 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10;

[0228] each R' independently is a C_{1-12} alkyl or C_{2-12} alkenyl;

[0229] m is selected from 1, 2, 3, 4, 5, 6, 7, 8, and 9;

[0230] l is selected from 1, 2, 3, 4, 5, 6, 7, 8, and 9.

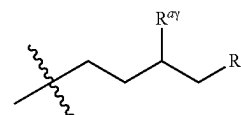
[0231] In some embodiments, the ionizable amino lipid is a compound of Formula (AII-b):



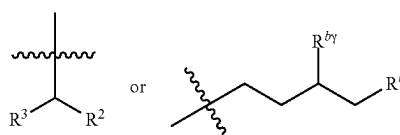
or its N-oxide, or a salt or isomer thereof,

[0232] wherein R^{1a} is $R^{branched}$ or R^{cyclic} , wherein

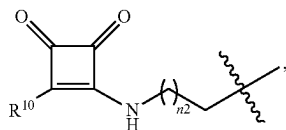
[0233] $R^{branched}$ is:



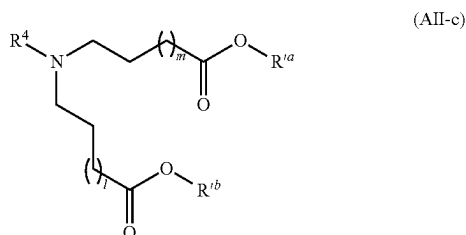
and R^{1b} is:



- [0234] wherein $\text{---}\zeta\text{---}$ denotes a point of attachment;
 [0235] $R^{\alpha\gamma}$ and $R^{b\gamma}$ are each independently selected from the group consisting of C_{1-12} alkyl and C_{2-12} alkenyl;
 [0236] R^2 and R^3 are each independently selected from the group consisting of C_{1-14} alkyl and C_{2-14} alkenyl;
 [0237] R^4 is selected from the group consisting of $\text{---}(\text{CH}_2)_n\text{OH}$ wherein n is selected from the group consisting of 1, 2, 3, 4, and 5, and

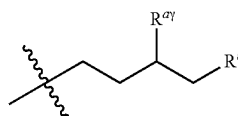


- [0238] wherein $\text{---}\zeta\text{---}$ denotes a point of attachment; wherein R^{10} is $N(R)_2$; each R is independently selected from the group consisting of C_{1-6} alkyl, C_{2-3} alkenyl, and H; and n_2 is selected from the group consisting of 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10;
 [0239] each R' independently is a C_{1-12} alkyl or C_{2-12} alkenyl;
 [0240] m is selected from 1, 2, 3, 4, 5, 6, 7, 8, and 9;
 [0241] l is selected from 1, 2, 3, 4, 5, 6, 7, 8, and 9.
 [0242] In some embodiments, the ionizable amino lipid is a compound of Formula (AII-c):

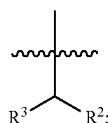


or its N-oxide, or a salt or isomer thereof,

- [0243] wherein R^{1a} is $R^{branched}$ or R^{cyclic} , wherein
 [0244] $R^{branched}$ is:

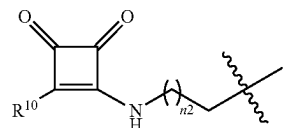


and R^{1b} is:

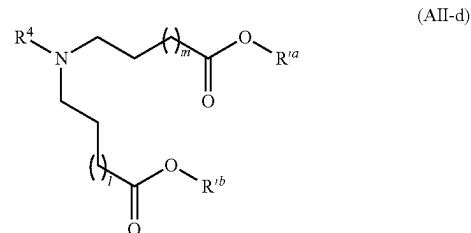


- [0245] wherein $\text{---}\zeta\text{---}$ denotes a point of attachment;

- [0246] wherein $R^{\alpha\gamma}$ is selected from the group consisting of C_{1-12} alkyl and C_{2-12} alkenyl;
 [0247] R^2 and R^3 are each independently selected from the group consisting of C_{1-14} alkyl and C_{2-14} alkenyl;
 [0248] R^4 is selected from the group consisting of $\text{---}(\text{CH}_2)_n\text{OH}$ wherein n is selected from the group consisting of 1, 2, 3, 4, and 5, and

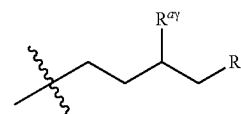


- [0249] wherein $\text{---}\zeta\text{---}$ denotes a point of attachment; wherein R^{10} is $N(R)_2$; each R is 5 independently selected from the group consisting of C_{1-6} alkyl, C_{2-3} alkenyl, and H; and
 [0250] n_2 is selected from the group consisting of 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10;
 [0251] R' is a C_{1-12} alkyl or C_{2-12} alkenyl;
 [0252] m is selected from 1, 2, 3, 4, 5, 6, 7, 8, and 9;
 [0253] l is selected from 1, 2, 3, 4, 5, 6, 7, 8, and 9.
 [0254] In some embodiments, the ionizable amino lipid is a compound of Formula (AII-d):

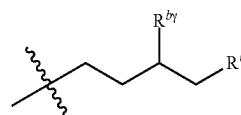


or its N-oxide, or a salt or isomer thereof,

- [0255] wherein R^{1a} is $R^{branched}$ or R^{cyclic} , wherein
 [0256] $R^{branched}$ is:

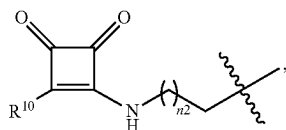


and R^{1b} is:



- [0257] wherein $\text{---}\zeta\text{---}$ denotes a point of attachment;
 [0258] wherein $R^{\alpha\gamma}$ and $R^{b\gamma}$ are each independently selected from the group consisting of C_{1-12} alkyl and C_{2-12} alkenyl;

[0259] R⁴ is selected from the group consisting of —(CH₂)_nOH wherein n is selected from the group consisting of 1, 2, 3, 4, and 5, and



[0260] wherein  denotes a point of attachment; wherein R¹⁰ is N(R)₂; each R is independently selected from the group consisting of C₁₋₆ alkyl, C₂₋₃ alkenyl, and H; and

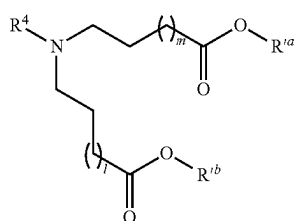
[0261] n2 is selected from the group consisting of 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10;

[0262] each R' independently is a C₁₋₁₂ alkyl or C₂₋₁₂ alkenyl;

[0263] m is selected from 1, 2, 3, 4, 5, 6, 7, 8, and 9;

[0264] 1 is selected from 1, 2, 3, 4, 5, 6, 7, 8, and 9.

[0265] In some embodiments, the ionizable amino lipid is a compound of Formula (AII-e):

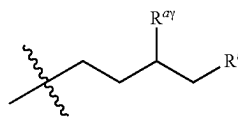


(AII-e)

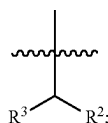
or its N-oxide, or a salt or isomer thereof,

[0266] wherein R^{1a} is R^{1branched} or R^{1cyclic}, wherein

[0267] R'^{branched} is:



and R'^b is:



[0268] wherein $\frac{\text{S}}{\text{S}}$ denotes a point of attachment;

[0269] wherein R^{ay} is selected from the group consisting of C₁₋₁₂ alkyl and C₂₋₁₂ alkenyl;

[0270] R^2 and R^3 are each independently selected from the group consisting of C_{1-14} alkyl and C_{7-14} alkenyl;

[0271] R⁴ is —(CH₂)_nOH wherein n is selected from the group consisting of 1, 2, 3, 4, and 5;

[0272] R' is a C₁-12 alkyl or C₂₋₁₂ alkenyl;

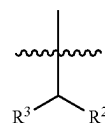
[0273] m is selected from 1, 2, 3, 4, 5, 6, 7, 8, and 9;

[0274] 1 is selected from 1, 2, 3, 4, 5, 6, 7, 8, and 9.

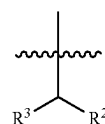
[0275] In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), m and l are each independently selected from 4, 5, and 6. In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), m and l are each 5.

[0276] In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), each R' independently is a C₁₋₁₂ alkyl. In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), each R' independently is a C₂₋₅ alkyl.

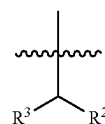
[0277] In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), R^b is:



and R² and R³ are each independently a C₁₋₁₄ alkyl. In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), R^b is:

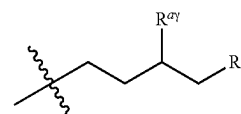


and R² and R³ are each independently a C₆₋₁₀ alkyl. In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), R^b is:

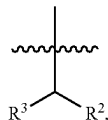


and R² and R³ are each a C₈ alkyl.

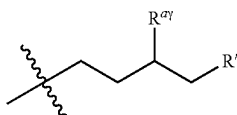
[0278] In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), R^{branched} is:



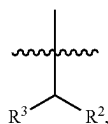
and R'^b is:



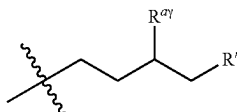
$R^{a\gamma}$ is a C_{1-12} alkyl and R^2 and R^3 are each independently a C_{6-10} alkyl. In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), $R'^{branched}$ is:



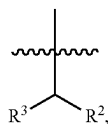
and R'^b is:



$R^{a\gamma}$ is a C_{2-6} alkyl and R^2 and R^3 are each independently a C_{6-10} alkyl. In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), $R'^{branched}$ is:

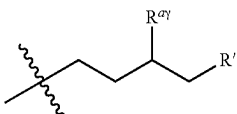


and R'^b is:

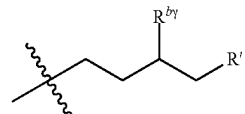


$R^{a\gamma}$ is a C_{2-6} alkyl, and R^2 and R^3 are each a C_8 alkyl.

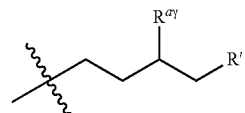
[0279] In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), $R'^{branched}$ is:



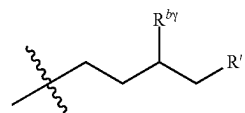
R'^b is:



and $R^{a\gamma}$ and $R^{b\gamma}$ are each a C_{1-12} alkyl. In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), $R'^{branched}$ is:



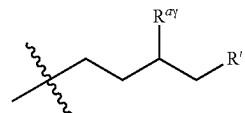
R'^b is:



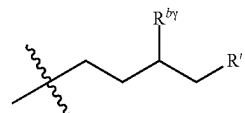
and $R^{a\gamma}$ and $R^{b\gamma}$ are each a C_{2-6} alkyl.

[0280] In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), m and l are each independently selected from 4, 5, and 6 and each R' independently is a C_{1-12} alkyl. In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), m and l are each 5 and each R' independently is a C_{2-5} alkyl.

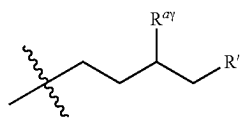
[0281] In some embodiments of the compound of (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), $R'^{branched}$ is:



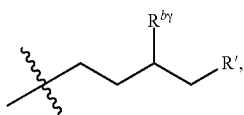
R'^b is:



m and l are each independently selected from 4, 5, and 6, each R' independently is a C_{1-12} alkyl, and $R^{a\gamma}$ and $R^{b\gamma}$ are each a C_{1-12} alkyl. In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), $R'^{branched}$ is:

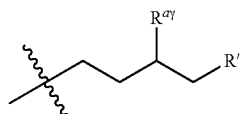


R'^b is:

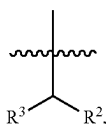


m and l are each 5, each R' independently is a C_{2-5} alkyl, and $R^{\alpha\gamma}$ and $R^{b\gamma}$ are each a C_{2-6} alkyl.

[0282] In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), $R^{branched}$ is:

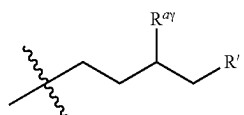


and R'^b is:

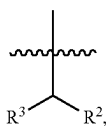


m and l are each independently selected from 4, 5, and 6, R' is a C_{1-12} alkyl, $R^{\alpha\gamma}$ is a C_{1-12} alkyl and R^2 and R^3 are each independently a C_{6-10} alkyl.

[0283] In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), $R^{branched}$ is:

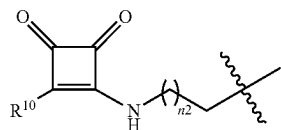


and R'^b is:

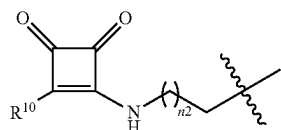


m and l are each 5, R' is a C_{2-5} alkyl, $R^{\alpha\gamma}$ is a C_{2-6} alkyl, and R^2 and R^3 are each a C_8 alkyl.

[0284] In some embodiments of the compound of (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), R^4 is

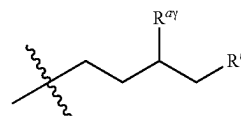


wherein R^{10} is $NH(C_{1-6}$ alkyl) and n2 is 2. In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), R^4 is

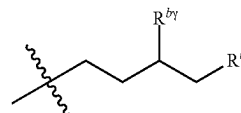


wherein R^{10} is $NH(CH_3)$ and n2 is 2.

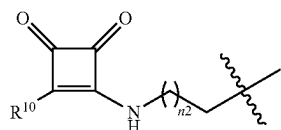
[0285] In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), $R^{branched}$ is:



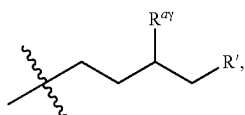
R'^b is:



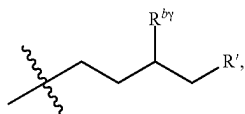
m and l are each independently selected from 4, 5, and 6, each R' independently is a C_{1-12} alkyl, $R^{\alpha\gamma}$ and $R^{b\gamma}$ are each a C_{1-12} alkyl, and R^4 is



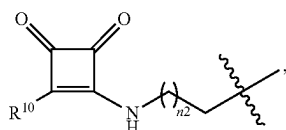
wherein R^{10} is $NH(C_{1-6}$ alkyl), and n2 is 2. In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), $R^{branched}$ is:



R^{tb} is:

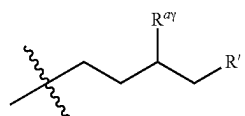


m and l are each 5, each R' independently is a C_{2-5} alkyl, $R^{\alpha\gamma}$ and $R^{b\gamma}$ are each a C_{2-6} alkyl, and R^4 is

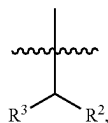


wherein R^{10} is $NH(CH_3)$ and n_2 is 2.

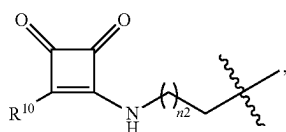
[0286] In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), $R^{branched}$ is:



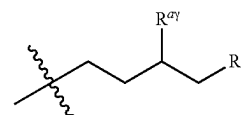
and R^{tb} is:



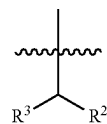
m and l are each independently selected from 4, 5, and 6, R' is a C_{1-12} alkyl, R^2 and R^3 are each independently a C_{6-10} alkyl, $R^{\alpha\gamma}$ is a C_{1-12} alkyl, and R^4 is



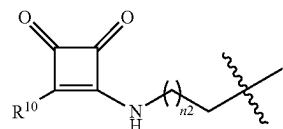
wherein R^{10} is $NH(C_{1-6} \text{ alkyl})$ and n_2 is 2. In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), $R^{branched}$ is:



and R^{tb} is:



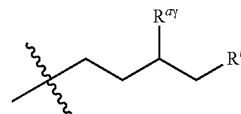
m and l are each 5, R' is a C_{2-5} alkyl, $R^{\alpha\gamma}$ is a C_{2-6} alkyl, R^2 and R^3 are each a C_8 alkyl, and R^4 is



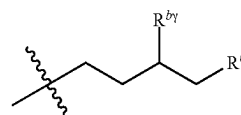
wherein R^{10} is $NH(CH_3)$ and n_2 is 2.

[0287] In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), R^4 is $-(CH_2)_nOH$ and n is 2, 3, or 4. In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), R^4 is $-(CH_2)_nOH$ and n is 2.

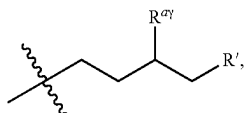
[0288] In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), $R^{branched}$ is:



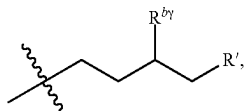
R^{tb} is:



m and l are each independently selected from 4, 5, and 6, each R' independently is a C_{1-12} alkyl, $R^{\alpha\gamma}$ and $R^{b\gamma}$ are each a C_{1-12} alkyl, R^4 is $-(CH_2)_nOH$, and n is 2, 3, or 4. In some embodiments of the compound of Formula (AII), (AII-a), (AII-b), (AII-c), (AII-d), or (AII-e), $R^{branched}$ is:

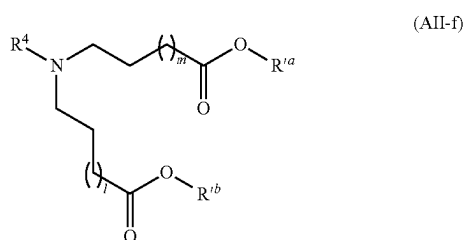


R'^b is:



m and l are each 5, each R' independently is a C_{2-5} alkyl, $R^{\alpha\gamma}$ and $R^{\beta\gamma}$ are each a C_{2-6} alkyl, R^4 is $-(CH_2)_nOH$, and n is 2.

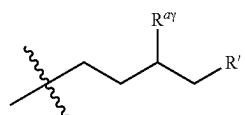
[0289] In some embodiments, the ionizable amino lipid is a compound of Formula (AII-f):



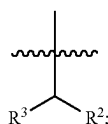
or its N-oxide, or a salt or isomer thereof,

[0290] wherein R'^a is $R'^{branched}$ or R'^{cyclic} ; wherein

[0291] $R'^{branched}$ is:



and R'^b is:



[0292] wherein $\text{---}\zeta\text{---}$ denotes a point of attachment;

[0293] $R^{\alpha\gamma}$ is a C_{1-12} alkyl;

[0294] R^2 and R^3 are each independently a C_{1-14} alkyl;

[0295] R^4 is $-(CH_2)_nOH$ wherein n is selected from the group consisting of 1, 2, 3, 4, and 5;

[0296] R' is a C_{1-12} alkyl;

[0297] m is selected from 4, 5, and 6; and

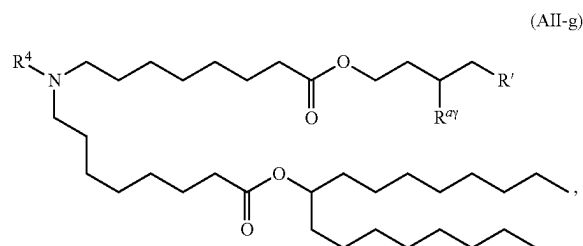
[0298] l is selected from 4, 5, and 6.

[0299] In some embodiments of the compound of Formula (AII-f), m and l are each 5, and n is 2, 3, or 4.

[0300] In some embodiments of the compound of Formula (AII-f) R' is a C_{2-5} alkyl, $R^{\alpha\gamma}$ is a C_{2-6} alkyl, and R^2 and R^3 are each a C_{6-10} alkyl.

[0301] In some embodiments of the compound of Formula (AII-f), m and l are each 5, n is 2, 3, or 4, R' is a C_{2-5} alkyl, $R^{\alpha\gamma}$ is a C_{2-6} alkyl, and R^2 and R^3 are each a C_{6-10} alkyl.

[0302] In some embodiments, the ionizable amino lipid is a compound of Formula (AII-g):



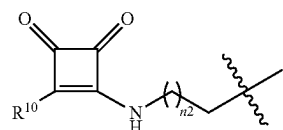
(AII-g)

wherein

[0303] $R^{\alpha\gamma}$ is a C_{2-6} alkyl;

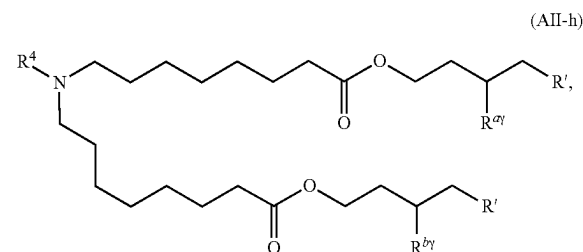
[0304] R' is a C_{2-5} alkyl; and

[0305] R^4 is selected from the group consisting of $-(CH_2)_nOH$ wherein n is selected from the group consisting of 3, 4, and 5, and



[0306] wherein $\text{---}\zeta\text{---}$ denotes a point of attachment, R^{10} is $NH(C_{1-6} \text{ alkyl})$, and n_2 is selected from the group consisting of 1, 2, and 3.

[0307] In some embodiments, the ionizable amino lipid is a compound of Formula (AII-h):



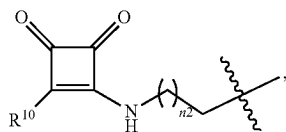
(AII-h)

wherein

[0308] $R^{\alpha\gamma}$ and $R^{\beta\gamma}$ are each independently a C_{2-6} alkyl;

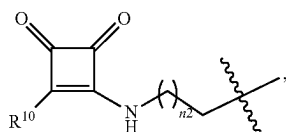
[0309] each R' independently is a C_{2-5} alkyl; and

[0310] R^4 is selected from the group consisting of $-(CH_2)_nOH$ wherein n is selected from the group consisting of 3, 4, and 5, and



[0311] wherein  denotes a point of attachment, R^{10} is $\text{NH}(\text{C}_{1-6} \text{ alkyl})$, and n_2 is selected from the group consisting of 1, 2, and 3.

[0312] In some embodiments of the compound of Formula (AII-g) or (AII-h), R^4 is

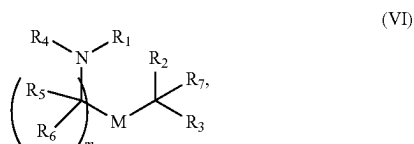


wherein

[0313] R^{10} is $\text{NH}(\text{CH}_3)$ and n_2 is 2.

[0314] In some embodiments of the compound of Formula (AII-g) or (AII-h), R^4 is $-(\text{CH}_2)_2\text{OH}$.

[0315] In some embodiments, the ionizable amino lipid may be one or more of compounds of Formula (VI):



[0316] or their N-oxides, or salts or isomers thereof, wherein:

[0317] R_1 is selected from the group consisting of C_{5-30} alkyl, C_{5-20} alkenyl, $-\text{R}^*\text{YR}''$, $-\text{YR}''$, and $-\text{R}''\text{M}'\text{R}'$;

[0318] R_2 and R_3 are independently selected from the group consisting of H, C_{1-14} alkyl, C_{2-14} alkenyl, $-\text{R}^*\text{YR}''$, $-\text{YR}''$, and $-\text{R}^*\text{OR}''$, or R_2 and R_3 , together with the atom to which they are attached, form a heterocycle or carbocycle;

[0319] R_4 is selected from the group consisting of hydrogen, a C_{3-6} carbocycle, $-(\text{CH}_2)_n\text{Q}$, $-(\text{CH}_2)_n\text{CHQR}$, $-\text{CHQR}$, $-\text{CQ}(\text{R})_2$, and unsubstituted C_{1-6} alkyl, where Q is selected from a carbocycle, heterocycle, $-\text{OR}$, $-\text{O}(\text{CH}_2)_n\text{N}(\text{R})_2$, $-\text{C}(\text{O})\text{OR}$, $-\text{OC}(\text{O})\text{R}$, $-\text{CX}_3$, $-\text{CX}_2\text{H}$, $-\text{CXH}_2$, $-\text{CN}$, $-\text{N}(\text{R})_2$, $-\text{C}(\text{O})\text{N}(\text{R})_2$, $-\text{N}(\text{R})\text{C}(\text{O})\text{R}$, $-\text{N}(\text{R})\text{S}(\text{O})_2\text{R}$, $-\text{N}(\text{R})\text{C}(\text{O})\text{N}(\text{R})_2$, $-\text{N}(\text{R})\text{C}(\text{S})\text{N}(\text{R})_2$, $-\text{N}(\text{R})\text{R}_8$, $-\text{N}(\text{R})\text{S}(\text{O})_2\text{R}_8$, $-\text{O}(\text{CH}_2)_n\text{OR}$, $-\text{N}(\text{R})\text{C}(=\text{NR}_9)\text{N}(\text{R})_2$, $-\text{N}(\text{R})\text{C}(=\text{CHR}_9)\text{N}(\text{R})_2$, $-\text{OC}(\text{O})\text{N}(\text{R})_2$, $-\text{N}(\text{R})\text{C}(\text{O})\text{OR}$, $-\text{N}(\text{OR})\text{C}(\text{O})\text{R}$, $-\text{N}(\text{OR})\text{S}(\text{O})_2\text{R}$, $-\text{N}(\text{OR})\text{C}(\text{O})\text{OR}$, $-\text{N}(\text{OR})\text{C}(\text{O})\text{N}(\text{R})_2$, $-\text{N}(\text{OR})\text{C}(\text{S})\text{N}(\text{R})_2$, $-\text{N}(\text{OR})\text{C}(=\text{NR}_9)\text{N}(\text{R})_2$, $-\text{N}(\text{OR})\text{C}(=\text{CHR}_9)\text{N}(\text{R})_2$, $-\text{C}(=\text{NR}_9)\text{N}(\text{R})_2$, $-\text{C}(=\text{NR}_9)\text{R}$, $-\text{C}(\text{O})\text{N}(\text{R})\text{OR}$, and $-\text{C}(\text{R})\text{N}(\text{R})_2\text{C}(\text{O})\text{OR}$, and each n is independently selected from 1, 2, 3, 4, and 5;

[0320] each R_5 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;

[0321] each R_6 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;

[0322] M and M' are independently selected from $-\text{C}(\text{O})\text{O}-$, $-\text{OC}(\text{O})-$, $-\text{OC}(\text{O})-\text{M}''-\text{C}(\text{O})\text{O}-$, $-\text{C}(\text{O})\text{N}(\text{R}')-$, $-\text{N}(\text{R}')\text{C}(\text{O})-$, $-\text{C}(\text{O})-$, $-\text{C}(\text{S})-$, $-\text{C}(\text{S})\text{S}-$, $-\text{SC}(\text{S})-$, $-\text{CH}(\text{OH})-$, $-\text{P}(\text{O})(\text{OR}')\text{O}-$, $-\text{S}(\text{O})_2-$, $-\text{S}-\text{S}-$, an aryl group, and a heteroaryl group, in which M'' is a bond, C_{1-13} alkyl or C_{2-13} alkenyl;

[0323] R_7 is selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;

[0324] R_8 is selected from the group consisting of C_{3-6} carbocycle and heterocycle;

[0325] R_9 is selected from the group consisting of H, CN, NO_2 , C_{1-6} alkyl, $-\text{OR}$, $-\text{S}(\text{O})_2\text{R}$, $-\text{S}(\text{O})_2\text{N}(\text{R})_2$, C_{2-6} alkenyl, C_{3-6} carbocycle and heterocycle;

[0326] each R is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;

[0327] each R' is independently selected from the group consisting of C_{1-18} alkyl, C_{2-18} alkenyl, $-\text{R}^*\text{YR}''$, $-\text{YR}''$, and H;

[0328] each R'' is independently selected from the group consisting of C_{3-15} alkyl and C_{3-15} alkenyl;

[0329] each R* is independently selected from the group consisting of C_{1-12} alkyl and C_{2-12} alkenyl;

[0330] each Y is independently a C_{3-6} carbocycle;

[0331] each X is independently selected from the group consisting of F, Cl, Br, and I; and

[0332] m is selected from 5, 6, 7, 8, 9, 10, 11, 12, and 13; and wherein when R_4 is $-(\text{CH}_2)_n\text{Q}$, $-(\text{CH}_2)_n\text{CHQR}$, $-\text{CHQR}$, or $-\text{CQ}(\text{R})_2$, then (i) Q is not $-\text{N}(\text{R})_2$ when n is 1, 2, 3, 4 or 5, or (ii) Q is not 5, 6, or 7-membered heterocycloalkyl when n is 1 or 2.

[0333] In some embodiments, another subset of compounds of Formula (VI) includes those in which:

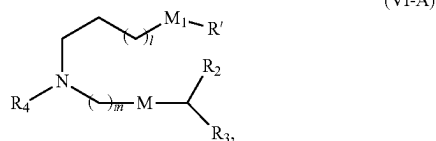
[0334] R^1 is selected from the group consisting of C_{5-30} alkyl, C_{5-20} alkenyl, $-\text{R}^*\text{YR}''$, $-\text{YR}''$, and $-\text{R}''\text{M}'\text{R}'$;

[0335] R_2 and R_3 are independently selected from the group consisting of H, C_{1-14} alkyl, C_{2-14} alkenyl, $-\text{R}^*\text{YR}''$, $-\text{YR}''$, and $-\text{R}^*\text{OR}''$, or R_2 and R_3 , together with the atom to which they are attached, form a heterocycle or carbocycle;

[0336] R_4 is selected from the group consisting of a C_{3-6} carbocycle, $-(\text{CH}_2)_n\text{Q}$, $-(\text{CH}_2)_n\text{CHQR}$, $-\text{CHQR}$, $-\text{CQ}(\text{R})_2$, and unsubstituted C_{1-6} alkyl, where Q is selected from a C_{3-6} carbocycle, a 5- to 14-membered heteroaryl having one or more heteroatoms selected from N, O, and S, $-\text{OR}$, $-\text{O}(\text{CH}_2)_n\text{N}(\text{R})_2$, $-\text{C}(\text{O})\text{OR}$, $-\text{OC}(\text{O})\text{R}$, $-\text{CX}_3$, $-\text{CX}_2\text{H}$, $-\text{CXH}_2$, $-\text{CN}$, $-\text{C}(\text{O})\text{N}(\text{R})_2$, $-\text{N}(\text{R})\text{C}(\text{O})\text{R}$, $-\text{N}(\text{R})\text{S}(\text{O})_2\text{R}$, $-\text{N}(\text{R})\text{C}(\text{O})\text{N}(\text{R})_2$, $-\text{N}(\text{R})\text{C}(\text{S})\text{N}(\text{R})_2$, $-\text{CRN}(\text{R})_2\text{C}(\text{O})\text{OR}$, $-\text{N}(\text{R})\text{R}_8$, $-\text{O}(\text{CH}_2)_n\text{OR}$, $-\text{N}(\text{R})\text{C}(=\text{NR}_9)\text{N}(\text{R})_2$, $-\text{N}(\text{R})\text{C}(=\text{CHR}_9)\text{N}(\text{R})_2$, $-\text{OC}(\text{O})\text{N}(\text{R})_2$, $-\text{N}(\text{R})\text{C}(\text{O})\text{OR}$, $-\text{N}(\text{OR})\text{C}(\text{O})\text{R}$, $-\text{N}(\text{OR})\text{S}(\text{O})_2\text{R}$, $-\text{N}(\text{OR})\text{C}(\text{O})\text{OR}$, $-\text{N}(\text{OR})\text{C}(\text{O})\text{N}(\text{R})_2$, $-\text{N}(\text{OR})\text{C}(\text{S})\text{N}(\text{R})_2$, $-\text{N}(\text{OR})\text{C}(=\text{NR}_9)\text{N}(\text{R})_2$, $-\text{N}(\text{OR})\text{C}(=\text{CHR}_9)\text{N}(\text{R})_2$, $-\text{C}(=\text{NR}_9)\text{N}(\text{R})_2$, $-\text{C}(=\text{NR}_9)\text{R}$, $-\text{C}(\text{O})\text{N}(\text{R})\text{OR}$, and a 5- to 14-membered heterocycloalkyl having one or more heteroatoms selected from N, O, and S which is substituted with one or more substituents selected from oxo ($=\text{O}$), OH, amino, mono- or di-alkylamino, and C_{1-3} alkyl, and each n is independently selected from 1, 2, 3, 4, and 5;

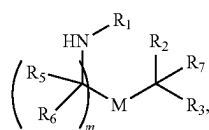
- [0337] each R_5 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;
- [0338] each R_6 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;
- [0339] M and M' are independently selected from $-C(O)O-$, $-OC(O)-$, $-C(O)N(R')$, $-N(R')C(O)-$, $-C(O)-$, $-C(S)-$, $-C(S)S-$, $-SC(S)-$, $-CH(OH)-$, $-P(O)(OR')O-$, $-S(O)_2-$, $-S-S-$, an aryl group, and a heteroaryl group;
- [0340] R_7 is selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;
- [0341] R_8 is selected from the group consisting of C_{3-6} carbocycle and heterocycle;
- [0342] R_9 is selected from the group consisting of H, CN, NO_2 , C_{1-6} alkyl, $-OR$, $-S(O)_2R$, $-S(O)_2N(R)_2$, C_{2-6} alkenyl, C_{3-6} carbocycle and heterocycle;
- [0343] each R is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;
- [0344] each R' is independently selected from the group consisting of C_{1-18} alkyl, C_{2-18} alkenyl, $-R^*YR''$, $-YR''$, and H;
- [0345] each R'' is independently selected from the group consisting of C_{3-14} alkyl and C_{3-14} alkenyl;
- [0346] each R* is independently selected from the group consisting of C_{1-12} alkyl and C_{2-12} alkenyl;
- [0347] each Y is independently a C_{3-6} carbocycle;
- [0348] each X is independently selected from the group consisting of F, Cl, Br, and I; and
- [0349] m is selected from 5, 6, 7, 8, 9, 10, 11, 12, and 13;
- [0350] or salts or isomers thereof.
- [0351] In some embodiments, another subset of compounds of Formula (VI) includes those in which:
- [0352] R_1 is selected from the group consisting of C_{5-30} alkyl, C_{5-20} alkenyl, $-R^*YR''$, $-YR''$, and $-R''M'R'$;
- [0353] R_2 and R_3 are independently selected from the group consisting of H, C_{1-14} alkyl, C_{2-14} alkenyl, $-R^*YR''$, $-YR''$, and $-R^*OR''$, or R_2 and R_3 , together with the atom to which they are attached, form a heterocycle or carbocycle;
- [0354] R_4 is selected from the group consisting of a C_{3-6} carbocycle, $-(CH_2)_nQ$, $-(CH_2)_nCHQR$, $-CHQR$, $-CQ(R)_2$, and unsubstituted C_{1-6} alkyl, where Q is selected from a C_{3-6} carbocycle, a 5- to 14-membered heterocycle having one or more heteroatoms selected from N, O, and S, $-OR$, $-O(CH_2)_nN(R)_2$, $-C(O)OR$, $-OC(O)R$, $-CX_3$, $-CX_2H$, $-CXH_2$, $-CN$, $-C(O)N(R)_2$, $-N(R)C(O)R$, $-N(R)S(O)_2R$, $-N(R)C(O)N(R)_2$, $-N(R)C(S)N(R)_2$, $-CRN(R)_2C(O)OR$, $-N(R)R_8$, $-O(CH_2)_nOR$, $-N(R)C(=NR_9)N(R)_2$, $-N(R)C(=CHR_9)N(R)_2$, $-OC(O)N(R)_2$, $-N(R)C(O)OR$, $-N(OR)C(O)R$, $-N(OR)S(O)_2R$, $-N(OR)C(O)OR$, $-N(OR)C(O)N(R)_2$, $-N(OR)C(S)N(R)_2$, $-N(OR)C(=NR_9)N(R)_2$, $-N(OR)C(=CHR_9)N(R)_2$, $-C(=NR_9)R$, $-C(O)N(R)OR$, and $-C(=NR_9)N(R)_2$, and each n is independently selected from 1, 2, 3, 4, and 5; and when Q is a 5- to 14-membered heterocycle and (i) R_4 is $-(CH_2)_nQ$ in which n is 1 or 2, or (ii) R_4 is $-(CH_2)_nCHQR$ in which n is 1, or (iii) R_4 is $-CHQR$, and $-CQ(R)_2$, then Q is either a 5- to 14-membered heteroaryl or 8- to 14-membered heterocycloalkyl;
- [0355] each R_5 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;
- [0356] each R_6 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;
- [0357] M and M' are independently selected from $-C(O)O-$, $-OC(O)-$, $-C(O)N(R')$, $-N(R')C(O)-$, $-C(O)-$, $-C(S)-$, $-C(S)S-$, $-SC(S)-$, $-CH(OH)-$, $-P(O)(OR')O-$, $-S(O)_2-$, $-S-S-$, an aryl group, and a heteroaryl group;
- [0358] R_7 is selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;
- [0359] R_8 is selected from the group consisting of C_{3-6} carbocycle and heterocycle;
- [0360] R_9 is selected from the group consisting of H, CN, NO_2 , C_{1-6} alkyl, $-OR$, $-S(O)_2R$, $-S(O)_2N(R)_2$, C_{2-6} alkenyl, C_{3-6} carbocycle and heterocycle;
- [0361] each R is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;
- [0362] each R' is independently selected from the group consisting of C_{1-18} alkyl, C_{2-18} alkenyl, $-R^*YR''$, $-YR''$, and H;
- [0363] each R'' is independently selected from the group consisting of C_{3-14} alkyl and C_{3-14} alkenyl;
- [0364] each R* is independently selected from the group consisting of C_{1-12} alkyl and C_{2-12} alkenyl;
- [0365] each Y is independently a C_{3-6} carbocycle;
- [0366] each X is independently selected from the group consisting of F, Cl, Br, and I; and
- [0367] m is selected from 5, 6, 7, 8, 9, 10, 11, 12, and 13;
- [0368] or salts or isomers thereof.
- [0369] In some embodiments, another subset of compounds of Formula (VI) includes those in which:
- [0370] R_1 is selected from the group consisting of C_{5-30} alkyl, C_{5-20} alkenyl, $-R^*YR''$, $-YR''$, and $-R''M'R'$;
- [0371] R_2 and R_3 are independently selected from the group consisting of H, C_{1-14} alkyl, C_{2-14} alkenyl, $-R^*YR''$, $-YR''$, and $-R^*OR''$, or R_2 and R_3 , together with the atom to which they are attached, form a heterocycle or carbocycle;
- [0372] R_4 is selected from the group consisting of a C_{3-6} carbocycle, $-(CH_2)_nQ$, $-(CH_2)_nCHQR$, $-CHQR$, $-CQ(R)_2$, and unsubstituted C_{1-6} alkyl, where Q is selected from a C_{3-6} carbocycle, a 5- to 14-membered heteroaryl having one or more heteroatoms selected from N, O, and S, $-OR$, $-O(CH_2)_nN(R)_2$, $-C(O)OR$, $-OC(O)R$, $-CX_3$, $-CX_2H$, $-CXH_2$, $-CN$, $-C(O)N(R)_2$, $-N(R)C(O)R$, $-N(R)S(O)_2R$, $-N(R)C(O)N(R)_2$, $-N(R)C(S)N(R)_2$, $-CRN(R)_2C(O)OR$, $-N(R)R_8$, $-O(CH_2)_nOR$, $-N(R)C(=NR_9)N(R)_2$, $-N(R)C(=CHR_9)N(R)_2$, $-OC(O)N(R)_2$, $-N(R)C(O)OR$, $-N(OR)C(O)R$, $-N(OR)S(O)_2R$, $-N(OR)C(O)OR$, $-N(OR)C(O)N(R)_2$, $-N(OR)C(S)N(R)_2$, $-N(OR)C(=NR_9)N(R)_2$, $-N(OR)C(=CHR_9)N(R)_2$, $-C(=NR_9)R$, $-C(O)N(R)OR$, and $-C(=NR_9)N(R)_2$, and each n is independently selected from 1, 2, 3, 4, and 5;
- [0373] each R_5 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;
- [0374] each R_6 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;
- [0375] M and M' are independently selected from $-C(O)O-$, $-OC(O)-$, $-C(O)N(R')$, $-N(R')C(O)-$, $-C(O)-$, $-C(S)-$, $-C(S)S-$, $-SC(S)-$, $-CH(OH)-$, $-P(O)(OR')O-$, $-S(O)_2-$, $-S-S-$, an aryl group, and a heteroaryl group;

- [0376] R_7 is selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;
- [0377] R_8 is selected from the group consisting of C_{3-6} carbocycle and heterocycle;
- [0378] R_9 is selected from the group consisting of H, CN, NO_2 , C_{1-6} alkyl, $-OR$, $-S(O)_2R$, $-S(O)_2N(R)_2$, C_{2-6} alkenyl, C_{3-6} carbocycle and heterocycle;
- [0379] each R is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;
- [0380] each R' is independently selected from the group consisting of C_{1-18} alkyl, C_{2-18} alkenyl, $-R^*YR''$, $-YR''$, and H;
- [0381] each R'' is independently selected from the group consisting of C_{3-14} alkyl and C_{3-14} alkenyl;
- [0382] each R^* is independently selected from the group consisting of C_{1-12} alkyl and C_{2-12} alkenyl;
- [0383] each Y is independently a C_{3-6} carbocycle;
- [0384] each X is independently selected from the group consisting of F, Cl, Br, and I; and
- [0385] m is selected from 5, 6, 7, 8, 9, 10, 11, 12, and 13,
- [0386] or salts or isomers thereof.
- [0387] In some embodiments, another subset of compounds of Formula (VI) includes those in which
- [0388] R_1 is selected from the group consisting of C_{5-30} alkyl, C_{5-20} alkenyl, $-R^*YR''$, $-YR''$, and $-R''M'R'$;
- [0389] R_2 and R_3 are independently selected from the group consisting of H, C_{2-14} alkyl, C_{2-14} alkenyl, $-R^*YR''$, $-YR''$, and $-R^*OR''$, or R_2 and R_3 , together with the atom to which they are attached, form a heterocycle or carbocycle;
- [0390] R_4 is $-(CH_2)_nQ$ or $-(CH_2)_nCHQR$, where Q is $-N(R)_2$, and n is selected from 3, 4, and 5;
- [0391] each R_5 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;
- [0392] each R_6 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;
- [0393] M and M' are independently selected from $-C(O)O-$, $-OC(O)-$, $-C(O)N(R')-$, $-N(R')C(O)-$, $-C(O)-$, $-C(S)-$, $-C(S)S-$, $-SC(S)-$, $-CH(OH)-$, $-P(O)(OR')O-$, $-S(O)_2-$, $-S-S-$, an aryl group, and a heteroaryl group;
- [0394] R_7 is selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;
- [0395] each R is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;
- [0396] each R' is independently selected from the group consisting of C_{1-18} alkyl, C_{2-18} alkenyl, $-R^*YR''$, $-YR''$, and H;
- [0397] each R'' is independently selected from the group consisting of C_{3-14} alkyl and C_{3-14} alkenyl;
- [0398] each R^* is independently selected from the group consisting of C_{1-12} alkyl and C_{1-12} alkenyl;
- [0399] each Y is independently a C_{3-6} carbocycle;
- [0400] each X is independently selected from the group consisting of F, Cl, Br, and I; and
- [0401] m is selected from 5, 6, 7, 8, 9, 10, 11, 12, and 13,
- [0402] or salts or isomers thereof.
- [0403] In some embodiments, another subset of compounds of Formula (VI) includes those in which
- [0404] R_1 is selected from the group consisting of C_{5-30} alkyl, C_{5-20} alkenyl, $-R^*YR''$, $-YR''$, and $-R''M'R'$;
- [0405] R_2 and R_3 are independently selected from the group consisting of C_{1-14} alkyl, C_{2-14} alkenyl, $-R^*YR''$, $-YR''$, and $-R^*OR''$, or R_2 and R_3 , together with the atom to which they are attached, form a heterocycle or carbocycle;
- [0406] R_4 is selected from the group consisting of $-(CH_2)_nQ$, $-(CH_2)_nCHQR$, $-CHQR$, and $-CQ(R)_2$, where Q is $-N(R)_2$, and n is selected from 1, 2, 3, 4, and 5;
- [0407] each R_5 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;
- [0408] each R_6 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;
- [0409] M and M' are independently selected from $-C(O)O-$, $-OC(O)-$, $-C(O)N(R')-$, $-N(R')C(O)-$, $-C(O)-$, $-C(S)-$, $-C(S)S-$, $-SC(S)-$, $-CH(OH)-$, $-P(O)(OR')O-$, $-S(O)_2-$, $-S-S-$, an aryl group, and a heteroaryl group;
- [0410] R_7 is selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;
- [0411] each R is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;
- [0412] each R' is independently selected from the group consisting of C_{1-18} alkyl, C_{2-18} alkenyl, $-R^*YR''$, $-YR''$, and H;
- [0413] each R'' is independently selected from the group consisting of C_{3-14} alkyl and C_{3-14} alkenyl;
- [0414] each R^* is independently selected from the group consisting of C_{1-12} alkyl and C_{1-12} alkenyl;
- [0415] each Y is independently a C_{3-6} carbocycle;
- [0416] each X is independently selected from the group consisting of F, Cl, Br, and I; and
- [0417] m is selected from 5, 6, 7, 8, 9, 10, 11, 12, and 13,
- [0418] or salts or isomers thereof.
- [0419] In certain embodiments, a subset of compounds of Formula (VI) includes those of Formula (VI-A):



- [0420] or its N-oxide, or a salt or isomer thereof, wherein l is selected from 1, 2, 3, 4, and 5; m is selected from 5, 6, 7, 8, and 9; M_1 is a bond or M' ; R_4 is hydrogen, unsubstituted C_{1-3} alkyl, or $-(CH_2)_nQ$, in which Q is OH, $-NHC(S)N(R)_2$, $-NHC(O)N(R)_2$, $-N(R)C(O)R$, $-N(R)S(O)_2R$, $-N(R)R_8$, $-NHC(=NR_9)N(R)_2$, $-NHC(=CHR_9)N(R)_2$, $-OC(O)N(R)_2$, $-N(R)C(O)OR$, heteroaryl or heterocycloalkyl; M and M' are independently selected from $-C(O)O-$, $-OC(O)-$, $-OC(O)-M''-C(O)O-$, $-C(O)N(R')-$, $-P(O)(OR')O-$, $-S-S-$, an aryl group, and a heteroaryl group; and R_2 and R_3 are independently selected from the group consisting of H, C_{1-14} alkyl, and C_{2-14} alkenyl. For example, m is 5, 7, or 9. For example, Q is OH, $-NHC(S)N(R)_2$, or $-NHC(O)N(R)_2$. For example, Q is $-N(R)C(O)R$, or $-N(R)S(O)_2R$.

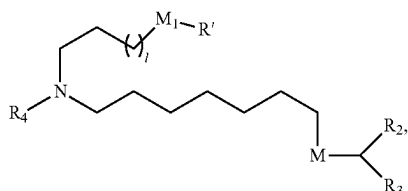
[0421] In certain embodiments, a subset of compounds of Formula (VI) includes those of Formula (VI-B):



(VI-B)

or its N-oxide, or a salt or isomer thereof in which all variables are as defined herein. For example, m is selected from 5, 6, 7, 8, and 9; R_4 is hydrogen, unsubstituted C_{1-3} alkyl, or $-(CH_2)_nQ$, in which Q is H, $-NHC(S)N(R)_2$, $-NHC(O)N(R)_2$, $-N(R)C(O)R$, $-N(R)S(O)_2R$, $-N(R)R_8$, $-NHC(=NR_9)N(R)_2$, $-NHC(=CHR_9)N(R)_2$, $-OC(O)N(R)_2$, $-N(R)C(O)OR$, heteroaryl or heterocycloalkyl; M and M' are independently selected from $-C(O)O-$, $-OC(O)-$, $-OC(O)-M''-C(O)O-$, $-C(O)N(R')-$, $-P(O)(OR')O-$, $-S-S-$, an aryl group, and a heteroaryl group; and R_2 and R_3 are independently selected from the group consisting of H, C_{1-14} alkyl, and C_{2-14} alkenyl. For example, m is 5, 7, or 9. For example, Q is OH, $-NHC(S)N(R)_2$, or $-NHC(O)N(R)_2$. For example, Q is $-N(R)C(O)R$, or $-N(R)S(O)_2R$.

[0422] In certain embodiments, a subset of compounds of Formula (VI) includes those of Formula (VII):



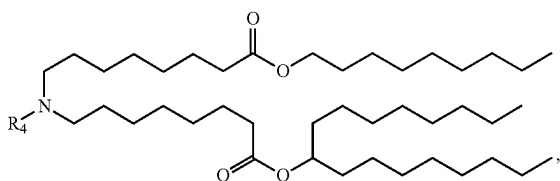
(VII)

[0423] or its N-oxide, or a salt or isomer thereof, wherein l is selected from 1, 2, 3, 4, and 5; M_1 is a bond or M' ; R_4 is hydrogen, unsubstituted C_{1-3} alkyl, or $-(CH_2)_nQ$, in which n is 2, 3, or 4, and Q is OH, $-NHC(S)N(R)_2$, $-NHC(O)N(R)_2$, $-N(R)C(O)R$, $-N(R)S(O)_2R$, $-N(R)R_8$, $-NHC(=NR_9)N(R)_2$, $-NHC(=CHR_9)N(R)_2$, $-OC(O)N(R)_2$, $-N(R)C(O)OR$, heteroaryl or heterocycloalkyl; M and M' are independently selected from $-C(O)O-$, $-OC(O)-$, $-OC(O)-M''-C(O)O-$, $-C(O)N(R')-$, $-P(O)(OR')O-$, $-S-S-$, an aryl group, and a heteroaryl group;

[0424] and R_2 and R_3 are independently selected from the group consisting of H, C_{1-14} alkyl, and C_{2-14} alkenyl.

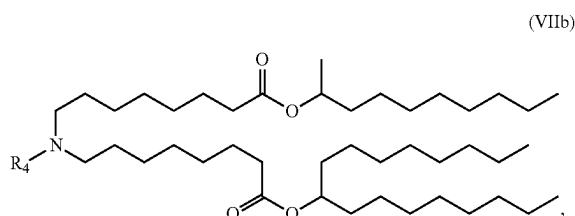
[0425] In one embodiment, the compounds of Formula (VI) are of Formula (VIIa),

(VIIa)



[0426] or their N-oxides, or salts or isomers thereof, wherein R^4 is as defined above.

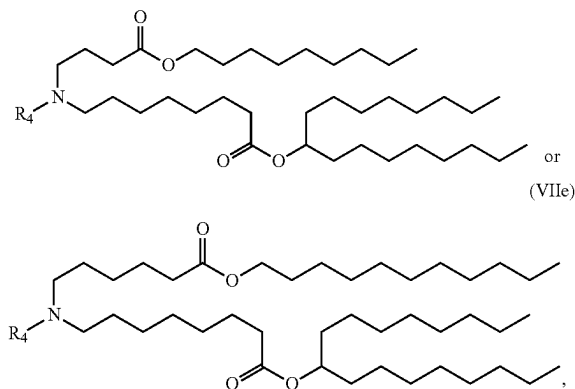
[0427] In another embodiment, the compounds of Formula (VI) are of Formula (VIIb),



(VIIb)

[0428] or their N-oxides, or salts or isomers thereof, wherein R^4 is as defined above.

[0429] In another embodiment, the compounds of Formula (VI) are of Formula (VIIc) or (VIIe):

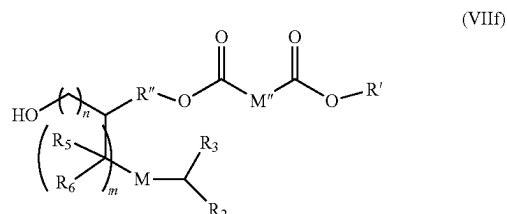


(VIIc)

(VIIe)

[0430] or their N-oxides, or salts or isomers thereof, wherein R^4 is as defined above.

[0431] In another embodiment, the compounds of Formula (VI) are of Formula (VIIf):

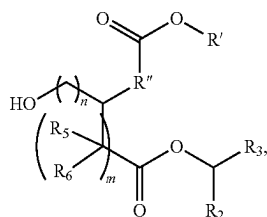


(VIIf)

or their N-oxides, or salts or isomers thereof,

[0432] wherein M is $-C(O)O-$ or $-OC(O)-$, M'' is C_{1-6} alkyl or C_{2-6} alkenyl, R_2 and R_3 are independently selected from the group consisting of C_{5-14} alkyl and C_{5-14} alkenyl, and n is selected from 2, 3, and 4.

[0433] In a further embodiment, the compounds of Formula (VI) are of Formula (VIIId),

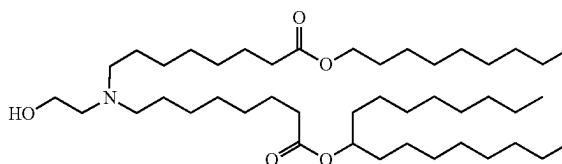


(VIIId)

(heptadecan-9-yl 8 ((2 hydroxyethyl)(6 oxo 6-(undecyloxy) hexyl)amino) octanoate).

[0436] In some embodiments, an ionizable amino lipid comprises a compound having structure:

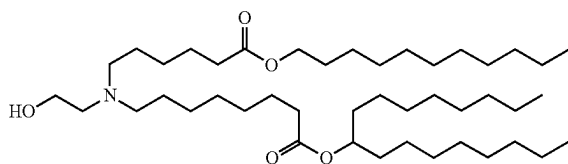
(Compound II)



[0434] or their N-oxides, or salts or isomers thereof, wherein n is 2, 3, or 4; and m, R', R'', and R₂ through R₆ are as defined above. For example, each of R₂ and R₃ may be independently selected from the group consisting of C₅₋₁₄ alkyl and C₅₋₁₄ alkenyl.

[0435] In some embodiments, an ionizable amino lipid comprises a compound having structure:

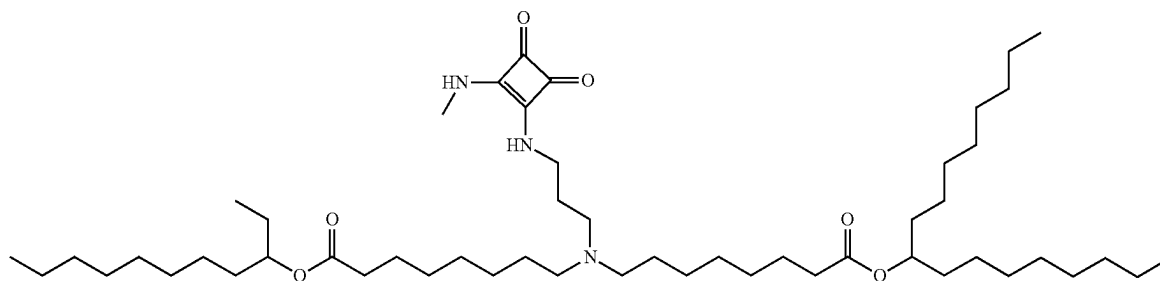
(Compound I)



(heptadecan-9-yl 8-((2-hydroxyethyl)(8-(nonyloxy)-8-oxooctyl)amino)octanoate).

[0437] In some embodiments, an ionizable amino lipid comprises a compound having structure:

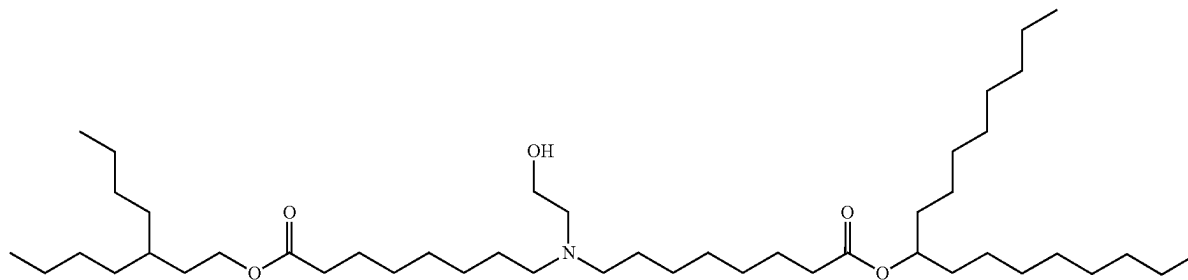
(Compound III)



((heptadecan-9-yl 8-((3-((2-(methylamino)-3,4-dioxocyclobut-1-en-1-yl)amino)propyl)(8-oxo-8-(undecan-3-yloxy) octyl)amino)octanoate)).

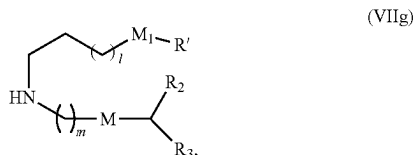
[0438] In some embodiments, an ionizable amino lipid comprises a compound having structure:

(Compound IV)



(3-butylheptyl 8-((8-(heptadecan-9-yloxy)-8-oxooctyl) (2-hydroxyethyl)amino)octanoate).

[0439] In a further embodiment, the compounds of Formula (VI) are of Formula (VIIg),

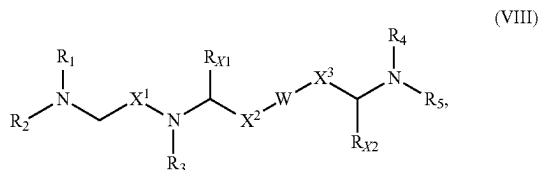


[0440] or their N-oxides, or salts or isomers thereof, wherein l is selected from 1, 2, 3, 4, and 5; m is selected from 5, 6, 7, 8, and 9; M_1 is a bond or M' ; M and M' are independently selected from $-\text{C}(\text{O})\text{O}-$, $-\text{OC}(\text{O})-$, $-\text{OC}(\text{O})-\text{M}''-\text{C}(\text{O})\text{O}-$, $-\text{C}(\text{O})\text{N}(\text{R}')-$, $-\text{P}(\text{O})(\text{OR}')\text{O}-$, $-\text{S}-\text{S}-$, an aryl group, and a heteroaryl group; and R_2 and R_3 are independently selected from the group consisting of H, C_{1-14} alkyl, and C_{2-14} alkenyl. For example, M'' is C_{1-6} alkyl (e.g., C_{1-4} alkyl) or C_{2-6} alkenyl (e.g., C_{2-4} alkenyl). For example, R_2 and R_3 are independently selected from the group consisting of C_{5-14} alkyl and C_{5-14} alkenyl.

[0441] In some embodiments, the ionizable amino lipids are one or more of the compounds described in U.S. Application Nos. 62/220,091, 62/252,316, 62/253,433, 62/266,460, 62/333,557, 62/382,740, 62/393,940, 62/471,937, 62/471,949, 62/475,140, and 62/475,166, and PCT Application No. PCT/US2016/052352.

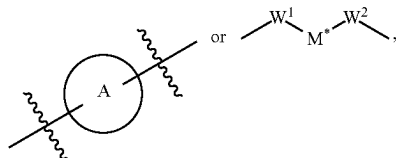
[0442] The central amine moiety of a lipid according to Formula (VI), (VI-A), (VI-B), (VII), (VIIa), (VIIb), (VIIc), (VIId), (VIIe), (VIIf), or (VIIg) may be protonated at a physiological pH. Thus, a lipid may have a positive or partial positive charge at physiological pH. Such amino lipids may be referred to as cationic lipids, ionizable lipids, cationic amino lipids, or ionizable amino lipids. Amino lipids may also be zwitterionic, i.e., neutral molecules having both a positive and a negative charge.

[0443] In some embodiments, the ionizable amino lipid may be one or more of compounds of formula (VIII),

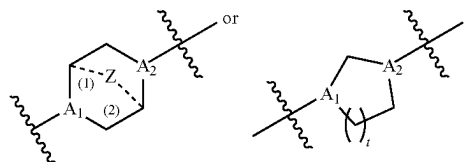


[0444] or salts or isomers thereof, wherein

[0445] W is



[0446] ring A is



[0447] t is 1 or 2;

[0448] A_1 and A_2 are each independently selected from CH or N;

[0449] Z is CH_2 or absent wherein when Z is CH_2 , the dashed lines (1) and (2) each represent a single bond; and when Z is absent, the dashed lines (1) and (2) are both absent;

[0450] R_1 , R_2 , R_3 , R_4 , and R_5 are independently selected from the group consisting of C_{5-20} alkyl, C_{5-20} alkenyl, $-\text{R}''\text{MR}'$, $-\text{R}^*\text{YR}''$, $-\text{YR}''$, and $-\text{R}^*\text{OR}''$;

[0451] R_{X1} and R_{X2} are each independently H or C_{1-3} alkyl;

[0452] each M is independently selected from the group consisting of $-\text{C}(\text{O})\text{O}-$, $-\text{OC}(\text{O})-$, $-\text{OC}(\text{O})\text{O}-$, $-\text{C}(\text{O})\text{N}(\text{R}')-$, $-\text{N}(\text{R}')\text{C}(\text{O})-$, $-\text{C}(\text{O})-$, $-\text{C}(\text{S})-$, $-\text{C}(\text{S})\text{S}-$, $-\text{SC}(\text{S})-$, $-\text{CH}(\text{OH})-$, $-\text{P}(\text{O})(\text{OR}')\text{O}-$, $-\text{S}(\text{O})_2-$, $-\text{C}(\text{O})\text{S}-$, $-\text{SC}(\text{O})-$, an aryl group, and a heteroaryl group;

[0453] M^* is C_1 - C_6 alkyl,

[0454] W^1 and W^2 are each independently selected from the group consisting of $-\text{O}-$ and $-\text{N}(\text{R}_6)-$;

[0455] each R_6 is independently selected from the group consisting of H and C_{1-5} alkyl;

[0456] X^1 , X^2 , and X^3 are independently selected from the group consisting of a bond, $-\text{CH}_2-$, $-(\text{CH}_2)_2-$, $-\text{CHR}-$, $-\text{CHY}-$, $-\text{C}(\text{O})-$, $-\text{C}(\text{O})\text{O}-$, $-\text{OC}(\text{O})-$, $-(\text{CH}_2)_n-\text{C}(\text{O})-$, $-\text{C}(\text{O})-(\text{CH}_2)_n-$, $-(\text{CH}_2)_n-\text{C}(\text{O})\text{O}-$, $-\text{OC}(\text{O})-(\text{CH}_2)_n-$, $-(\text{CH}_2)_n-\text{OC}(\text{O})-$, $-\text{C}(\text{O})\text{O}-(\text{CH}_2)_n-$, $-\text{CH}(\text{OH})-$, $-\text{C}(\text{S})-$, and $-\text{CH}(\text{SH})-$;

[0457] each Y is independently a C_{3-6} carbocycle;

[0458] each R^* is independently selected from the group consisting of C_{1-12} alkyl and C_{2-12} alkenyl;

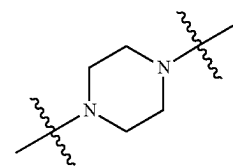
[0459] each R is independently selected from the group consisting of C_{1-3} alkyl and a C_{3-6} carbocycle;

[0460] each R' is independently selected from the group consisting of C_{1-12} alkyl, C_{2-12} alkenyl, and H;

[0461] each R'' is independently selected from the group consisting of C_{3-12} alkyl, C_{3-12} alkenyl and $-\text{R}^*\text{MR}'$; and

[0462] n is an integer from 1-6;

[0463] wherein when ring A is

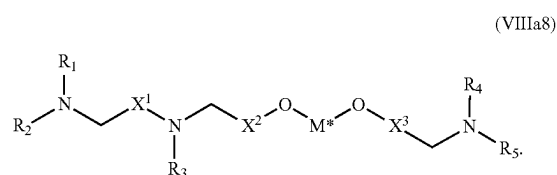
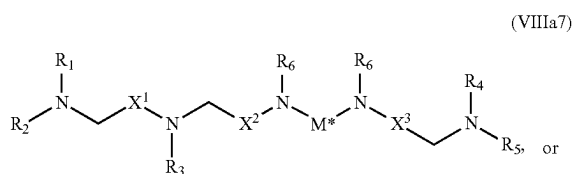
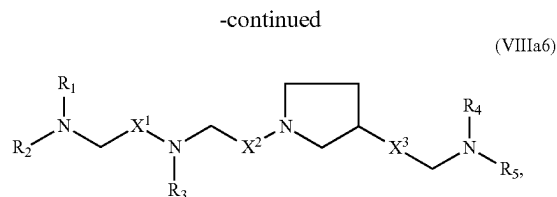
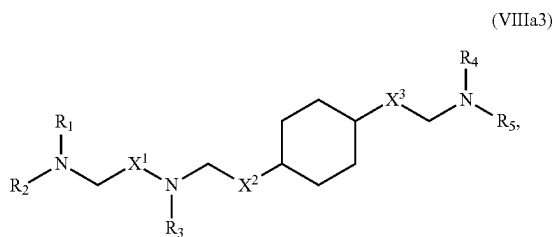
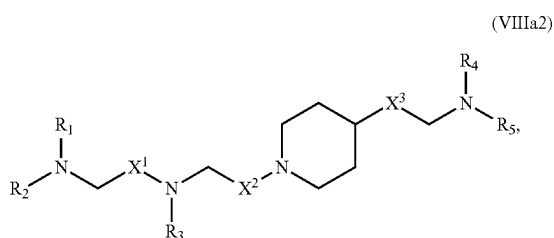
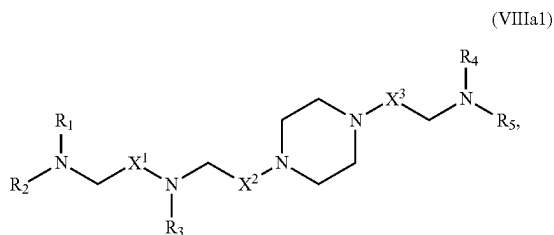


then

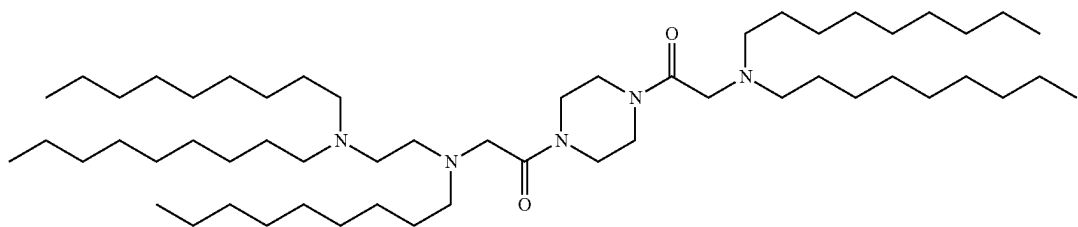
[0464] i) at least one of X^1 , X^2 , and X^3 is not $-\text{CH}_2-$; and/or

[0465] ii) at least one of R_1 , R_2 , R_3 , R_4 , and R_5 is $-R''MR'$.

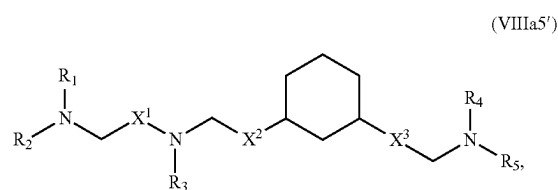
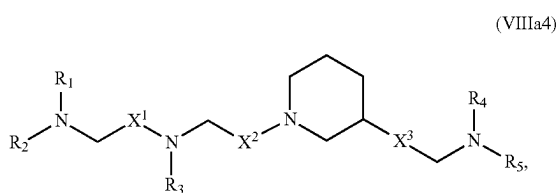
[0466] In some embodiments, the compound is of any of formulae (VIIIa1)-(VIIIa8):



[0467] In some embodiments, the ionizable amino lipid is



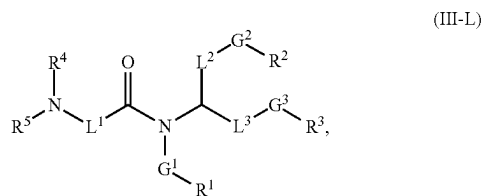
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or a salt thereof.

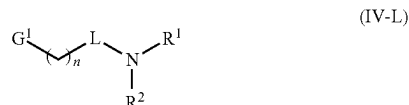
[0468] The central amine moiety of a lipid according to Formula (VIII), (VIIIa1), (VIIIa2), (VIIIa3), (VIIIa4), (VIIIa5), (VIIIa6), (VIIIa7), or (VIIIa8) may be protonated at a physiological pH. Thus, a lipid may have a positive or partial positive charge at physiological pH.

[0469] In some embodiments, the lipid nanoparticle comprises a lipid having the structure:



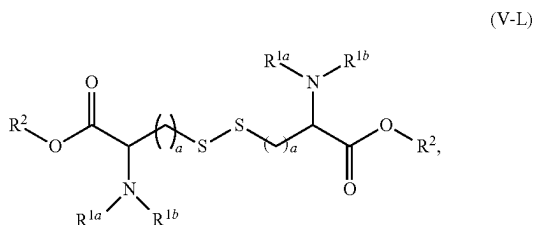
or a pharmaceutically acceptable salt, tautomer, or stereoisomer thereof, wherein:

- [0470]** R¹ is optionally substituted C₁-C₂₄ alkyl or optionally substituted C₂-C₂₄ alkenyl;
- [0471]** R² and R³ are each independently optionally substituted C₁-C₃₆ alkyl;
- [0472]** R⁴ and R⁵ are each independently optionally substituted C₁-C₆ alkyl, or R⁴ and R⁵ join, along with the N to which they are attached, to form a heterocyclyl or heteroaryl;
- [0473]** L¹, L², and L³ are each independently optionally substituted C₁-C₁₈ alkylene;
- [0474]** G¹ is a direct bond, —(CH₂)_nO(C=O)—, —(CH₂)_n(C=O)O—, or —(C=O)—;
- [0475]** G² and G³ are each independently —(C=O)O— or —O(C=O)—; and n is an integer greater than 0.
- [0476]** In some embodiments, the lipid nanoparticle comprises a lipid having the structure:



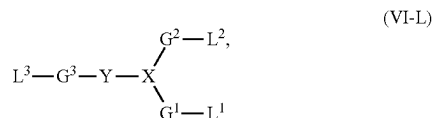
or a pharmaceutically acceptable salt, tautomer, or stereoisomer thereof, wherein:

- [0477] R^1 is $-\text{N}(\text{R}^3)\text{R}^4$ or $-\text{OR}^5$;
 [0478] R^1 is optionally substituted branched, saturated or unsaturated $\text{C}_{12}\text{-C}_{36}$ alkyl;
 [0479] R^2 is optionally substituted branched or unbranched, saturated or unsaturated $\text{C}_{12}\text{-C}_{36}$ alkyl when L is $-\text{C}(=\text{O})-$; or R^2 is optionally substituted branched or unbranched, saturated or unsaturated $\text{C}_4\text{-C}_{36}$ alkyl when L is $\text{C}_6\text{-C}_{12}$ alkylenylene, $\text{C}_6\text{-C}_{12}$ alkynylene, or $\text{C}_2\text{-C}_6$ alkynylene;
 [0480] R^3 and R^4 are each independently H, optionally substituted branched or unbranched, saturated or unsaturated $\text{C}_1\text{-C}_6$ alkyl; or R^3 and R^4 are each independently optionally substituted branched or unbranched, saturated or unsaturated $\text{C}_1\text{-C}_6$ alkyl when L is $\text{C}_6\text{-C}_{12}$ alkylenylene, $\text{C}_6\text{-C}_{12}$ alkynylene, or $\text{C}_2\text{-C}_6$ alkynylene; or R^3 and R^4 , together with the nitrogen to which they are attached, join to form a heterocyclyl;
 [0481] R^5 is H or optionally substituted $\text{C}_1\text{-C}_6$ alkyl;
 [0482] L is $-\text{C}(=\text{O})-$, $\text{C}_6\text{-C}_{12}$ alkylenylene, $\text{C}_6\text{-C}_{12}$ alkynylene, or $\text{C}_2\text{-C}_6$ alkynylene; and
 [0483] n is an integer from 1 to 12.
 [0484] In some embodiments, the lipid nanoparticle comprises a lipid having the structure:



or a pharmaceutically acceptable salt thereof, wherein:

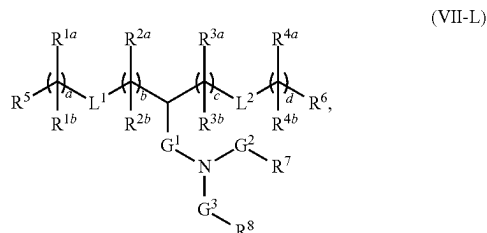
- [0485] each R^{1a} is independently hydrogen, R^{1c} , or R^{1d} ;
 [0486] each R^{1b} is independently R^{1c} or R^{1d} ;
 [0487] each R^{1c} is independently $-\text{[CH}_2\text{]}_2\text{C(O)X}^1\text{R}^3$;
 [0488] each R^{1d} is independently $-\text{C(O)R}^4$;
 [0489] each R_2 is independently $-\text{[C(R}^{2a}\text{)]}_2\text{R}^{2b}$;
 [0490] each R^{2a} is independently hydrogen or $\text{C}_1\text{-C}_6$ alkyl;
 [0491] R^{2b} is $-\text{N(L}_1\text{-B)}_2$; $-(\text{OCH}_2\text{CH}_2)_6\text{OH}$; or $-(\text{OCH}_2\text{CH}_2)_b\text{OCH}_3$;
 [0492] each R^3 and R^4 is independently $\text{C}_6\text{-C}_{30}$ aliphatic;
 [0493] each I_3 is independently $\text{C}_1\text{-C}_{10}$ alkylene;
 [0494] each B is independently hydrogen or an ionizable nitrogen-containing group;
 [0495] each X^1 is independently a covalent bond or O;
 [0496] each a is independently an integer of 1-10;
 [0497] each b is independently an integer of 1-10; and
 [0498] each c is independently an integer of 1-10.
 [0499] In some embodiments, the lipid nanoparticle comprises a lipid having the structure:



or a pharmaceutically acceptable salt, prodrug or stereoisomer thereof, wherein:

- [0500] X is N, and Y is absent; or X is CR, and Y is NR;
 [0501] L¹ is —O(C=O)R¹, —(C=O)OR¹, —C(=O)R¹, —OR¹, —S(O)_xR¹, —S—SR¹, —C(=O)SR¹, —SC(=O)R¹, —NR^aC(=O)R¹, —C(=O)NR^bR^c, —NR^aC(=O)NR^bR^c, —OC(=O)NR^bR^c, or —NR^aC(=O)OR¹;
 [0502] L² is —O(C=O)R², —(C=O)OR², —C(=O)R², —OR², —S(O)_xR², —S—SR², —C(=O)SR², —SC(=O)R², —NR^dC(=O)R², —C(=O)NR^eR^f, —NR^dC(=O)NR^eR^f, —OC(=O)NR^eR^f, —NR^dC(=O)OR² or a direct bond to R²;
 [0503] L³ is —O(C=O)R³ or —(C=O)OR³;
 [0504] G¹ and G² are each independently C₂-C₁₂ alkenylene or C₂-C₁₂ alkenylene;
 [0505] G³ is C₁-C₂₄ alkenylene, C₂-C₂₄ alkenylene, C₁-C₂₄ heteroalkylene or C₂-C₂₄ heteroalkenylene when X is CR, and Y is NR; and G³ is C₁-C₂₄ heteroalkylene or C₂-C₂₄ heteroalkenylene when X is N, and Y is absent;
 [0506] R^a, R^b, R^d and R^e are each independently H or C₁-C₁₂ alkyl or C₁-C₁₂ alkenyl;
 [0507] R^c and R^f are each independently C₁-C₁₂ alkyl or C₂-C₁₂ alkenyl;
 [0508] each R is independently H or C₁-C₁₂ alkyl;
 [0509] R¹, R² and R³ are each independently C₁-C₂₄ alkyl or C₂-C₂₄ alkenyl; and x is 0, 1 or 2, and
 [0510] wherein each alkyl, alkenyl, alkenylene, alkenylene, heteroalkylene and heteroalkenylene is independently substituted or unsubstituted unless otherwise specified.

[0511] In some embodiments, the lipid nanoparticle comprises a lipid having the structure:



or a pharmaceutically acceptable salt, tautomer, prodrug or stereoisomer thereof, wherein:

[0512] L^1 and L^2 are each independently $-\text{O}(\text{C}=\text{O})-$, $-(\text{C}=\text{O})\text{O}-$, $-\text{C}(\text{O})-$, $-\text{O}-$, $-\text{S}(\text{O})_x-\text{S}-$, $-\text{S}-$, $-\text{C}(\text{O})\text{S}-$, $-\text{SC}(\text{O})-$, $-\text{NR}^a\text{C}(\text{O})-$, $-\text{C}(\text{O})\text{NR}^a-$, $-\text{NR}^a\text{C}(\text{O})\text{NR}^a-$, $-\text{OC}(\text{O})\text{NR}^a-$, $-\text{NR}^a\text{C}(\text{O})\text{O}-$ or a direct bond;

[0513] G^1 is C , $-\text{C}_2$ alkylene, $-(\text{C}=\text{O})-$, $-\text{O}(\text{C}=\text{O})-$, $-\text{SC}(\text{O})-$, $-\text{NR}^a\text{C}(\text{O})-$ or a direct bond;

[0514] G^2 is $-\text{C}(\text{O})-$, $-(\text{CO})\text{O}-$, $-\text{C}(\text{O})\text{S}-$, $-\text{C}(\text{O})\text{NR}^a-$ or a direct bond;

[0515] G^3 is C_1 - C_6 alkylene;

[0516] R^a is H or C_1 - C_{12} alkyl;

[0517] R^{1a} and R^{1b} are, at each occurrence, independently either: (a) H or C_1 - C_{12} alkyl; or (b) R^{1a} is H or C_1 - C_{12} alkyl, and R^{1b} together with the carbon atom to which it is bound is taken together with an adjacent R^{1b} and the carbon atom to which it is bound to form a carbon-carbon double bond;

[0518] R^{2a} and R^{2b} are, at each occurrence, independently either: (a) H or C_1 - C_{12} alkyl; or (b) R^{2a} is H or C_1 - C_{12} alkyl, and R^{2b} together with the carbon atom to which it is bound is taken together with an adjacent R^{2b} and the carbon atom to which it is bound to form a carbon-carbon double bond;

[0519] R^{3a} and R^{3b} are, at each occurrence, independently either: (a) H or C_1 - C_{12} alkyl; or (b) R^{3a} is H or C_1 - C_{12} alkyl, and R^{3b} together with the carbon atom to which it is bound is taken together with an adjacent R and the carbon atom to which it is bound to form a carbon-carbon double bond;

[0520] R^{4a} and R^{4b} are, at each occurrence, independently either: (a) H or C_1 - C_{12} alkyl; or (b) R^{4a} is H or C_1 - C_{12} alkyl, and R^{4b} together with the carbon atom to which it is bound is taken together with an adjacent R^{4b} and the carbon atom to which it is bound to form a carbon-carbon double bond;

[0521] R^5 and R^6 are each independently H or methyl;

[0522] R^7 is H or C_1 - C_{20} alkyl;

[0523] R^8 is OH, $-\text{N}(\text{R}^9)(\text{C}=\text{O})\text{R}^{10}$, $-(\text{C}=\text{O})\text{NR}^9\text{R}^{10}$, $-\text{NR}^9\text{R}^{10}$, $-(\text{C}=\text{O})\text{OR}^{11}$ or $-\text{O}(\text{C}=\text{O})\text{R}^{11}$,

[0524] provided that G^3 is C_4 - C_6 alkylene when R^8 is $-\text{NR}^9\text{R}^{10}$;

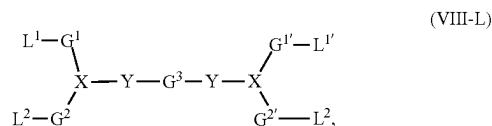
[0525] R^9 and R^{10} are each independently H or C_1 - C_{12} alkyl;

[0526] R^{11} is aralkyl;

[0527] a, b, c and d are each independently an integer from 1 to 24; and x is 0, 1 or 2,

[0528] wherein each alkyl, alkylene and aralkyl is optionally substituted.

[0529] In some embodiments, the lipid nanoparticle comprises a lipid having the structure:



or a pharmaceutically acceptable salt, prodrug or stereoisomer thereof, wherein:

[0530] X and X' are each independently N or CR;

[0531] Y and Y' are each independently absent, $-\text{O}(\text{C}=\text{O})-$, $-(\text{C}=\text{O})\text{O}-$ or NR, provided that:

[0532] a) Y is absent when X is N;

[0533] b) Y' is absent when X' is N;

[0534] c) Y is $-\text{O}(\text{C}=\text{O})-$, $-(\text{C}=\text{O})\text{O}-$ or NR when X is CR; and

[0535] d) Y' is $-\text{O}(\text{C}=\text{O})-$, $-(\text{C}=\text{O})\text{O}-$ or NR when X' is CR,

[0536] L^1 and $L^{1'}$ are each independently $-\text{O}(\text{C}=\text{O})\text{R}^1$, $-(\text{C}=\text{O})\text{OR}^1$, $-\text{C}(\text{O})\text{R}^1$, $-\text{OR}^1$, $-\text{S}(\text{O})_2\text{R}^1$, $-\text{S}-\text{SR}^1$, $-\text{C}(\text{O})\text{SR}^1$, $-\text{SC}(\text{O})\text{R}^1$, $-\text{NR}^a\text{C}(\text{O})\text{R}^1$, $-\text{C}(\text{O})\text{NR}^b\text{R}^c$, $-\text{NR}^a\text{C}(\text{O})\text{NR}^b\text{R}^c$, $-\text{OC}(\text{O})\text{NR}^b\text{R}^c$ or $-\text{NR}^a\text{C}(\text{O})\text{OR}^1$;

[0537] L^2 and $L^{2'}$ are each independently $-\text{O}(\text{C}=\text{O})\text{R}^2$, $-(\text{C}=\text{O})\text{OR}^2$, $-\text{C}(\text{O})\text{R}^2$, $-\text{OR}^2$, $-\text{S}(\text{O})_2\text{R}^2$, $-\text{S}-\text{SR}^2$, $-\text{C}(\text{O})\text{SR}^2$, $-\text{SC}(\text{O})\text{R}^2$, $-\text{NR}^d\text{C}(\text{O})\text{R}^2$, $-\text{C}(\text{O})\text{NR}^e\text{R}^f$, $-\text{NR}^d\text{C}(\text{O})\text{NR}^e\text{R}^f$, $-\text{OC}(\text{O})\text{NR}^e\text{R}^f$, $-\text{NR}^d\text{C}(\text{O})\text{OR}^2$ or a direct bond to R^2 ;

[0538] G^1 , $G^{1'}$, G^2 and $G^{2'}$ are each independently C_2 - C_{12} alkylene or C_2 - C_{12} alkenylene;

[0539] G is C_2 - C_{24} heteroalkylene or C_2 - C_{24} heteroalkenylene;

[0540] R^a , R^b , R^d and R^e are, at each occurrence, independently H, C_1 - C_{12} alkyl or C_2 - C_{12} alkenyl;

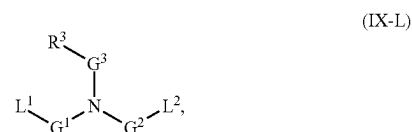
[0541] R^c and R^f are, at each occurrence, independently C_1 - C_{12} alkyl or C_2 - C_{12} alkenyl;

[0542] R is, at each occurrence, independently H or C_1 - C_{12} alkyl;

[0543] R^1 and R^2 are, at each occurrence, independently branched C_6 - C_{24} alkyl or branched C_6 - C_{24} alkenyl;

[0544] z is 0, 1 or 2, and wherein each alkyl, alkenyl, alkylene, alkenylene, heteroalkylene and heteroalkenylene is independently substituted or unsubstituted unless otherwise specified.

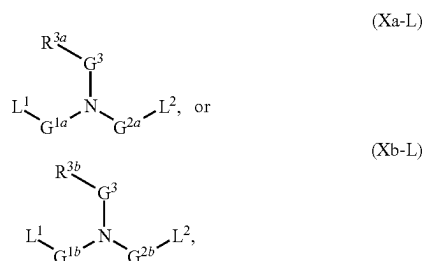
[0545] In some embodiments, the lipid nanoparticle comprises a lipid having the structure:



or a pharmaceutically acceptable salt, prodrug or stereoisomer thereof, wherein:

[0546] L^1 is $-\text{O}(\text{C}=\text{O})\text{R}^1$, $-(\text{C}=\text{O})\text{OR}^1$, $-\text{C}(\text{O})\text{R}^1$, $-\text{OR}^1$, $-\text{S}(\text{O})_2\text{R}^1$, $-\text{S}-\text{SR}^1$, $-\text{C}(\text{O})\text{SR}^1$,

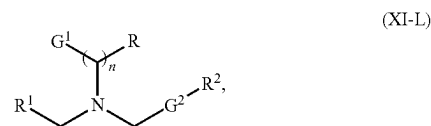
- SC(=O)R¹, —NR^aC(=O)R¹, —C(=O)NR^bR^c, —NR^aC(=O)NR^bR^c, —OC(=O)NR^bR^c or —NR^aC(=O)OR¹;
- [0547] L² is —O(C=O)R², —(C=O)OR², —C(=O)R², —OR², —S(O)_xR², —S—SR², —C(=O)SR², —SC(=O)R², —NR^dC(=O)R², —C(=O)NR^eR^f, —NR^dC(=O)NR^eR^f, —OC(=O)NR^eR^f; —NR^dC(=O)OR² or a direct bond to R²;
- [0548] G¹ and G² are each independently C₂-C₁₂ alkylene or C₂-C₁₂ alkenylene;
- [0549] G³ is C₁-C₂₄ alkylene, C₂-C₂₄ alkenylene, C₃-C₈ cycloalkylene or C₃-C₈ cycloalkenylene;
- [0550] R^a, R^b, R^d and R^e are each independently H or C₁-C₁₂ alkyl or C₁-C₁₂ alkenyl;
- [0551] R^e and R^f are each independently C₁-C₁₂ alkyl or C₂-C₁₂ alkenyl;
- [0552] R¹ and R² are each independently branched C₆-C₂₄ alkyl or branched C₆-C₂₄ alkenyl;
- [0553] R³ is —N(R⁴)R⁵;
- [0554] R⁴ is C₁-C₁₂ alkyl;
- [0555] R⁵ is substituted C₁-C₁₂ alkyl; and
- [0556] x is 0, 1 or 2, and
- [0557] wherein each alkyl, alkenyl, alkylene, alkenylene, cycloalkylene, cycloalkenylene, aryl and aralkyl is independently substituted or unsubstituted unless otherwise specified.
- [0558] In some embodiments, the lipid nanoparticle comprises a lipid having the structure:



or a pharmaceutically acceptable salt, prodrug or stereoisomer thereof, wherein:

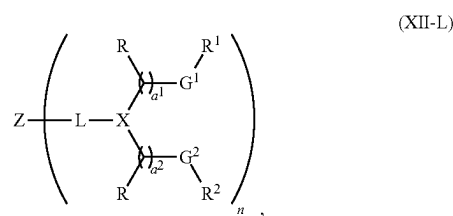
- [0559] L¹ is —O(C=O)R¹, —(C=O)OR¹, —C(=O)R¹, —OR¹, —S(O)_xR¹, —S—SR¹, —C(=O)SR¹, —SC(=O)R¹, —NR^aC(=O)R¹, —C(=O)NR^bR^c, —NR^aC(=O)NR^bR^c, —OC(=O)NR^bR^c or —NR^aC(=O)OR¹;
- [0560] L² is —O(C=O)R², —(C=O)OR², —C(=O)R², —OR², —S(O)_xR², —S—SR², —C(=O)SR², —SC(=O)R², —NR^dC(=O)R², —C(=O)NR^eR^f, —NR^dC(=O)NR^eR^f, —OC(=O)NR^eR^f; —NR^dC(=O)OR² or a direct bond to R²;
- [0561] G^{1a} and G^{2b} are each independently C₂-C₁₂ alkylene or C₂-C₁₂ alkenylene;
- [0562] G^{1b} and G^{2b} are each independently C₁-C₁₂ alkylene or C₂-C₁₂ alkenylene;
- [0563] G³ is C₁-C₂₄ alkylene, C₂-C₂₄ alkenylene, C₃-C₈ cycloalkylene or C₃-C₈ cycloalkenylene;
- [0564] R^a, R^b, R^d and R^e are each independently H or C₁-C₁₂ alkyl or C₂-C₁₂ alkenyl;
- [0565] R^e and R^f are each independently C₁-C₁₂ alkyl or C₂-C₁₂ alkenyl;

- [0566] R¹ and R² are each independently branched C₆-C₂₄ alkyl or branched C₆-C₂₄ alkenyl;
- [0567] R^{3a} is —C(=O)N(R^{4a})R^{5a} or —C(=O)OR⁶;
- [0568] R^{3b} is —NR^{4b}C(=O)R^{5b};
- [0569] R^{4a} is C₁-C₁₂ alkyl;
- [0570] R^{4b} is H, C₁-C₁₂ alkyl or C₂-C₁₂ alkenyl;
- [0571] R^{5a} is H, C₁-C₈ alkyl or C₂-C₈ alkenyl;
- [0572] R^{5b} is C₂-C₁₂ alkyl or C₂-C₁₂ alkenyl when R^{4b} is H; or R^{5b} is C₁-C₁₂ alkyl or C₂-C₁₂ alkenyl when R^{4b} is C₁-C₁₂ alkyl or C₂-C₁₂ alkenyl;
- [0573] R⁶ is H, aryl or aralkyl; and
- [0574] x is 0, 1 or 2, and
- [0575] wherein each alkyl, alkenyl, alkylene, alkenylene, cycloalkylene, cycloalkenylene, aryl and aralkyl is independently substituted or unsubstituted.
- [0576] In some embodiments, the lipid nanoparticle comprises a lipid having the structure:



or a pharmaceutically acceptable salt, prodrug or stereoisomer thereof, wherein:

- [0577] G¹ is —OH, —R³R⁴, —(C=O)R⁵ or —R³(C=O)R⁵;
- [0578] G² is —CH₂— or —(C=O)—;
- [0579] R is, at each occurrence, independently H or OH;
- [0580] R¹ and R² are each independently optionally substituted branched, saturated or unsaturated C₁₂-C₃₆ alkyl;
- [0581] R³ and R⁴ are each independently H or optionally substituted straight or branched, saturated or unsaturated C₁-C₆ alkyl;
- [0582] R⁵ is optionally substituted straight or branched, saturated or unsaturated C₁-C₆ alkyl; and
- [0583] n is an integer from 2 to 6.
- [0584] In some embodiments, the lipid nanoparticle comprises a lipid having the structure:



or a pharmaceutically acceptable salt, prodrug or stereoisomer thereof, wherein:

- [0585] one of G¹ or G² is, at each occurrence, —O(C=O)—, —(C=O)O—, —C(=O)—, —O—, —S(O)—, —S—S—, —C(=O)S—, —SC(=O)—, —N(R^a)C(=O)—, —C(=O)N(R^a)—, —N(R^a)C(=O)N(R^a)—, —OC(=O)N(R^a)— or —N(R^a)C(=O)O—, and the other of G¹ or G² is, at each occurrence, —O(C=O)—, —(C=O)O—, —C(=O)—, —O—, —S(O)—, —S—S—, —C(=O)S—, —SC(=O)—, —N(R^a)C(=O)—, —C(=O)N(R^a)—

(R^a)—, —N(R^a)C(=O)N(R^a)—, —OC(=O)N(R^a)— or —N(R^a)C(=O)O— or a direct bond;

[0586] L is, at each occurrence, —O(C=O)—, wherein ~ represents a covalent bond to X;

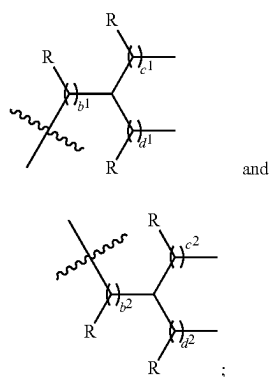
X is CR^a;

[0587] Z is alkyl, cycloalkyl or a monovalent moiety comprising at least one polar functional group when n is 1; or Z is alkylene, cycloalkylene or a polyvalent moiety comprising at least one polar functional group when n is greater than 1;

[0588] R^a is, at each occurrence, independently H, C₁-C₁₂ alkyl, C₁-C₁₂ hydroxylalkyl, C₁-C₁₂ aminoalkyl, C₁-C₁₂ alkylaminylalkyl, C₁-C₁₂ alkoxyalkyl, C₁-C₁₂ alkoxyalkyl, C₁-C₁₂ alkylcarbonyloxy, C₁-C₁₂ alkylcarbonyloxyalkyl or C₁-C₁₂ alkylcarbonyl;

[0589] R is, at each occurrence, independently either: (a) H or C₁-C₁₂ alkyl; or (b) R together with the carbon atom to which it is bound is taken together with an adjacent R and the carbon atom to which it is bound to form a carbon-carbon double bond;

[0590] R¹ and R² have, at each occurrence, the following structure, respectively:



R¹

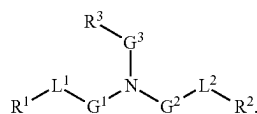
R²

[0591] a¹ and a² are, at each occurrence, independently an integer from 3 to 12; b¹ and b² are, at each occurrence, independently 0 or 1;

[0592] c¹ and c² are, at each occurrence, independently an integer from 5 to 10; d¹ and d² are, at each occurrence, independently an integer from 5 to 10; y is, at each occurrence, independently an integer from 0 to 2; and n is an integer from 1 to 6,

[0593] wherein each alkyl, alkylene, hydroxylalkyl, aminoalkyl, alkylaminylalkyl, alkoxyalkyl, alkoxyalkyl, alkylcarbonyloxy, alkylcarbonyloxyalkyl and alkylcarbonyl is optionally substituted with one or more substituent.

[0594] In some embodiments, the lipid nanoparticle comprises a lipid having the structure:



(XIII-L)

or a pharmaceutically acceptable salt, prodrug or stereoisomer thereof, wherein:

[0595] one of L¹ or L² is —O(C=O)—, —(C=O)O—, —C(=O)—, —O—, —S(O)_x—, —S—S—, —C(=O)S—, —SC(=O)—, —R^aC(=O)—, —C(=O)R^a, R^aC(=O)R^a, —OC(=O)R^a or —R^aC(=O)O—, and the other of L¹ or L² is —O(C=O)—, —(C=O)O—, —C(=O)—, —O—, —S(O)_x—, —S—S—, —C(=O)S—, —SC(=O)—, —R^aC(=O)—, —C(=O)R^a, R^aC(=O)R^a, —OC(=O)R^a or —NR^aC(=O)O— or a direct bond;

[0596] G¹ and G² are each independently unsubstituted C₁-C₁₂ alkylene or C₁-C₁₂ alkenylene;

[0597] G³ is C₁-C₂₄ alkylene, C₁-C₂₄ alkenylene, C₃-C₈ cycloalkylene, C₃-C₈ cycloalkenylene;

[0598] R^a is H or C₁-C₁₂ alkyl;

[0599] R¹ and R² are each independently C₆-C₂₄ alkyl or C₆-C₂₄ alkenyl;

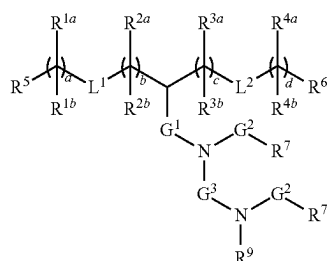
[0600] R³ is H, OR⁵, CN, —C(=O)OR⁴, —OC(=O)R⁴ or —R⁵C(=O)R⁴;

[0601] R⁴ is C₁-C₁₂ alkyl;

[0602] R⁵ is H or C₁-C₆ alkyl; and

[0603] x is 0, 1 or 2.

[0604] In some embodiments, the lipid nanoparticle comprises a lipid having the structure:



(XIV-L)

or a pharmaceutically acceptable salt, tautomer, prodrug or stereoisomer thereof, wherein:

[0605] L¹ and L² are each independently —O(C=O)—, —(C=O)O—, —C(=O)—, —O—, —S(O)_x—, —S—S—, —C(=O)S—, —SC(=O)—, —R^aC(=O)—, —C(=O)R^a, —R^aC(=O)R^a, —OC(=O)R^a, —R^aC(=O)O— or a direct bond;

[0606] G¹ is C₁-C₂ alkylene, —C(=O)—, —O(C=O)—, —SC(=O)—, —R^aC(=O)— or a direct bond;

[0607] G² is —C(=O)—, —(C=O)O—, —C(=O)S—, —C(=O)NR^a— or a direct bond;

[0608] G³ is C₁-C₆ alkylene;

[0609] R^a is H or C₁-C₁₂ alkyl;

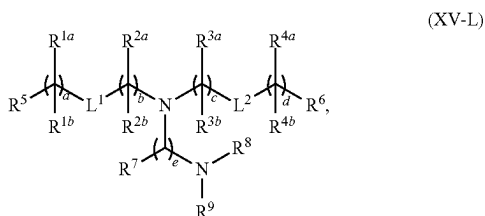
[0610] R^{1a} and R^{1b} are, at each occurrence, independently either: (a) H or C₁-C₁₂ alkyl; or (b)

[0611] R^{1a} is H or C₁-C₁₂ alkyl, and R^{1b} together with the carbon atom to which it is bound is taken together with an adjacent R^{1b} and the carbon atom to which it is bound to form a carbon-carbon double bond;

[0612] R^{2a} and R^{2b} are, at each occurrence, independently either: (a) H or C₁-C₁₂ alkyl; or (b) R^{2a} is H or C₁-C₁₂ alkyl, and R^{2b} together with the carbon atom to which it is bound is taken together with an adjacent R^{2b}

and the carbon atom to which it is bound to form a carbon-carbon double bond;

- [0613] R^{3a} and R^{3b} are, at each occurrence, independently either (a) H or C_1 - C_{12} alkyl; or (b) R^{3a} is H or C_1 - C_{12} alkyl, and R^{3b} together with the carbon atom to which it is bound is taken together with an adjacent R and the carbon atom to which it is bound to form a carbon-carbon double bond;
- [0614] R^{4a} and R^{4b} are, at each occurrence, independently either: (a) H or C_1 - C_{12} alkyl; or (b) R^{4a} is H or C_1 - C_{12} alkyl, and R^{4b} together with the carbon atom to which it is bound is taken together with an adjacent R^{4b} and the carbon atom to which it is bound to form a carbon-carbon double bond;
- [0615] R^5 and R^6 are each independently H or methyl;
- [0616] R^7 is C_4 - C_{20} alkyl;
- [0617] R^8 and R^9 are each independently C_1 - C_{12} alkyl; or R^8 and R^9 , together with the nitrogen atom to which they are attached, form a 5, 6 or 7-membered heterocyclic ring;
- [0618] a, b, c and d are each independently an integer from 1 to 24; and x is 0, 1 or 2.
- [0619] In some embodiments, the lipid nanoparticle comprises a lipid having the structure:

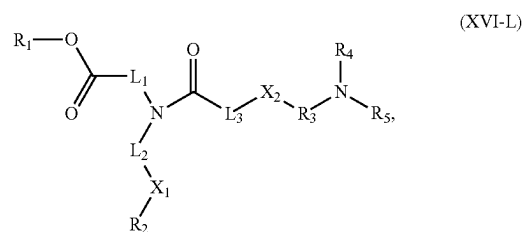


or a pharmaceutically acceptable salt, tautomer, prodrug or stereoisomer thereof, wherein:

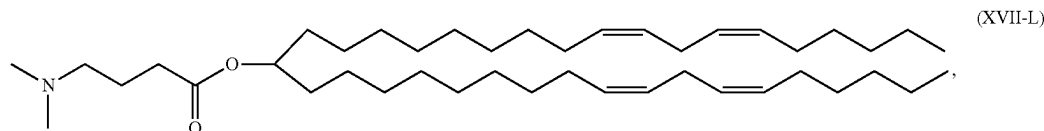
- [0620] L^1 and L^2 are each independently $-O(C=O)-$, $-(C=O)O-$ or a carbon-carbon double bond;
- [0621] R^{1a} and R^{1b} are, at each occurrence, independently either (a) H or C_1 - C_{12} alkyl, or (b) R^{1a} is H or C_1 - C_{12} alkyl, and R^{1b} together with the carbon atom to which it is bound is taken together with an adjacent R^{1b} and the carbon atom to which it is bound to form a carbon-carbon double bond;
- [0622] R^{2a} and R^{2b} are, at each occurrence, independently either (a) H or C_1 - C_{12} alkyl, or (b) R^{2a} is H or C_1 - C_{12} alkyl, and R^{2b} together with the carbon atom to which it is bound is taken together with an adjacent R^{2b} and the carbon atom to which it is bound to form a carbon-carbon double bond;
- [0623] R^{3a} and R^{3b} are, at each occurrence, independently either (a) H or C_1 - C_{12} alkyl, or (b) R^{3a} is H or C_1 - C_{12} alkyl, and R^{3b} together with the carbon atom to which it is bound is taken together with an adjacent R^{3b}

and the carbon atom to which it is bound to form a carbon-carbon double bond;

- [0624] R^{4a} and R^{4b} are, at each occurrence, independently either (a) H or C_1 - C_{12} alkyl, or (b) R^{4a} is H or C_1 - C_{12} alkyl, and R^{4b} together with the carbon atom to which it is bound is taken together with an adjacent R^{4b} and the carbon atom to which it is bound to form a carbon-carbon double bond;
- [0625] R^5 and R^6 are each independently methyl or cycloalkyl;
- [0626] R^7 is, at each occurrence, independently H or C_1 - C_{12} alkyl; R^8 and R^9 are each independently unsubstituted C_1 - C_{12} alkyl; or R^8 and R^9 , together with the nitrogen atom to which they are attached, form a 5, 6 or 7-membered heterocyclic ring comprising one nitrogen atom; a and d are each independently an integer from 0 to 24; b and c are each independently an integer from 1 to 24; and e is 1 or 2,
- [0627] provided that:
- [0628] at least one of R^{1a} , R^{2a} , R^{3a} or R^{4a} is C_1 - C_{12} alkyl, or at least one of L^1 or L^2 is $-O(C=O)-$ or $-(C=O)O-$; and
- [0629] R^{1a} and R^{1b} are not isopropyl when a is 6 or n-butyl when a is 8.
- [0630] In some embodiments, the lipid nanoparticle comprises a lipid having the structure:

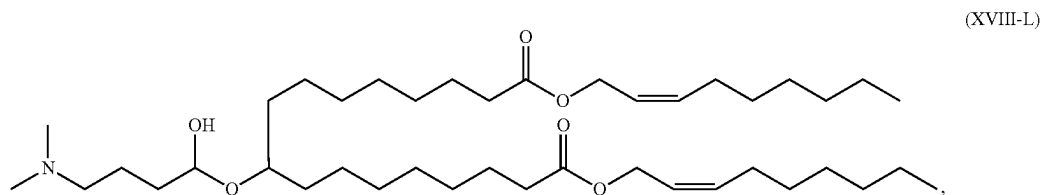


- [0631] or a pharmaceutically acceptable salt thereof, wherein
- [0632] R_1 and R_2 are the same or different, each a linear or branched alkyl with 1-9 carbons, or as alkenyl or alkynyl with 2 to 11 carbon atoms,
- [0633] L_1 and L_2 are the same or different, each a linear alkyl having 5 to 18 carbon atoms, or form a heterocycle with N,
- [0634] X_1 is a bond, or is $-CG-G-$ whereby $L_2-CO-O-R_2$ is formed,
- [0635] X_2 is S or O,
- [0636] L_3 is a bond or a lower alkyl, or form a heterocycle with N,
- [0637] R_3 is a lower alkyl, and
- [0638] R_4 and R_5 are the same or different, each a lower alkyl.
- [0639] In some embodiments, the lipid nanoparticle comprises an ionizable lipid having the structure:



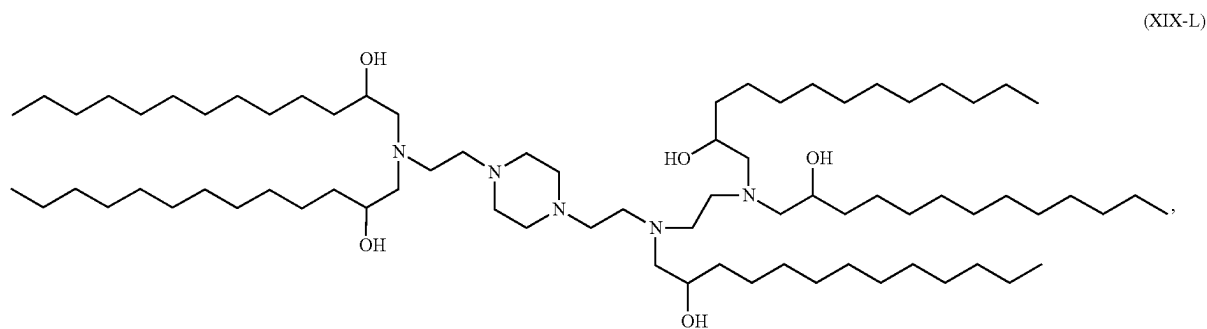
or a pharmaceutically acceptable salt thereof.

[0640] In some embodiments, the lipid nanoparticle comprises a lipid having the structure:



or a pharmaceutically acceptable salt thereof.

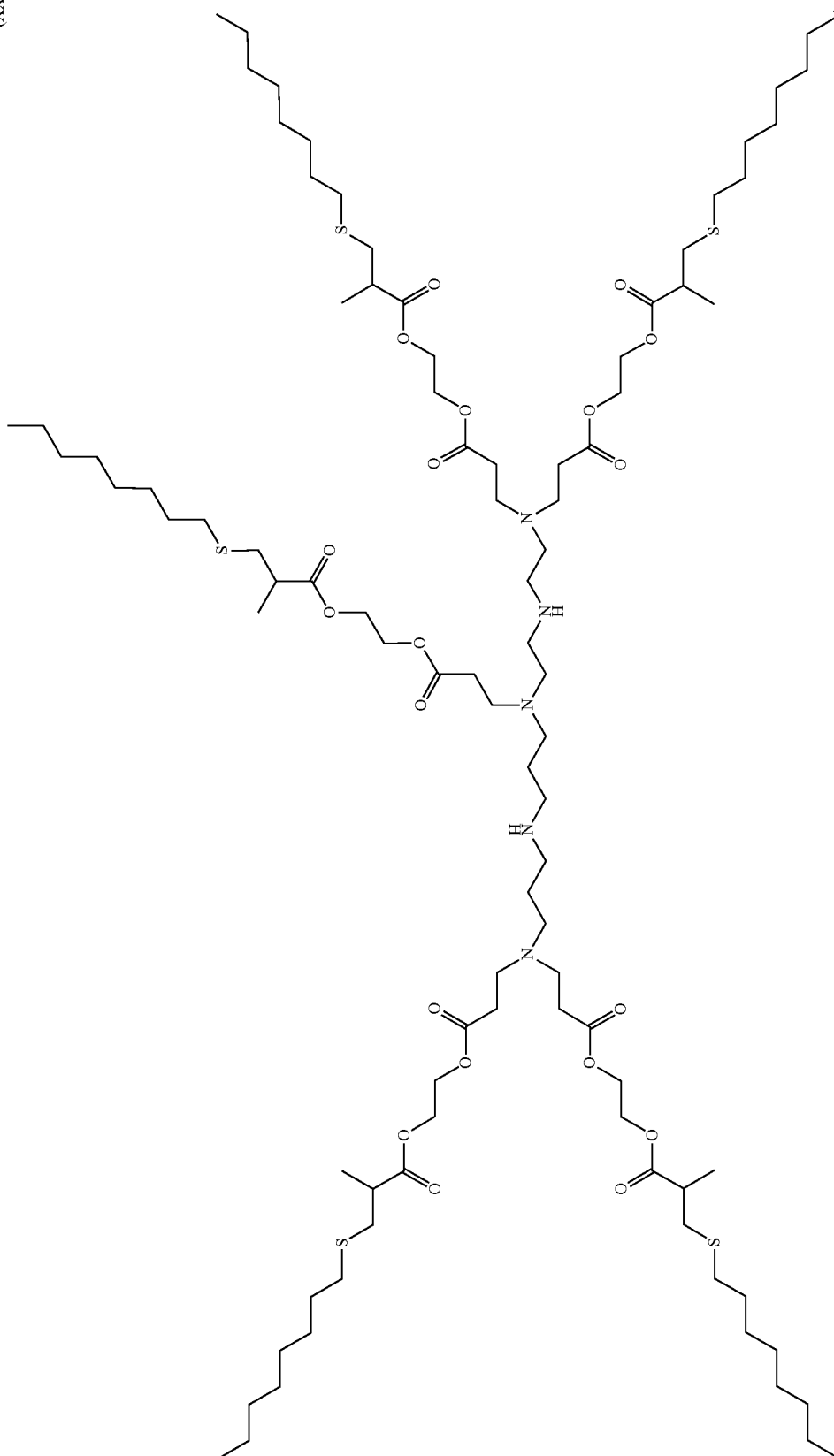
[0641] In some embodiments, the lipid nanoparticle comprises a lipid having the structure:



or a pharmaceutically acceptable salt thereof.

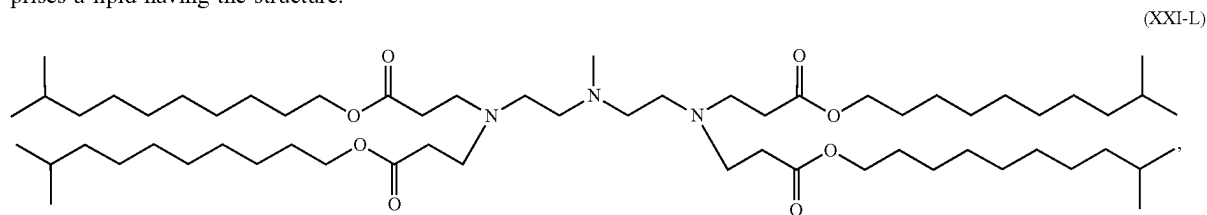
[0642] In some embodiments, the lipid nanoparticle comprises a lipid having the structure:

(XX-L)



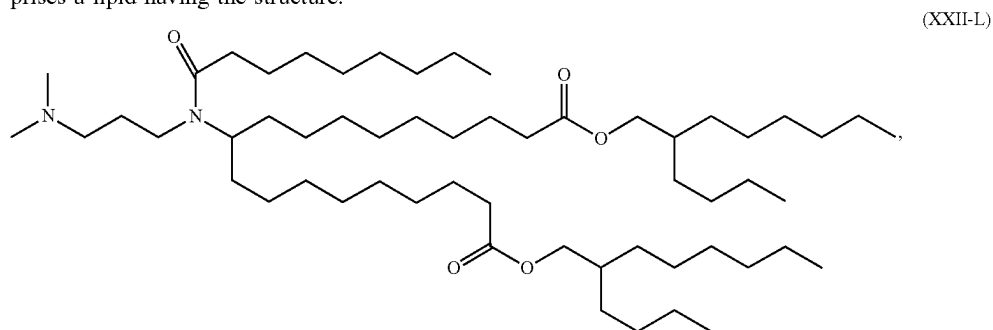
or a pharmaceutically acceptable salt thereof.

[0643] In some embodiments, the lipid nanoparticle comprises a lipid having the structure:



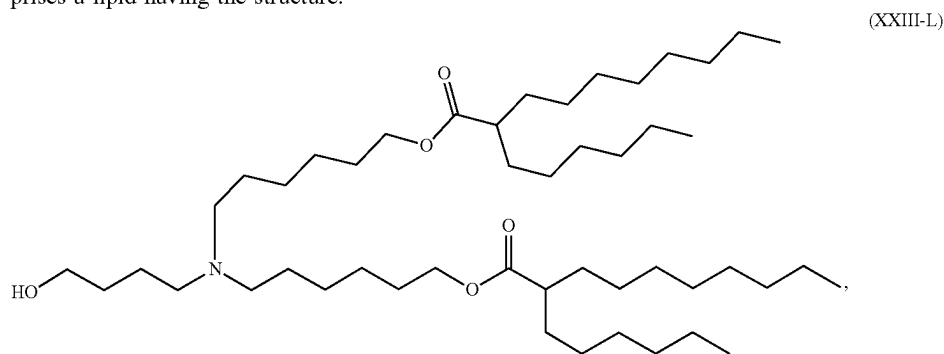
or a pharmaceutically acceptable salt thereof.

[0644] In some embodiments, the lipid nanoparticle comprises a lipid having the structure:



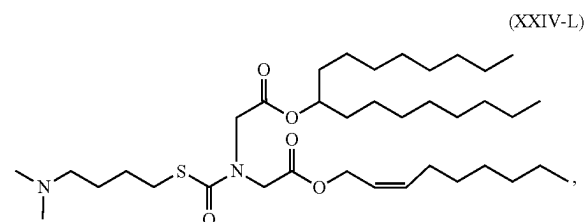
or a pharmaceutically acceptable salt thereof.

[0645] In some embodiments, the lipid nanoparticle comprises a lipid having the structure:



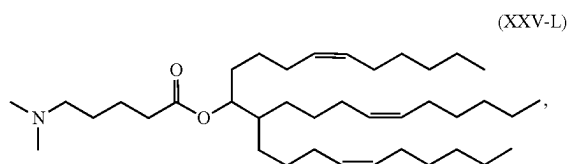
or a pharmaceutically acceptable salt thereof.

[0646] In some embodiments, the lipid nanoparticle comprises a lipid having the structure:



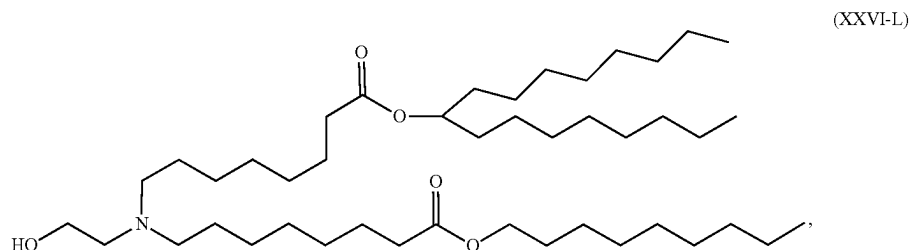
or a pharmaceutically acceptable salt thereof.

[0647] In some embodiments, the lipid nanoparticle comprises a lipid having the structure:



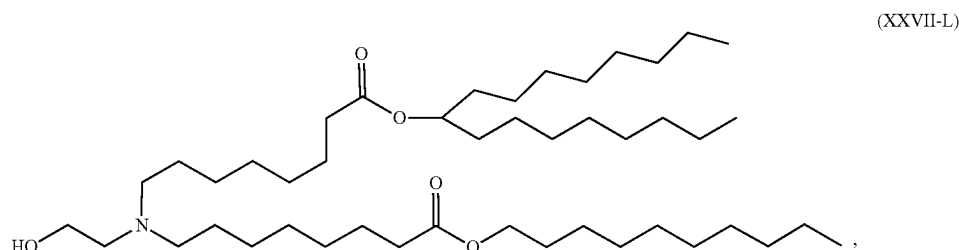
or a pharmaceutically acceptable salt thereof.

[0648] In some embodiments, the lipid nanoparticle comprises a lipid having the structure:



or a pharmaceutically acceptable salt thereof.

[0649] In some embodiments, the lipid nanoparticle comprises a lipid having the structure:



or a pharmaceutically acceptable salt thereof.

Neutral Lipids

[0650] In certain embodiments, the lipid nanoparticles provided herein comprise one or more neutral lipids. Neutral lipids may be phospholipids.

[0651] In some embodiments, a neutral lipid comprises 1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC), 1,2-dioleoyl-sn-glycero-3-phosphoethanolamine (DOPE), 1,2-dilinoleoyl-sn-glycero-3-phosphocholine (DLPC), 1,2-dimyristoyl-sn-glycero-3-phosphocholine (DMPC), 1,2-dioleoyl-sn-glycero-3-phosphocholine (DOPC), 1,2-dipalmitoyl-sn-glycero-3-phosphocholine (DPPC), 1,2-diundecanoyl-sn-glycero-3-phosphocholine (DUPC), 1-palmitoyl-2-oleoyl-sn-glycero-3-phosphocholine (POPC), 1,2-di-O-octadecenyl-sn-glycero-3-phosphocholine (18:0 Diether PC), 1-oleoyl-2-cholesterylhemisuccinoyl-sn-glycero-3-phosphocholine (OChemPC), 1-hexadecyl-sn-glycero-3-phosphocholine (C16 Lyso PC), 1,2-dilinolenoyl-sn-glycero-3-phosphocholine, 1,2-diarachidonoyl-sn-glycero-3-phosphocholine, 1,2-didocosahexaenoyl-sn-glycero-3-phosphocholine, 1,2-diphytanoyl-sn-glycero-3-phosphoethanolamine (ME 16.0 PE), 1,2-distearoyl-sn-glycero-3-phosphoethanolamine, 1,2-dilinoleoyl-sn-glycero-3-phosphoethanolamine, 1,2-dilinolenoyl-sn-glycero-3-phosphoethanolamine, 1,2-diarachidonoyl-sn-glycero-3-phosphoethanolamine, 1,2-didocosahexaenoyl-sn-glycero-3-phosphoethanolamine, 1,2-dioleoyl-sn-glycero-3-phospho-rac-(1-glycerol) sodium salt (DOPG), sphingomyelin, or mixtures thereof.

[0652] In some embodiments, the lipid nanoparticle comprises 5-15 mol %, 5-10 mol %, or 10-15 mol % DSPC. For

example, the lipid nanoparticle may comprise 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 mol % DSPC.

[0653] In certain embodiments, the lipid composition of the lipid nanoparticle disclosed herein can comprise one or more phospholipids, for example, one or more saturated or (poly) unsaturated phospholipids or a combination thereof. In general, phospholipids comprise a phospholipid moiety and one or more fatty acid moieties.

[0654] A phospholipid moiety can be selected, for example, from the non-limiting group consisting of phosphatidyl choline, phosphatidyl ethanolamine, phosphatidyl glycerol, phosphatidyl serine, phosphatidic acid, 2-lyso-phosphatidyl choline, and a sphingomyelin.

[0655] A fatty acid moiety can be selected, for example, from the non-limiting group consisting of lauric acid, myristic acid, myristoleic acid, palmitic acid, palmitoleic acid, stearic acid, oleic acid, linoleic acid, alpha-linolenic acid, erucic acid, phytanoic acid, arachidic acid, arachidonic acid, eicosapentaenoic acid, behenic acid, docosapentaenoic acid, and docosahexaenoic acid.

[0656] Particular phospholipids can facilitate fusion to a membrane. For example, a cationic phospholipid can interact with one or more negatively charged phospholipids of a membrane (e.g., a cellular or intracellular membrane). Fusion of a phospholipid to a membrane can allow one or more elements (e.g., a therapeutic agent) of a lipid-containing composition (e.g., LNPs) to pass through the membrane permitting, e.g., delivery of the one or more elements to a target tissue.

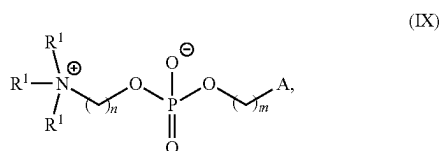
[0657] Non-natural phospholipid species including natural species with modifications and substitutions including branching, oxidation, cyclization, and alkynes are also con-

templated. For example, a phospholipid can be functionalized with or cross-linked to one or more alkynes (e.g., an alkenyl group in which one or more double bonds is replaced with a triple bond). Under appropriate reaction conditions, an alkyne group can undergo a copper-catalyzed cycloaddition upon exposure to an azide. Such reactions can be useful in functionalizing a lipid bilayer of a nanoparticle composition to facilitate membrane permeation or cellular recognition or in conjugating a nanoparticle composition to a useful component such as a targeting or imaging moiety (e.g., a dye).

[0658] Phospholipids include, but are not limited to, glycerophospholipids such as phosphatidylcholines, phosphatidylethanolamines, phosphatidylserines, phosphatidylinositols, phosphatidyl glycerols, and phosphatidic acids. Phospholipids also include phosphosphingolipids, such as sphingomyelin.

[0659] In some embodiments, a phospholipid is selected from 1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC), 1,2-Distearoyl-sn-glycero-3-phosphoethanolamine (DSPE), 1,2-dioloyleoyl-sn-glycero-3-phosphoethanolamine (DOPE), 1,2-dilinoleoyl-sn-glycero-3-phosphocholine (DLPC), 1,2-dimyristoyl-sn-glycero-3-phosphocholine (DMPC), 1,2-dioloyleoyl-sn-glycero-3-phosphocholine (DOPC), 1,2-dipalmitoyl-sn-glycero-3-phosphocholine (DPPC), 1,2-diundecanoyl-sn-glycero-3-phosphocholine (DUPC), 1-palmitoyl-2-oleoyl-sn-glycero-3-phosphocholine (POPC), 1,2-di-O-octadecenyl-sn-glycero-3-phosphocholine (18:0 Diether PC), 1-oleoyl-2-cholesterylhemisuccinoyl-sn-glycero-3-phosphocholine (OChemSPC), 1-hexadecyl-sn-glycero-3-phosphocholine (C16 Lyso PC), 1,2-dilinolenoyl-sn-glycero-3-phosphocholine, 1,2-diarachidonoyl-sn-glycero-3-phosphocholine, 1,2-didocosahexaenoyl-sn-glycero-3-phosphocholine, 1,2-diphytanoyl-sn-glycero-3-phosphoethanolamine (ME 16.0 PE), 1,2-distearoyl-sn-glycero-3-phosphoethanolamine, 1,2-dilinoleoyl-sn-glycero-3-phosphoethanolamine, 1,2-dilinolenoyl-sn-glycero-3-phosphoethanolamine, 1,2-diarachidonoyl-sn-glycero-3-phosphoethanolamine, 1,2-didocosahexaenoyl-sn-glycero-3-phosphoethanolamine, 1,2-dioleoyl-sn-glycero-3-phospho-rac-(1-glycerol) sodium salt (DOPG), sphingomyelin, and mixtures thereof.

[0660] In certain embodiments, a phospholipid is an analog or variant of DSPC. In certain embodiments, a phospholipid is a compound of Formula (IX):



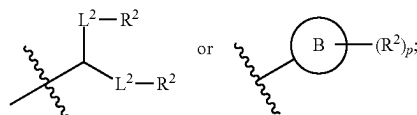
[0661] or a salt thereof, wherein:

[0662] each R^1 is independently optionally substituted alkyl; or optionally two R^1 are joined together with the intervening atoms to form optionally substituted monocyclic carbocyclyl or optionally substituted monocyclic heterocyclyl; or optionally three R^1 are joined together with the intervening atoms to form optionally substituted bicyclic carbocyclyl or optionally substituted bicyclic heterocyclyl;

[0663] n is 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10;

[0664] m is 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10;

[0665] A is of the formula:



[0666] each instance of L^2 is independently a bond or optionally substituted C_{1-6} alkylene, wherein one methylene unit of the optionally substituted C_{1-6} alkylene is optionally replaced with O, $N(R^N)$, S, $C(O)$, $C(O)N(R^N)$, $NR^N C(O)$, $C(O)O$, $OC(O)$, $OC(O)O$, $OC(O)N(R^N)$, $—NR^N C(O)O$, or $NR^N C(O)N(R^N)$;

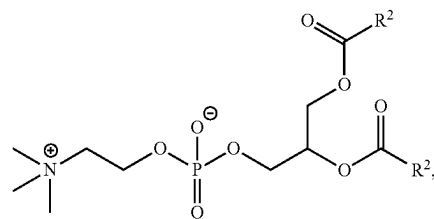
[0667] each instance of R^2 is independently optionally substituted C_{1-30} alkyl, optionally substituted C_{1-30} alkenyl, or optionally substituted C_{1-30} alkynyl; optionally wherein one or more methylene units of R^2 are independently replaced with optionally substituted carbocyclylene, optionally substituted heterocyclylene, optionally substituted arylene, optionally substituted heteroarylene, $N(R^N)$, O, S, $C(O)$, $C(O)N(R^N)$, $NR^N C(O)$, $NR^N C(O)N(R^N)$, $C(O)O$, $OC(O)$, $—OC(O)O$, $OC(O)N(R^N)$, $NR^N C(O)O$, $C(O)S$, $SC(O)$, $C(=NR^N)$, $C(=NR^N)N(R^N)$, $NR^N C(=NR^N)$, $NR^N C(=NR^N)N(R^N)$, $C(S)$, $C(S)N(R^N)$, $NR^N C(S)$, $NR^N C(S)N(R^N)$, $S(O)$, $OS(O)$, $S(O)O$, $—OS(O)O$, $OS(O)_2$, $S(O)_2O$, $OS(O)_2O$, $N(R^N)S(O)$, $S(O)N(R^N)$, $N(R^N)S(O)N(R^N)$, $OS(O)N(R^N)$, $N(R^N)S(O)O$, $S(O)_2$, $N(R^N)S(O)_2$, $S(O)_2N(R^N)$, $N(R^N)S(O)_2N(R^N)$, $OS(O)_2N(R^N)$, or $—N(R^N)S(O)_2O$;

[0668] each instance of R^N is independently hydrogen, optionally substituted alkyl, or a nitrogen protecting group;

[0669] Ring B is optionally substituted carbocyclyl, optionally substituted heterocyclyl, optionally substituted aryl, or optionally substituted heteroaryl; and

[0670] p is 1 or 2;

[0671] provided that the compound is not of the formula:



[0672] wherein each instance of R^2 is independently unsubstituted alkyl, unsubstituted alkenyl, or unsubstituted alkynyl.

[0673] In some embodiments, the phospholipids may be one or more of the phospholipids described in PCT Application No. PCT/US2018/037922.

Structural Lipids

[0674] The lipid composition of a pharmaceutical composition disclosed herein can comprise one or more structural

lipids. As used herein, the term “structural lipid” includes sterols and also lipids containing sterol moieties.

[0675] Incorporation of structural lipids in the lipid nanoparticle may help mitigate aggregation of other lipids in the particle. Structural lipids can be selected from the group including but not limited to, cholesterol, fecosterol, sitosterol, ergosterol, campesterol, stigmasterol, brassicasterol, tomatidine, tomatine, ursolic acid, alpha-tocopherol, hopanoids, phytosterols, steroids, and mixtures thereof. In some embodiments, the structural lipid is a sterol. As used herein, “sterols” are a subgroup of steroids consisting of steroid alcohols. In certain embodiments, the structural lipid is a steroid. In certain embodiments, the structural lipid is cholesterol. In certain embodiments, the structural lipid is an analog of cholesterol. In certain embodiments, the structural lipid is alpha-tocopherol.

[0676] In some embodiments, the structural lipids may be one or more of the structural lipids described in U.S. application Ser. No. 16/493,814.

Lipid Nanoparticle Compositions

[0677] As described herein, the cargo molecule or molecules (e.g., mRNA) of the disclosure is formulated in a lipid nanoparticle (LNP). Lipid nanoparticles typically comprise ionizable amino (cationic) lipid, neutral lipid, sterol and PEG lipid components along with the nucleic acid cargo of interest. The LNPs and nucleic acid form a stabilized composition. As used herein, a stabilized composition comprising an LNP and nucleic acid is able to maintain its size (e.g., diameter) and potency (e.g., immunogenicity of the mRNA) during storage over a period of time. As used herein, storage may be at any temperature, such as room temperature or refrigerated temperatures. For example, a stabilized composition may be stored for 1 hour, 2 hours, 3 hours, 4 hours, 5 hours, 6 hours, 12 hours, 18 hours, 1 day, 2 days, 3 days, 4 days, 5 days, 6 days, 1 week, 1.5 weeks, 2 weeks, 2.5 weeks, 3 weeks, 3.5 weeks, 1 month, 1.5 months, 2 months, 2.5 months, 3 months, 3.5 months, 4 months, 4.5 months, 5 months, 5.5 months, 6 months, 7 months, 8 months, 9 months, 10 months, 11 months, 1 year, 1.5 years, 2 years, 2.5 years, 3 years, 3.5 years, 4 years, 4.5 years, 5 years, or longer. The stabilized composition, for example, may be stored at temperatures below 0° C. or temperatures above 0° C., for example, at 1° C., 2° C., 3° C., 4° C., 5° C., 6° C., 7° C., 8° C., 9° C., 10° C., 11° C., 12° C., 13° C., 14° C., 15° C., 16° C., 17° C., 18° C., 19° C., 20° C., 21° C., 22° C., 23° C., 24° C., 25° C. or warmer. In some embodiments, the stabilized composition is stored at room temperature (approximately 20° C.). In some embodiments, the stabilized composition is refrigerated (e.g., kept at a temperature between 2-8° C., 2-6° C., 2-4° C., 3-8° C., 3-6° C., 4-8° C., or at 2° C., 3° C., 4° C., 5° C., 6° C., 7° C., or 8° C.).

[0678] In some embodiments, the stabilized composition is formulated in an aqueous solution. An aqueous solution is one in which water is the dissolution medium or solvent. In some embodiments, the aqueous solution may comprise a buffer, such as a phosphate buffer, a tris buffer, an acetate buffer, a histidine buffer, a citrate buffer, or any combination of buffers. In some embodiments, the pH of the aqueous solution is about 5 to 8, such as about 5.5, about 6, about 6.5, about 7, about 7.5, or about 8.

[0679] Size of an LNP (e.g., diameter) may be measured, in some embodiments, by suspending the synthetic nanocarriers in a liquid (usually aqueous) media and using

dynamic light scattering (DLS) (e.g., using a Brookhaven ZetaPALS instrument). As an example, a suspension of LNPs can be diluted from an aqueous buffer into purified water to achieve a final synthetic nanocarrier suspension concentration of approximately 0.01 to 0.5 mg/mL. The diluted suspension may be prepared directly inside, or transferred to, a suitable cuvette for DLS analysis. The cuvette may then be placed in the DLS, allowed to equilibrate to the controlled temperature, and then scanned for sufficient time to acquire a stable and reproducible distribution based on appropriate inputs for viscosity of the medium and refractive indices of the sample. The effective diameter, or mean of the distribution, is then reported. Determining the effective sizes of high aspect ratio, or non-spheroidal, synthetic nanocarriers may require augmentative techniques, such as electron microscopy, to obtain more accurate measurements. “Dimension” or “size” or “diameter” of LNPs means the mean of a particle size distribution, for example, obtained using dynamic light scattering.

[0680] In some embodiments, the diameter of the LNPs is measured over time (e.g., prior to, during, and after storage). As used herein, a “threshold diameter value” is the diameter of the LNP prior to storage or prior to filtration. In some embodiments, the diameter of the LNPs in the stabilized composition after storage varies 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 12%, 14%, 15%, 16%, 18%, 20%, 35%, 30%, 35%, 40%, 45%, 50%, 55%, 60% or more from the threshold diameter value. In some embodiments, the diameter of the LNPs in the stabilized composition increases 50 nm or less during processing or storage (e.g., 50 nm, 45 nm, 40 nm, 35 nm, 34 nm, 33 nm, 32 nm, 31 nm, 30 nm, 29 nm, 28 nm, 27 nm, 26 nm, 25 nm, 24 nm, 23 nm, 22 nm, 21 nm, 20 nm, 19 nm, 18 nm, 17 nm, 16 nm, 15 nm, 14 nm, 13 nm, 12 nm, 11 nm, 10 nm, 9 nm, 8 nm, 7 nm, 6 nm, 5 nm, 4 nm, 3 nm, 2 nm, 1 nm, or less than 1 nm). In some embodiments, the processing comprises tangential flow filtration (TFF).

[0681] As used herein, the term “tangential-flow filtration” refers to a process in which the fluid mixture containing the components to be separated by filtration (e.g., LNPs and extraneous components) is recirculated at high velocities tangential to the plane of the membrane, increasing the mass-transfer coefficient for back diffusion. During the filtration process, a pressure differential is applied along the length of the membrane, causing the fluid and filterable solutes to flow through the filter. This filtration may be a continuous-flow process or a batch process. As an example, the solution may be passed repeatedly over the membrane while that fluid which passes through the filter is continually drawn off into a separate unit or the solution is passed once over the membrane and the fluid passing through the filter is continually processed downstream.

[0682] In some embodiments, the lipid nanoparticle comprises 1-5% PEG-modified lipid, optionally 1-3 mol %, for example 1.5 to 2.5 mol %, 1-2 mol %, 2-3 mol %, 2.5-3.5%, 3-4 mol %, or 4-5 mol %. In some embodiments, the lipid nanoparticle comprises 0.5-15 mol % PEG-modified lipid. For example, the lipid nanoparticle may comprise 0.5-10 mol %, 0.5-5 mol %, 1-15 mol %, 1-10 mol %, 1-5 mol %, 2-15 mol %, 2-10 mol %, 2-5 mol %, 5-15 mol %, 5-10 mol %, or 10-15 mol %. In some embodiments, the lipid nanoparticle comprises 0.5 mol %, 1 mol %, 2 mol %, 2.5 mol %, 3 mol %, 3.5 mol %, 4 mol %, 4.5 mol %, 5 mol %, 6

mol %, 7 mol %, 8 mol %, 9 mol %, 10 mol %, 11 mol %, 12 mol %, 13 mol %, 14 mol %, or 15 mol % PEG-modified lipid.

[0683] In some embodiments, the lipid nanoparticle comprises 5-25 mol % neutral lipid. For example, the lipid nanoparticle may comprise 5-20 mol %, 5-15 mol %, 5-10 mol %, 10-25 mol %, 10-20 mol %, 10-15 mol %, 15-25 mol %, 15-20 mol %, or 20-25 mol % neutral lipid. In some embodiments, the lipid nanoparticle comprises 5 mol %, 10 mol %, 15 mol %, 20 mol %, or 25 mol % neutral lipid.

[0684] In some embodiments, the lipid nanoparticle comprises 40-50 mol % ionizable lipid, optionally 45-50 mol %, for example, 45-46 mol %, 46-47 mol %, 47-48 mol %, 48-49 mol %, or 49-50 mol % for example about 45 mol %, 45.5 mol %, 46 mol %, 46.5 mol %, 47 mol %, 47.5 mol %, 48 mol %, 48.5 mol %, 49 mol %, or 49.5 mol %.

[0685] In some embodiments, the lipid nanoparticle comprises 30-45 mol % sterol, optionally 35-40 mol %, for example, 30-31 mol %, 31-32 mol %, 32-33 mol %, 33-34 mol %, 35-35 mol %, 35-36 mol %, 36-37 mol %, 38-38 mol %, 38-39 mol %, or 39-40 mol %. In some embodiments, the lipid nanoparticle comprises 25-55 mol % sterol. For example, the lipid nanoparticle may comprise 25-50 mol %, 25-45 mol %, 25-40 mol %, 25-35 mol %, 25-30 mol %, 30-55 mol %, 30-50 mol %, 30-45 mol %, 30-40 mol %, 30-35 mol %, 35-55 mol %, 35-50 mol %, 35-45 mol %, 35-40 mol %, 40-55 mol %, 40-50 mol %, 40-45 mol %, 45-55 mol %, 45-50 mol %, or 50-55 mol % sterol. In some embodiments, the lipid nanoparticle comprises 25 mol %, 30 mol %, 35 mol %, 40 mol %, 45 mol %, 50 mol %, or 55 mol % sterol.

[0686] In some embodiments, the lipid nanoparticle comprises 35-40 mol % cholesterol. For example, the lipid nanoparticle may comprise 35, 35.5, 36, 36.5, 37, 37.5, 38, 38.5, 39, 39.5, or 40 mol % cholesterol.

[0687] In some embodiments, the lipid nanoparticle comprises 20-60 mol % ionizable amino lipid. For example, the lipid nanoparticle may comprise 20-50 mol %, 20-40 mol %, 20-30 mol %, 30-60 mol %, 30-50 mol %, 30-40 mol %, 40-60 mol %, 40-50 mol %, or 50-60 mol % ionizable amino lipid. In some embodiments, the lipid nanoparticle comprises 20 mol %, 30 mol %, 40 mol %, 50 mol %, or 60 mol % ionizable amino lipid. In some embodiments, the lipid nanoparticle comprises 35 mol %, 36 mol %, 37 mol %, 38 mol %, 39 mol %, 40 mol %, 41 mol %, 42 mol %, 43 mol %, 44 mol %, 45 mol %, 46 mol %, 47 mol %, 48 mol %, 49 mol %, 50 mol %, 51 mol %, 52 mol %, 53 mol %, 54 mol %, or 55 mol % ionizable amino lipid.

[0688] In some embodiments, the lipid nanoparticle comprises 45-55 mole percent (mol %) ionizable amino lipid. For example, lipid nanoparticle may comprise 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, or 55 mol % ionizable amino lipid.

[0689] In some embodiments, the percentage (e.g., by weight, or by mole) of a single type of ionizable molecule (e.g., amino lipid or ionizable lipid) and/or of all the ionizable molecules within a particle may be greater than or equal to about 15%, greater than or equal to about 16%, greater than or equal to about 17%, greater than or equal to about 18%, greater than or equal to about 19%, greater than or equal to about 20%, greater than or equal to about 21%, greater than or equal to about 22%, greater than or equal to about 23%, greater than or equal to about 24%, greater than or equal to about 25%, greater than or equal to about 30%, greater than or equal to about 35%, greater than or equal to

about 40%, greater than or equal to about 42%, greater than or equal to about 45%, greater than or equal to about 48%, greater than or equal to about 50%, greater than or equal to about 52%, greater than or equal to about 55%, greater than or equal to about 58%, greater than or equal to about 60%, greater than or equal to about 62%, greater than or equal to about 65%, or greater than or equal to about 68%. In some instances, the percentage (e.g., by weight, or by mole) may be less than or equal to about 70%, less than or equal to about 68%, less than or equal to about 65%, less than or equal to about 62%, less than or equal to about 60%, less than or equal to about 58%, less than or equal to about 55%, less than or equal to about 52%, less than or equal to about 50%, or less than or equal to about 48%. Combinations of the above referenced ranges are also possible (e.g., greater than or equal to 20% and less than or equal to about 60%, greater than or equal to 40% and less than or equal to about 55%, etc.). In embodiments in which more than one type of ionizable molecule is present in a particle, each type of ionizable molecule may independently have a percentage (e.g., by weight, or by mole) in one or more of the ranges described above. The percentage (e.g., by weight, or by mole) may be determined by extracting the ionizable molecule(s) from the dried particles using, e.g., organic solvents, and measuring the quantity of the agent using high pressure liquid chromatography (i.e., HPLC), liquid chromatography-mass spectrometry (LC-MS), nuclear magnetic resonance (NMR), or mass spectrometry (MS). Those of ordinary skill in the art would be knowledgeable of techniques to determine the quantity of a component using the above-referenced techniques. For example, HPLC may be used to quantify the amount of a component, by, e.g., comparing the area under the curve of a HPLC chromatogram to a standard curve.

[0690] In some embodiments, the lipid nanoparticle comprises 30-45 mol % sterol, optionally 35-40 mol %, for example, 30-31 mol %, 31-32 mol %, 32-33 mol %, 33-34 mol %, 35-35 mol %, 35-36 mol %, 36-37 mol %, 38-38 mol %, 38-39 mol %, or 39-40 mol %. In some embodiments, the lipid nanoparticle comprises 25-55 mol % sterol. For example, the lipid nanoparticle may comprise 25-50 mol %, 25-45 mol %, 25-40 mol %, 25-35 mol %, 25-30 mol %, 30-55 mol %, 30-50 mol %, 30-45 mol %, 30-40 mol %, 30-35 mol %, 35-55 mol %, 35-50 mol %, 35-45 mol %, 35-40 mol %, 40-55 mol %, 40-50 mol %, 40-45 mol %, 45-55 mol %, 45-50 mol %, or 50-55 mol % sterol. In some embodiments, the lipid nanoparticle comprises 25 mol %, 30 mol %, 35 mol %, 40 mol %, 45 mol %, 50 mol %, or 55 mol % sterol.

[0691] In some embodiments, the lipid nanoparticle comprises 35-40 mol % cholesterol. For example, the lipid nanoparticle may comprise 35, 35.5, 36, 36.5, 37, 37.5, 38, 38.5, 39, 39.5, or 40 mol % cholesterol.

[0692] In some embodiments, the lipid nanoparticle comprises a molar ratio of 5-25% phospholipid relative to the other lipid components. For example, the lipid nanoparticle may comprise a molar ratio of 5-30%, 5-15%, 5-10%, 10-25%, 10-20%, 10-15%, 15-25%, 15-20%, 20-25%, or 25-30% phospholipid. In some embodiments, the lipid nanoparticle comprises a molar ratio of 5%, 10%, 15%, 20%, 25%, or 30% neutral lipid.

[0693] In some embodiments, the lipid nanoparticle comprises a molar ratio of 25-55% structural lipid relative to the other lipid components. For example, the lipid nanoparticle

may comprise a molar ratio of 10-55%, 25-50%, 25-45%, 25-40%, 25-35%, 25-30%, 30-55%, 30-50%, 30-45%, 30-40%, 30-35%, 35-55%, 35-50%, 35-45%, 35-40%, 40-55%, 40-50%, 40-45%, 45-55%, 45-50%, or 50-55% structural lipid. In some embodiments, the lipid nanoparticle comprises a molar ratio of 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, or 55% structural lipid.

[0694] In some embodiments, the lipid nanoparticle comprises a molar ratio of 0.5-15% PEG lipid relative to the other lipid components. For example, the lipid nanoparticle may comprise a molar ratio of 0.5-10%, 0.5-5%, 1-15%, 1-10%, 1-5%, 2-15%, 2-10%, 2-5%, 5-15%, 5-10%, or 10-15% PEG lipid. In some embodiments, the lipid nanoparticle comprises a molar ratio of 0.5%, 1%, 1.5%, 2%, 2.5%, 3%, 3.5%, 4%, 4.5%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, or 15% PEG-lipid.

[0695] In some embodiments, the lipid nanoparticle comprises a molar ratio of 20-60% amino lipid, 5-25% phospholipid, 25-55% structural lipid, and 0.5-15% PEG lipid.

[0696] In some embodiments, the lipid nanoparticle comprises a molar ratio of 20-60% amino lipid, 5-30% phospholipid, 10-55% structural lipid, and 0.5-15% PEG lipid.

[0697] In some embodiments, the lipid nanoparticle comprises 20-60 mol % ionizable amino lipid, 5-25 mol % neutral lipid, 25-55 mol % sterol, and 0.5-15 mol % PEG-modified lipid.

[0698] In some embodiments, a LNP comprises an ionizable amino lipid of Compound I, wherein the neutral lipid is DSPC, the structural lipid that is cholesterol, and the PEG lipid is DMG-PEG.

[0699] In some embodiments, a LNP comprises an ionizable amino lipid of Compound II, wherein the neutral lipid is DSPC, the structural lipid that is 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96, 99,102,105,108,111,114,117,120,123,126,129,132-tetratetracontaoxatetriacontahexyl stearate, and the PEG lipid is DMG-PEG.

[0700] In some embodiments, a LNP comprises an ionizable amino lipid of Compound III, wherein the neutral lipid is DSPC, the structural lipid that is 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetracontaoxatetriacontahexyl stearate, and the PEG lipid is DMG-PEG.

[0701] In some embodiments, a LNP comprises an ionizable amino lipid of Compound IV, wherein the neutral lipid is DSPC, the structural lipid that is 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetracontaoxatetriacontahexyl stearate, and the PEG lipid is DMG-PEG.

[0702] In some embodiments, the lipid nanoparticle comprises 49 mol % ionizable amino lipid, 10 mol % DSPC, 38.5 mol % cholesterol, and 2.5 mol % DMG-PEG.

[0703] In some embodiments, the lipid nanoparticle comprises 49 mol % ionizable amino lipid, 11 mol % DSPC, 38.5 mol % cholesterol, and 1.5 mol % DMG-PEG.

[0704] In some embodiments, the lipid nanoparticle comprises 48 mol % ionizable amino lipid (e.g., Compound I), 11 mol % DSPC, 38.5 mol % cholesterol, and 2.5 mol % DMG-PEG.

[0705] In some embodiments, the lipid nanoparticle comprises 47 mol % ionizable amino lipid (e.g., Compounds III, IV, or V), 11 mol % DSPC, 39 mol % cholesterol, and 3 mol

% 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetratetracontaoxatetriacontahexyl stearate.

[0706] In some embodiments, the lipid nanoparticles comprise one or more of ionizable molecules, polynucleotides, and optional components, such as structural lipids, sterols, neutral lipids, phospholipids and a molecule capable of reducing particle aggregation (e.g., polyethylene glycol (PEG), PEG-modified lipid), such as those described above.

[0707] Some aspects provide lipid compositions comprising a lipid and a compound of compound of Formula I or of Formula II, or a tautomer or solvate thereof.

[0708] In some embodiments, a LNP comprises a lipid-to-oligonucleotide phosphate (N:P) ratio of from about 2:1 to about 30:1.

[0709] In some embodiments, a LNP comprises an N:P ratio of about 6:1.

[0710] In some embodiments, a LNP comprises an N:P ratio of about 3:1, 4:1, or 5:1.

[0711] In some embodiments, a LNP comprises a wt/wt ratio of the ionizable amino lipid component to the RNA of from about 10:1 to about 100:1.

[0712] In some embodiments, a LNP comprises a wt/wt ratio of the ionizable amino lipid component to the RNA of about 20:1.

[0713] In some embodiments, a LNP comprises a wt/wt ratio of the ionizable amino lipid component to the RNA of about 10:1.

[0714] In some embodiments, a LNP has a mean diameter from about 30 nm to about 150 nm.

[0715] In some embodiments, a LNP has a mean diameter from about 60 nm to about 120 nm. In some embodiments, a LNP has a mean diameter from about 80 nm to 140 nm, 80 nm to 130 nm, 80 nm to 120 nm, 80 nm to 110 nm, 80 nm to 100 nm, 80 nm to 90 nm, 80 nm to 90 nm, 90 nm to 140 nm, 90 nm to 130 nm, 90 nm to 120 nm, 90 nm to 110 nm, 90 nm to 100 nm, 100 nm to 140 nm, 100 nm to 130 nm, 100 nm to 120 nm, 100 nm to 110 nm, 110 nm to 140 nm, 110 nm to 130 nm, 110 nm to 120 nm. In some embodiments, a LNP has a mean diameter of about 80 nm, 85 nm, 90 nm, 95 nm, 100 nm, 110 nm, 111 nm, 112 nm, 113 nm, 114 nm, 115 nm, 116 nm, 117 nm, 118 nm, 119 nm, 120 nm, 125 nm, 130 nm, 135 nm, or 140 nm.

[0716] In some embodiments, a LNP has a mean diameter from about 60 nm to about 120 nm.

[0717] In some embodiments, the lipid nanoparticle has a diameter of at most 80 nm, at most 70 nm, at most 60 nm, at most 50 nm, at most 40 nm, at most 30 nm, or at most 20 nm.

[0718] In some embodiments, the lipid nanoparticle has a diameter of at most 30 nm.

[0719] In some embodiments, the lipid nanoparticle has a diameter of at most 20 nm.

[0720] In some embodiments, a lipid nanoparticle may comprise two or more components (e.g., amino lipid and nucleic acid, PEG-lipid, phospholipid, structural lipid). For instance, a lipid nanoparticle may comprise an amino lipid and a nucleic acid. Compositions comprising the lipid nanoparticles may be used for a wide variety of applications, including the stealth delivery of therapeutic payloads with minimal adverse innate immune response.

Nucleic Acids

[0721] Also provided are stabilized nucleic acids. The term “nucleic acid” refers to multiple nucleotides (i.e., molecules comprising a sugar (e.g., ribose or deoxyribose) linked to a phosphate group and to an exchangeable organic base, which is either a substituted pyrimidine (e.g., cytosine (C), thymine (T) or uracil (U)) or a substituted purine (e.g., adenine (A) or guanine (G))). As used herein, the term nucleic acid refers to polyribonucleotides as well as polydeoxyribonucleotides. The term nucleic acid also includes polynucleosides (i.e., a polynucleotide minus the phosphate) and any other organic base containing polymer. Non-limiting examples of nucleic acids include chromosomes, genomic loci, genes or gene segments that encode polynucleotides or polypeptides, coding sequences, non-coding sequences (e.g., intron, 5'-UTR, or 3'-UTR) of a gene, pri-mRNA, pre-mRNA, cDNA, RNA, etc. In some embodiments, the nucleic acid is mRNA. A nucleic acid may include a substitution and/or modification. In some embodiments, the substitution and/or modification is in one or more bases and/or sugars. For example, in some embodiments a nucleic acid includes nucleic acids having backbone sugars that are covalently attached to low molecular weight organic groups other than a hydroxyl group at the 2' position and other than a phosphate group or hydroxy group at the 5' position. Thus, in some embodiments, a substituted or modified nucleic acid includes a 2'-O-alkylated ribose group. In some embodiments, a modified nucleic acid includes sugars such as hexose, 2'-F hexose, 2'-amino ribose, constrained ethyl (cEt), locked nucleic acid (LNA), arabinose or 2'-fluoroarabinose instead of ribose. Thus, in some embodiments, a nucleic acid is heterogeneous in backbone composition thereby containing any possible combination of polymer units linked together such as peptide-nucleic acids (which have an amino acid backbone with nucleic acid bases).

[0722] In some embodiments, a nucleic acid is DNA, RNA, PNA, cEt, LNA, ENA or hybrids including any chemical or natural modification thereof. Chemical and natural modifications are well known in the art. Non-limiting examples of modifications include modifications designed to increase translation of the nucleic acid, to increase cell penetration or sub-cellular distribution of the nucleic acid, to stabilize the nucleic acid against nucleases and other enzymes that degrade or interfere with the structure or activity of the nucleic acid, and to improve the pharmacokinetic properties of the nucleic acid.

[0723] In some embodiments, the compositions comprise a RNA having an open reading frame (ORF) encoding a polypeptide. In some embodiments, the RNA is a messenger RNA (mRNA). In some embodiments, the RNA (e.g., mRNA) further comprises a 5' UTR, 3' UTR, a poly(A) tail and/or a 5' cap analog.

[0724] Messenger RNA (mRNA) is any RNA that encodes a (at least one) protein (e.g., a naturally-occurring, non-naturally-occurring, or modified polymer of amino acids) and can be translated to produce the encoded protein in vitro, in vivo, in situ, or ex vivo. The skilled artisan will appreciate that, except where otherwise noted, nucleic acid sequences set forth in the instant application may recite “T”s in a representative DNA sequence but where the sequence represents RNA (e.g., mRNA), the “T”s would be substituted for “U”s. Thus, any of the DNAs disclosed and identified by a particular sequence identification number herein also dis-

close the corresponding RNA (e.g., mRNA) sequence complementary to the DNA, where each “T” of the DNA sequence is substituted with “U.”

[0725] An open reading frame (ORF) is a continuous stretch of DNA or RNA beginning with a start codon (e.g., methionine (ATG or AUG)) and ending with a stop codon (e.g., TAA, TAG or TGA, or UAA, UAG or UGA). An ORF typically encodes a protein. It will be understood that the sequences disclosed herein may further comprise additional elements, e.g., 5' and 3' UTRs, but that those elements, unlike the ORF, need not necessarily be present in an RNA polynucleotide.

[0726] Naturally-occurring eukaryotic mRNA molecules can contain stabilizing elements, including, but not limited to untranslated regions (UTR) at their 5'-end (5' UTR) and/or at their 3'-end (3' UTR), in addition to other structural features, such as a 5'-cap structure or a 3'-poly(A) tail. Both the 5' UTR and the 3' UTR are typically transcribed from the genomic DNA and are elements of the premature mRNA. Characteristic structural features of mature mRNA, such as the 5'-cap and the 3'-poly(A) tail are usually added to the transcribed (premature) mRNA during mRNA processing.

[0727] In some embodiments, a composition includes an RNA polynucleotide having an open reading frame encoding at least one polypeptide having at least one modification, at least one 5' terminal cap, and is formulated within a lipid nanoparticle along with the stabilizing compound (e.g., a compound of Formula I, of Formula II, or a tautomer or solvate thereof).

[0728] 5' terminal caps can include endogenous caps or cap analogs. According to the present invention, a 5' terminal cap can comprise a guanine analog. Useful guanine analogs include, but are not limited to, inosine, N1-methyl-guanosine, 2'-fluoro-guanosine, 7-deaza-guanosine, 8-oxo-guanosine, 2-amino-guanosine, LNA-guanosine, and 2-azido-guanosine.

[0729] Also provided herein are exemplary caps including those that can be used in co-transcriptional capping methods for ribonucleic acid (RNA) synthesis, using RNA polymerase, e.g., wild type RNA polymerase or variants thereof, e.g., such as those variants described herein.

[0730] In one embodiment, caps can be added when RNA is produced in a “one-pot” reaction, without the need for a separate capping reaction. Thus, the methods, in some embodiments, comprise reacting a polynucleotide template with a RNA polymerase variant, nucleoside triphosphates, and a cap analog under in vitro transcription reaction conditions to produce RNA transcript.

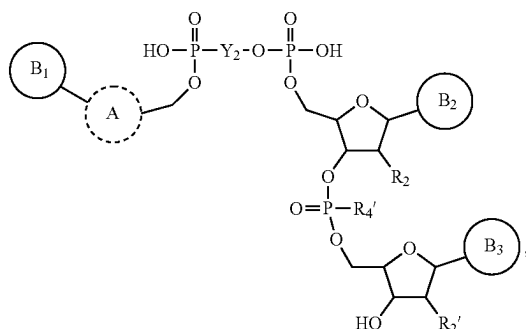
[0731] In some embodiments, the cap analog binds to a polynucleotide template that comprises a promoter region comprising a transcriptional start site having a first nucleotide at nucleotide position +1, a second nucleotide at nucleotide position +2, and a third nucleotide at nucleotide position +3. In some embodiments, the cap analog hybridizes to the polynucleotide template at least at nucleotide position +1, such as at the +1 and +2 positions, or at the +1, +2, and +3 positions.

[0732] A cap analog may be, for example, a dinucleotide cap, a trinucleotide cap, or a tetranucleotide cap. In some embodiments, a cap analog is a dinucleotide cap. In some embodiments, a cap analog is a trinucleotide cap. In some embodiments, a cap analog is a tetranucleotide cap. As used here the term “cap” includes the inverted G nucleotide and

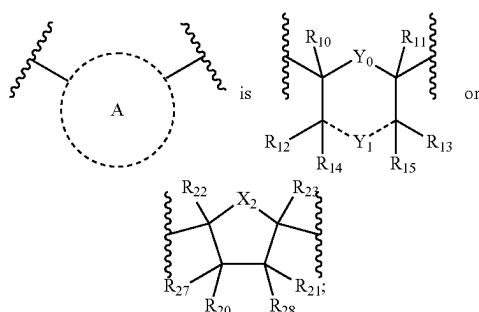
can comprise additional nucleotides 3' of the inverted G, e.g., 1, 2, or more nucleotides 3' of the inverted G and 5' to the 5' UTR.

[0733] Exemplary caps comprise a sequence GG, GA, or GGA wherein the underlined, italicized G is an inverted G.

[0734] A trinucleotide cap, in some embodiments, comprises a compound of formula (I)



or a stereoisomer, tautomer or salt thereof, wherein



[0735] ring B₁ is a modified or unmodified Guanine;

[0736] ring B₂ and ring B₃ each independently is a nucleobase or a modified nucleobase;

[0737] X_2 is O, S(O)_p, NR₂₄ or CR₂₅R₂₆ in which p is 0, 1, or 2;

[0738] Y_0 is O or CR_6R_7 ;

[0739] Y1 is O, S(O)_n, CR₆R₇, or NR₈, in which n is 0, 1, or 2;

[0740] each - - - is a single bond or absent, wherein when each - - - is a single bond, Y_i is O, S(O)_n, CR₆R₇, or NR₈; and when each - - - is absent, Y₁ is void;

[0741] Y_2 is $(OP(O)R_4)_m$ in which m is 0, 1, or 2, or $-O-(CR_{40}R_{41})_u-Q_0-(CR_{42}R_{43})_v-$, in which Q_0 is a bond, O, S(O), NR_{44} , or $CR_{45}R_{46}$, r is 0, 1, or 2, and each of u and v independently is 1, 2, 3 or 4;

[0742] each R₂ and R₂' independently is halo, LNA, or OR₃;

[0743] each R₃ independently is H, C₁-C₆ alkyl, C₂-C₆ alkenyl, or C₂-C₆ alkynyl and R₃, when being C₁-C₆ alkyl, C₂-C₆ alkenyl, or C₂-C₆ alkynyl, is optionally substituted with one or more of halo, OH and C₁-C₆ alkoxyl that is optionally substituted with one or more OH or OC(O)—C₁-C₆ alkyl;

[0744] each R₄ and R₄' independently is H, halo, C₁-C₆ alkyl, OH, SH, ScH, or BH₃;

[0745] each of R₆, R₇, and R₈, independently, is -Q₁-T₁, in which Q₁ is a bond or C₁-C₃ alkyl linker optionally substituted with one or more of halo, cyano, OH and C₁-C₆ alkoxy, and T₁ is H, halo, OH, COOH, cyano, or R_{s1}, in which R_{s1} is C₁-C₃ alkyl, C₂-C₆ alkenyl, C₂-C₆ alkynyl, C₁-C₆ alkoxy, C(O)O—C₁-C₆ alkyl, C₃-C₈ cycloalkyl, C₆-C₁₀ aryl, NR₃₁R₃₂, (NR₃₁R₃₂R₃₃)⁺, 4 to 12-membered heterocycloalkyl, or 5- or 6-membered heteroaryl, and R_{s1} is optionally substituted with one or more substituents selected from the group consisting of halo, OH, oxo, C₁-C₆ alkyl, COOH, C(O)O—C₁-C₆ alkyl, cyano, C₁-C₆ alkoxy, NR₃₁R₃₂, (NR₃₁R₃₂R₃₃)⁺, C₃-C₈ cycloalkyl, C₆-C₁₀ aryl, 4 to 12-membered heterocycloalkyl, and 5- or 6-membered heteroaryl;

[0746] each of R₁₀, R₁₁, R₁₂, R₁₃, R₁₄, and R₁₅, independently, is -Q₂-T₂, in which Q₂ is a bond or C₁-C₃ alkyl linker optionally substituted with one or more of halo, cyano, OH and C₁-C₆ alkoxy, and T₂ is H, halo, OH, NH₂, cyano, NO₂, N₃, R_{s2}, or OR_{s2}, in which R_{s2} is C₁-C₆ alkyl, C₂-C₆ alkenyl, C₂-C₆ alkynyl, C₃-C₈ cycloalkyl, C₆-C₁₀ aryl, NHC(O)-C₁-C₆ alkyl, NR₃₁R₃₂, (NR₃₁R₃₂R₃₃)⁺, 4 to 12-membered heterocycloalkyl, or 5- or 6-membered

[0747] heteroaryl, and R_{x2} is optionally substituted with one or more substituents selected from the group consisting of halo, OH, oxo, C_1 - C_6 alkyl, COOH, $C(O)$ O- C_1 - C_6 alkyl, cyano, C_1 - C_6 alkoxy, $NR_{31}R_{32}$, $(NR_{31}R_{32}R_{33})^+$, C_3 - C_8 cycloalkyl, C_6 - C_{10} aryl, 4 to 12-membered

[0748] heterocycloalkyl, and 5- or 6-membered heteroaryl; or alternatively R₁₂ together with R₁₄ is oxo, or R₁₃ together with R₁₅ is oxo,

[0749] each of R₂₀, R₂₁, R₂₂, and R₂₃ independently is -Q₃-T₃, in which Q₃ is a bond or C₁-C₃ alkyl linker optionally substituted with one or more of halo, cyano, OH and C₁-C₆ alkoxy, and T₃ is H, halo, OH, NH₂, cyano, NO₂, N₃, R_{s3}, or OR_{s3}, in which R_{s3} is C₁-C₆ alkyl, C₂-C₆ alkenyl, C₂-C₆ alkynyl, C₃-C₈ cycloalkyl, C₆-C₁₀ aryl, NHC(O)-C₁-C₆ alkyl, mono-C₁-C₆ alkylamino, di-C₁-C₆ alkylamino, 4 to 12-membered heterocycloalkyl, or 5- or 6-membered heteroaryl, and R_{s3} is optionally substituted with one or more substituents selected from the group consisting of halo, OH, oxo, C₁-C₆ alkyl, COOH, C(O)O-C₁-C₆ alkyl, cyano, C₁-C₆ alkoxy, amino, mono-C₁-C₆ alkylamino, di-C₁-C₆ alkylamino, C₃-C₈ cycloalkyl, C₆-C₁₀ aryl, 4 to 12-membered heterocycloalkyl, and 5- or 6-membered heteroaryl;

[0750] each of R₂₄, R₂₅, and R₂₆ independently is H or C₁-C₆ alkyl;

[0751] each of R₂₇ and R₂₈ independently is H or OR₂₉;
or R₂₇ and R₂₈ together form O—R₃₀—O;

[0752] each R₂₉ independently is H, C₁-C₆ alkyl, C₂-C₆ alkenyl, or C₂-C₆ alkynyl and R₂₉, when being C₁-C₆ alkyl, C₂-C₆ alkenyl, or C₂-C₆ alkynyl, is optionally substituted with one or more of halo, OH and C₁-C₆ alkoxy that is optionally substituted with one or more OH or OC(O)—C₁-C₆ alkyl:

[0753] R₃₀ is C₁-C₆ alkylene optionally substituted with one or more of halo, OH and C₁-C₆ alkoxy:

[0754] each of R₃₁, R₃₂, and R₃₃, independently is H, C₁-C₆ alkyl, C₃-C₈ cycloalkyl, C₆-C₁₀ aryl, 4 to 12-membered heterocycloalkyl, or 5- or 6-membered heteroaryl];

[0755] each of R₄₀, R₄₁, R₄₂, and R₄₃ independently is H, halo, OH, cyano, N₃, OP(O)R₄₇R₄₈, or C₁-C₆ alkyl optionally substituted with one or more OP(O)R₄₇R₄₈, or one R₄₁ and one R₄₃, together with the carbon atoms to which they are attached and Q₀, form C₄-C₁₀ cycloalkyl, 4- to 14-membered heterocycloalkyl, C₆-C₁₀ aryl, or 5- to 14-membered heteroaryl, and each of the cycloalkyl, heterocycloalkyl, phenyl, or 5- to 6-membered heteroaryl is optionally substituted with one or more of OH, halo, cyano, N₃, oxo, OP(O)R₄₇R₄₈, C₁-C₆ alkyl, C₁-C₆ haloalkyl, COOH, C(O)O—C₁-C₆ alkyl, C₁-C₆ alkoxy, C₁-C₆ haloalkoxy, amino, mono-C₁-C₆ alkylamino, and di-C₁-C₆ alkylamino;

[0756] R₄₄ is H, C₁-C₆ alkyl, or an amine protecting group;

[0757] each of R₄₅ and R₄₆ independently is H, OP(O)R₄₇R₄₈, or C₁-C₆ alkyl optionally substituted with one or more OP(O)R₄₇R₄₈, and

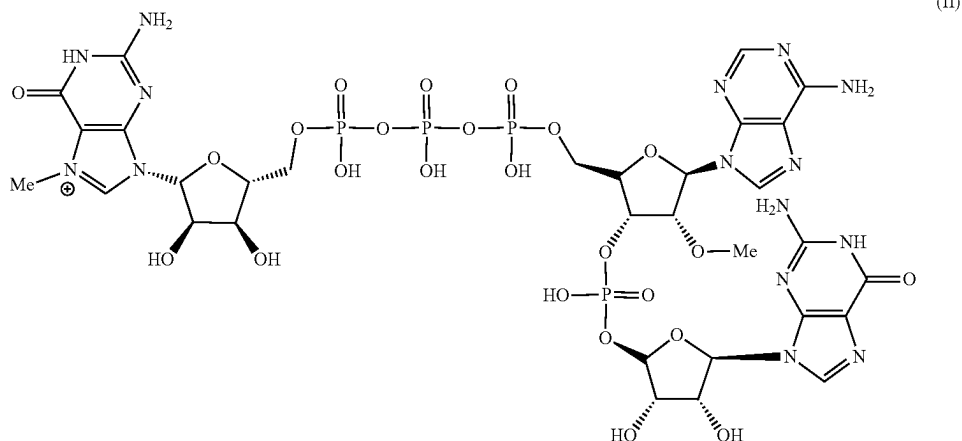
[0758] each of R_{47} and R_{48} , independently is H, halo, C_1 - C_6 alkyl, OH, SH, SeH, or BH_3 .

[0759] It should be understood that a cap analog, as provided herein, may include any of the cap analogs described in international publication WO 2017/066797, published on 20 Apr. 2017, incorporated by reference herein in its entirety.

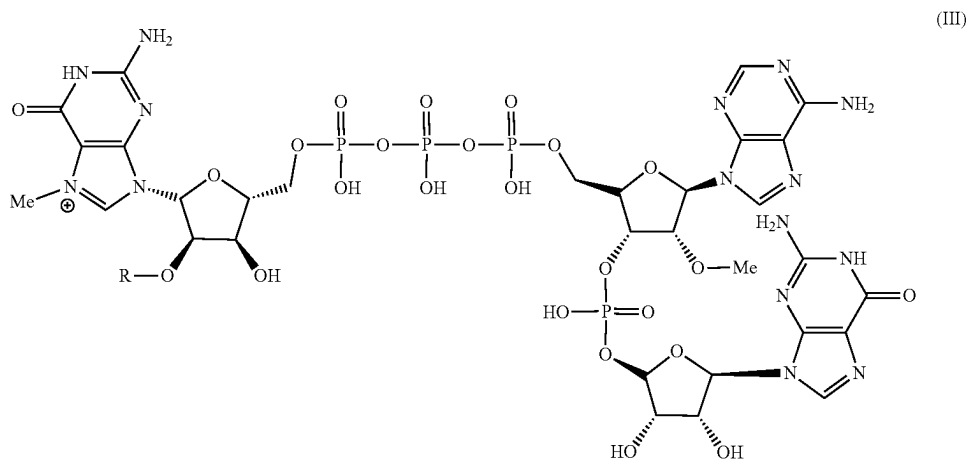
[0760] In some embodiments, the B₂ middle position can be a non-ribose molecule, such as arabinose.

[0761] In some embodiments R₂ is ethyl-based.

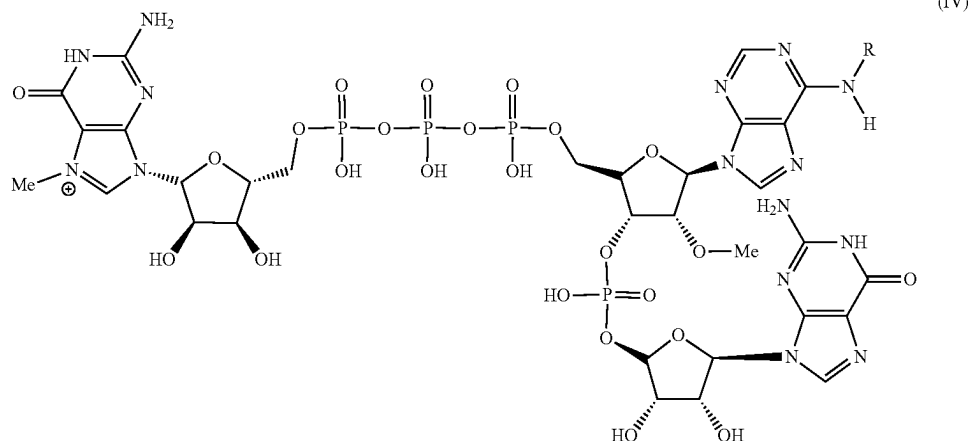
[0762] Thus, in some embodiments, a trinucleotide cap comprises the following structure:



[0763] In yet other embodiments, a trinucleotide cap comprises the following structure



[0764] In still other embodiments, a trinucleotide cap comprises the following structure:



[0765] In some embodiments, R is an alkyl (e.g., C₁-C₆ alkyl). In some embodiments, R is a methyl group (e.g., C₁ alkyl). In some embodiments, R is an ethyl group (e.g., C₂ alkyl).

[0766] A trinucleotide cap, in some embodiments, comprises a sequence selected from the following sequences: GAA, GAC, GAG, GAU, GCA, GCC, GCG, GCU, GGA, GGC, GGG, GGU, GUA, GUC, GUG, and GUU. In some embodiments, a trinucleotide cap comprises GAA. In some embodiments, a trinucleotide cap comprises GAC. In some embodiments, a trinucleotide cap comprises GAG. In some embodiments, a trinucleotide cap comprises GAU. In some embodiments, a trinucleotide cap comprises GCA. In some embodiments, a trinucleotide cap comprises GCC. In some embodiments, a trinucleotide cap comprises GCG. In some embodiments, a trinucleotide cap comprises GCU. In some embodiments, a trinucleotide cap comprises GGA. In some embodiments, a trinucleotide cap comprises GGC. In some embodiments, a trinucleotide cap comprises GGG. In some embodiments, a trinucleotide cap comprises GGU. In some embodiments, a trinucleotide cap comprises GUA. In some embodiments, a trinucleotide cap comprises GUC. In some embodiments, a trinucleotide cap comprises GUG. In some embodiments, a trinucleotide cap comprises GUU.

[0767] In some embodiments, a trinucleotide cap comprises a sequence selected from the following sequences: m⁷GpppApA, m⁷GpppApC, m⁷GpppApG, m⁷GpppApU, m⁷GpppCpA, m⁷GpppCpC, m⁷GpppCpG, m⁷GpppCPU, m⁷GpppGpA, m⁷GpppGpC, m⁷GpppGpG, m⁷GpppGpU, m⁷GpppUpA, m⁷GpppUpC, m⁷GpppUpG, and m⁷GpppUpU.

[0768] In some embodiments, a trinucleotide cap comprises m⁷GpppApA. In some embodiments, a trinucleotide cap comprises m⁷GpppApC. In some embodiments, a trinucleotide cap comprises m⁷GpppApG. In some embodiments, a trinucleotide cap comprises m⁷GpppApU. In some embodiments, a trinucleotide cap comprises m⁷GpppCpA. In some embodiments, a trinucleotide cap comprises m⁷GpppCpC. In some embodiments, a trinucleotide cap comprises m⁷GpppCpG. In some embodiments, a trinucleotide cap comprises m⁷GpppCPU. In some embodiments, a trinucleotide cap comprises m⁷GpppGpA. In some embodiments, a trinucleotide cap comprises m⁷GpppGpC. In some

embodiments, a trinucleotide cap comprises m⁷GpppGpG. In some embodiments, a trinucleotide cap comprises m⁷GpppGpU. In some embodiments, a trinucleotide cap comprises m⁷GpppUpA. In some embodiments, a trinucleotide cap comprises m⁷GpppUpC. In some embodiments, a trinucleotide cap comprises m⁷GpppUpG. In some embodiments, a trinucleotide cap comprises m⁷GpppUpU.

[0769] A trinucleotide cap, in some embodiments, comprises a sequence selected from the following sequences: m⁷G_{3'OMe}pppApA, m⁷G_{3'OMe}pppApC, m⁷G_{3'OMe}pppApG, m⁷G_{3'OMe}pppApU, m⁷G_{3'OMe}pppCpA, m⁷G_{3'OMe}pppCpC, m⁷G_{3'OMe}pppCpG, m⁷G_{3'OMe}pppCpU, m⁷G_{3'OMe}pppGpA, m⁷G_{3'OMe}pppGpC, m⁷G_{3'OMe}pppGpG, m⁷G_{3'OMe}pppGpU, m⁷G_{3'OMe}pppUpA, m⁷G_{3'OMe}pppUpC, m⁷G_{3'OMe}pppUpG, and m⁷G_{3'OMe}pppUpU.

[0770] In some embodiments, a trinucleotide cap comprises m⁷G_{3'OMe}pppApA. In some embodiments, a trinucleotide cap comprises m⁷G_{3'OMe}pppApC. In some embodiments, a trinucleotide cap comprises m⁷G_{3'OMe}pppApG. In some embodiments, a trinucleotide cap comprises m⁷G_{3'OMe}pppApU. In some embodiments, a trinucleotide cap comprises m⁷G_{3'OMe}pppCpA. In some embodiments, a trinucleotide cap comprises m⁷G_{3'OMe}pppCpC. In some embodiments, a trinucleotide cap comprises m⁷G_{3'OMe}pppCpG. In some embodiments, a trinucleotide cap comprises m⁷G_{3'OMe}pppCpU. In some embodiments, a trinucleotide cap comprises m⁷G_{3'OMe}pppGpA. In some embodiments, a trinucleotide cap comprises m⁷G_{3'OMe}pppGpC. In some embodiments, a trinucleotide cap comprises m⁷G_{3'OMe}pppGpG. In some embodiments, a trinucleotide cap comprises m⁷G_{3'OMe}pppGpU. In some embodiments, a trinucleotide cap comprises m⁷G_{3'OMe}pppUpA. In some embodiments, a trinucleotide cap comprises m⁷G_{3'OMe}pppUpC. In some embodiments, a trinucleotide cap comprises m⁷G_{3'OMe}pppUpG. In some embodiments, a trinucleotide cap comprises m⁷G_{3'OMe}pppUpU.

[0771] A trinucleotide cap, in other embodiments, comprises a sequence selected from the following sequences: m⁷G_{3'OMe}pppA_{2'OMe}pA, m⁷G_{3'OMe}pppA_{2'OMe}pC, m⁷G_{3'OMe}pppA_{2'OMe}pG, m⁷G_{3'OMe}pppA_{2'OMe}pU, m⁷G_{3'OMe}pppU_{2'OMe}pA, m⁷G_{3'OMe}pppU_{2'OMe}pC, m⁷G_{3'OMe}pppU_{2'OMe}pG, and m⁷G_{3'OMe}pppU_{2'OMe}pU.

[0772] In some embodiments, a trinucleotide cap comprises $m^7G_{3'OMe}pppA_{2'OMe}pA$. In some embodiments, a trinucleotide cap comprises $m^7G_{3'OMe}pppA_{2'OMe}pC$. In some embodiments, a trinucleotide cap comprises $m^7G_{3'OMe}pppA_{2'OMe}pG$. In some embodiments, a trinucleotide cap comprises $m^7G_{3'OMe}pppA_{2'OMe}pU$. In some embodiments, a trinucleotide cap comprises $m^7G_{3'OMe}pppC_{2'OMe}pA$. In some embodiments, a trinucleotide cap comprises $m^7G_{3'OMe}pppC_{2'OMe}pC$. In some embodiments, a trinucleotide cap comprises $m^7G_{3'OMe}pppC_{2'OMe}pG$. In some embodiments, a trinucleotide cap comprises $m^7G_{3'OMe}pppC_{2'OMe}pU$. In some embodiments, a trinucleotide cap comprises $m^7G_{3'OMe}pppG_{2'OMe}pA$. In some embodiments, a trinucleotide cap comprises $m^7G_{3'OMe}pppG_{2'OMe}pC$. In some embodiments, a trinucleotide cap comprises $m^7G_{3'OMe}pppG_{2'OMe}pG$. In some embodiments, a trinucleotide cap comprises $m^7G_{3'OMe}pppG_{2'OMe}pU$. In some embodiments, a trinucleotide cap comprises $m^7G_{3'OMe}pppU_{2'OMe}pA$. In some embodiments, a trinucleotide cap comprises $m^7G_{3'OMe}pppU_{2'OMe}pC$. In some embodiments, a trinucleotide cap comprises $m^7G_{3'OMe}pppU_{2'OMe}pG$. In some embodiments, a trinucleotide cap comprises $m^7G_{3'OMe}pppU_{2'OMe}pU$.

[0773] A trinucleotide cap, in still other embodiments, comprises a sequence selected from the following sequences: $m^7GpppA_{2'OMe}pA$, $m^7GpppA_{2'OMe}pC$, $m^7GpppA_{2'OMe}pG$, $m^7GpppA_{2'OMe}pU$, $m^7GpppC_{2'OMe}pA$, $m^7GpppC_{2'OMe}pC$, $m^7GpppC_{2'OMe}pG$, $m^7GpppC_{2'OMe}pU$, $m^7GpppG_{2'OMe}pA$, $m^7GpppG_{2'OMe}pC$, $m^7GpppG_{2'OMe}pG$,

$m^7GpppG_{2'OMe}pU$, $m^7GpppU_{2'OMe}pA$, $m^7GpppU_{2'OMe}pC$, $m^7GpppU_{2'OMe}pG$, and $m^7GpppU_{2'OMe}pU$.

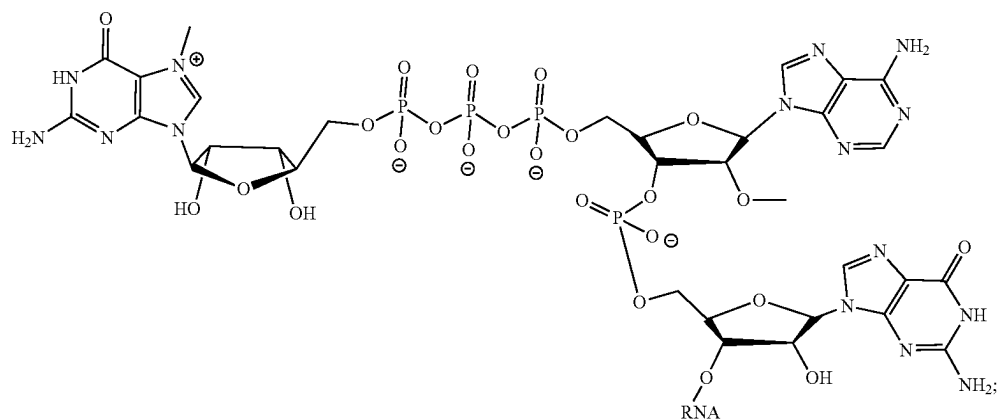
[0774] In some embodiments, a trinucleotide cap comprises $m^7GpppA_{2'OMe}pA$. In some embodiments, a trinucleotide cap comprises $m^7GpppA_{2'OMe}pC$. In some embodiments, a trinucleotide cap comprises $m^7GpppA_{2'OMe}pG$. In some embodiments, a trinucleotide cap comprises $m^7GpppA_{2'OMe}pU$. In some embodiments, a trinucleotide cap comprises $m^7GpppC_{2'OMe}pA$. In some embodiments, a trinucleotide cap comprises $m^7GpppC_{2'OMe}pC$. In some embodiments, a trinucleotide cap comprises $m^7GpppC_{2'OMe}pG$. In some embodiments, a trinucleotide cap comprises $m^7GpppC_{2'OMe}pU$. In some embodiments, a trinucleotide cap comprises $m^7GpppG_{2'OMe}pA$. In some embodiments, a trinucleotide cap comprises $m^7GpppG_{2'OMe}pC$. In some embodiments, a trinucleotide cap comprises $m^7GpppG_{2'OMe}pG$. In some embodiments, a trinucleotide cap comprises $m^7GpppG_{2'OMe}pU$. In some embodiments, a trinucleotide cap comprises $m^7GpppU_{2'OMe}pA$. In some embodiments, a trinucleotide cap comprises $m^7GpppU_{2'OMe}pC$. In some embodiments, a trinucleotide cap comprises $m^7GpppU_{2'OMe}pG$. In some embodiments, a trinucleotide cap comprises $m^7GpppU_{2'OMe}pU$.

[0775] In some embodiments, a trinucleotide cap comprises $m^7Gpppm^6A_{2'OMe}pG$. In some embodiments, a trinucleotide cap comprises $m^7Gpppm^6A_{2'OMe}pG$.

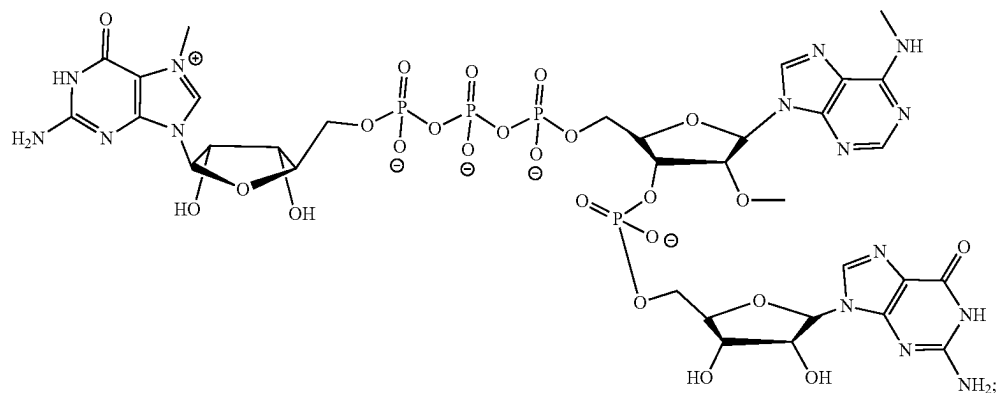
[0776] In some embodiments, a trinucleotide cap comprises GAG. In some embodiments, a trinucleotide cap comprises GCG. In some embodiments, a trinucleotide cap comprises GUG. In some embodiments, a trinucleotide cap comprises GGG.

[0777] In some embodiments, a trinucleotide cap comprises any one of the following structures:

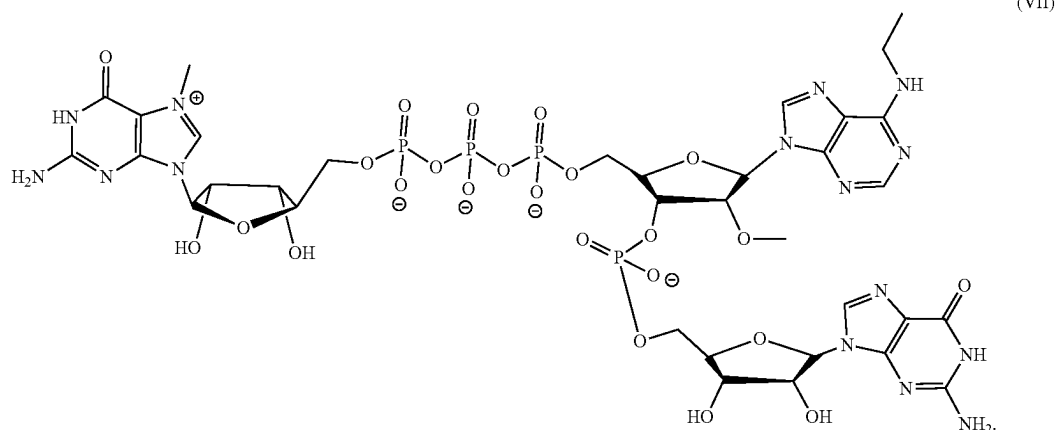
(V)



(VI)



-continued



GUUA. In some embodiments the tetranucleotide cap comprises a sequence selected from the following sequences: GAAG, GACG, GAGG, GAUG, GCAG, GCCG, GCGG, GCUG, GGAG, GGCG, GGGG, GGUG, GUCC, GUGG, and GUUG. In some embodiments the tetranucleotide cap comprises a sequence selected from the following sequences: GAAU, GACU, GAGU, GAUU, GCAU, GCCU, GCGU, GCUU, GGAU, GGCU, GGGU, GGUU, GUAU, GUCU, GUGU, and GUUU. In some embodiments the tetranucleotide cap comprises a sequence selected from the following sequences: GAAC, GACC, GAGC, GAUC, GCAC, GCCC, GCGC, GCUC, GGAC, GGCC, GGGC, GGUC, GUAC, GUCC, GUGC, and GUUC.

[0783] A tetranucleotide cap, in some embodiments, comprises a sequence selected from the following sequences:

$m^7G_{3'OMe}pppApApN$,	$m^7G_{3'OMe}pppApCpN$,
$m^7G_{3'OMe}pppApGpN$,	$m^7G_{3'OMe}pppApUpN$,
$m^7G_{3'OMe}pppCpApN$,	$m^7G_{3'OMe}pppCpCpN$,
$m^7G_{3'OMe}pppCpGpN$,	$m^7G_{3'OMe}pppCpUpN$,
$m^7G_{3'OMe}pppGpApN$,	$m^7G_{3'OMe}pppGpCpN$,
$m^7G_{3'OMe}pppGpGpN$,	$m^7G_{3'OMe}pppGpUpN$,
$m^7G_{3'OMe}pppUpApN$,	$m^7G_{3'OMe}pppUpCpN$,
$m^7G_{3'OMe}pppUpGpN$, and	$m^7G_{3'OMe}pppUpUpN$, where N is a natural, a modified, or an unnatural nucleoside base.

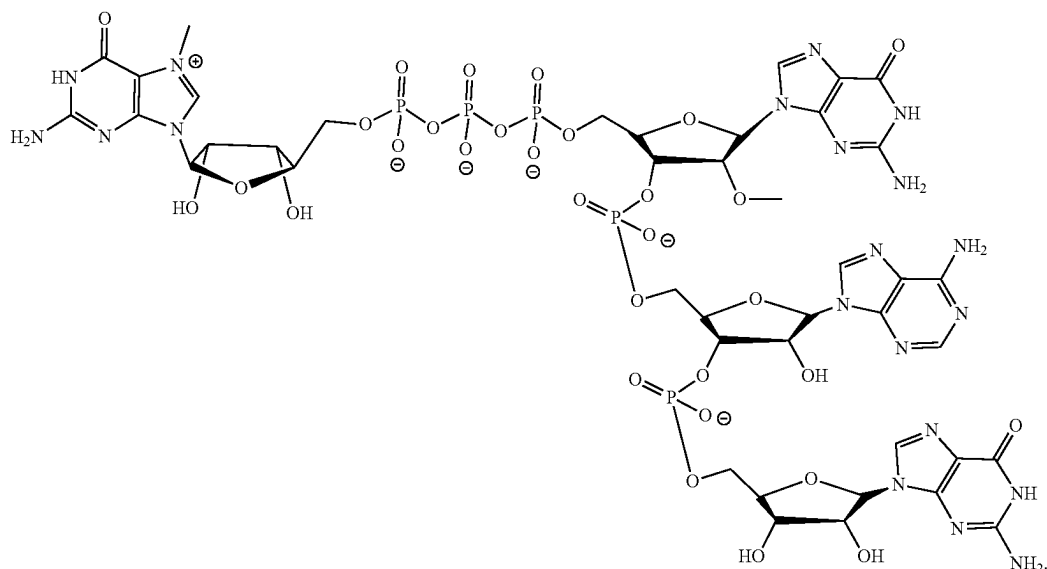
$m^7GpppC_{2'OMe}pApN$, $m^7GpppC_{2'OMe}pCpN$,
 $m^7GpppC_{2'OMe}pGpN$, $m^7GpppC_{2'OMe}pUpN$,
 $m^7GpppG_{2'OMe}pApN$, $m^7GpppG_{2'OMe}pCpN$,
 $m^7GpppG_{2'OMe}pGpN$, $m^7GpppU_{2'OMe}pUpN$, where N is a natural, a modified, or an unnatural nucleoside base.

[0786] A tetranucleotide cap, in other embodiments, comprises a sequence selected from the following sequence: $m^7G_{3'OMe}pppA_{2'OMe}PA_{2'OMe}pN$, $m^7G_{3'OMe}pppA_{2'OMe}PC_{2'OMe}pN$, $m^7G_{3'OMe}pppU_{2'OMe}pG_{2'OMe}pN$, and $m^7G_{3'OMe}pppU_{2'OMe}pU_{2'OMe}pN$, where N is a natural, a modified, or an unnatural nucleoside base.

[0787] A tetranucleotide cap, in still other embodiments, comprises a sequence selected from the following sequences:

$m^7GpppA_{2'OMe}pA_{2'OMe}pN$,	$m^7GpppA_{2'OMe}pC_{2'OMe}pN$,
$m^7GpppA_{2'OMe}pG_{2'OMe}pN$,	$m^7GpppA_{2'OMe}pU_{2'OMe}pN$,
$m^7GpppC_{2'OMe}pA_{2'OMe}pN$,	$m^7GpppC_{2'OMe}pC_{2'OMe}pN$,
$m^7GpppC_{2'OMe}pG_{2'OMe}pN$,	$m^7GpppC_{2'OMe}pU_{2'OMe}pN$,
$m^7GpppU_{2'OMe}pA_{2'OMe}pN$,	$m^7GpppU_{2'OMe}pC_{2'OMe}pN$,
$m^7GpppU_{2'OMe}pG_{2'OMe}pN$, and	$m^7GpppU_{2'OMe}pU_{2'OMe}pN$, where N is a natural, a modified, or an unnatural nucleoside base.

[0788] In some embodiments, a tetranucleotide cap comprises GGAG. In some embodiments, a tetranucleotide cap comprises the following structure:



[0784] A tetranucleotide cap, in other embodiments, comprises a sequence selected from the following sequences:

$m^7G_{3'OMe}pppA_{2'OMe}pApN$,	$m^7G_{3'OMe}pppA_{2'OMe}pCpN$,
$m^7G_{3'OMe}pppA_{2'OMe}pGpN$,	$m^7G_{3'OMe}pppA_{2'OMe}pUpN$,
$m^7G_{3'OMe}pppC_{2'OMe}pApN$,	$m^7G_{3'OMe}pppC_{2'OMe}pCpN$,
$m^7G_{3'OMe}pppC_{2'OMe}pGpN$,	$m^7G_{3'OMe}pppC_{2'OMe}pUpN$,
$m^7G_{3'OMe}pppU_{2'OMe}pGpN$, and	$m^7G_{3'OMe}pppU_{2'OMe}pUpN$, where N is a natural, a modified, or an unnatural nucleoside base.

[0785] A tetranucleotide cap, in still other embodiments, comprises a sequence selected from the following sequences: $m^7GpppA_{2'OMe}pApN$, $m^7GpppA_{2'OMe}pCpN$, $m^7GpppA_{2'OMe}pGpN$, and $m^7GpppA_{2'OMe}pUpN$,

[0789] The capping efficiency of a post-transcriptional or co-transcriptional capping reaction may vary. As used herein “capping efficiency” refers to the amount (e.g., expressed as a percentage) of mRNAs comprising a cap structure relative to the total mRNAs in a mixture (e.g., a post-transcriptional capping reaction or a co-transcriptional capping reaction). In some embodiments, the capping efficiency of a capping reaction is at least 60%, 70%, 80%, 90%, 95%, 99%, or 99.9% (e.g., after the capping reaction at least 60%, 70%, 80%, 90%, 95%, 99%, or 99.9% of the input mRNAs comprise a cap). In some embodiments, multivalent co-IVT reactions described herein do not affect the capping efficiency of the mRNAs resulting from the IVT reaction.

[0790] A 3'-poly(A) tail is typically a stretch of adenine nucleotides added to the 3'-end of the transcribed mRNA. It can, in some instances, comprise up to about 400 adenine nucleotides. In some embodiments, the length of the 3'-poly (A) tail may be an essential element with respect to the stability of the individual mRNA.

[0791] In some embodiments, a composition comprises an RNA (e.g., mRNA) having an ORF that encodes a signal peptide fused to the expressed polypeptide. Signal peptides, usually comprising the N-terminal 15-60 amino acids of proteins, are typically needed for the translocation across the membrane on the secretory pathway and, thus, universally control the entry of most proteins both in eukaryotes and prokaryotes to the secretory pathway. A signal peptide may have a length of 15-60 amino acids.

[0792] In some embodiments, an ORF encoding a polypeptide is codon optimized. Codon optimization methods are known in the art. For example, an ORF of any one or more of the sequences provided herein may be codon optimized. Codon optimization, in some embodiments, may be used to match codon frequencies in target and host organisms to ensure proper folding; bias GC content to increase mRNA stability or reduce secondary structures; minimize tandem repeat codons or base runs that may impair gene construction or expression; customize transcriptional and translational control regions; insert or remove protein trafficking sequences; remove/add post translation modification sites in encoded protein (e.g., glycosylation sites); add, remove or shuffle protein domains; insert or delete restriction sites; modify ribosome binding sites and mRNA degradation sites; adjust translational rates to allow the various domains of the protein to fold properly; or reduce or eliminate problem secondary structures within the polynucleotide. Codon optimization tools, algorithms and services are known in the art—non-limiting examples include services from GeneArt (Life Technologies), DNA2.0 (Menlo Park CA) and/or proprietary methods. In some embodiments, the open reading frame (ORF) sequence is optimized using optimization algorithms.

[0793] In some embodiments, an RNA (e.g., mRNA) is not chemically modified and comprises the standard ribonucleotides consisting of adenosine, guanosine, cytosine and uridine. In some embodiments, nucleotides and nucleosides comprise standard nucleoside residues such as those present in transcribed RNA (e.g. A, G, C, or U). In some embodiments, nucleotides and nucleosides comprise standard deoxyribonucleosides such as those present in DNA (e.g. dA, dG, dC, or dT).

[0794] The compositions can comprise, in some embodiments, an RNA having an open reading frame encoding a polypeptide, wherein the nucleic acid comprises nucleotides and/or nucleosides that can be standard (unmodified) or modified as is known in the art. In some embodiments, nucleotides and nucleosides comprise modified nucleotides or nucleosides. Such modified nucleotides and nucleosides can be naturally-occurring modified nucleotides and nucleosides or non-naturally occurring modified nucleotides and nucleosides. Such modifications can include those at the sugar, backbone, or nucleobase portion of the nucleotide and/or nucleoside as are recognized in the art.

[0795] In some embodiments, a naturally-occurring modified nucleotide or nucleotide is one as is generally known or recognized in the art. Non-limiting examples of such natu-

rally occurring modified nucleotides and nucleotides can be found, inter alia, in the widely recognized MODOMICS database.

[0796] Also provided are modified nucleosides and nucleotides of a nucleic acid (e.g., RNA nucleic acids, such as mRNA nucleic acids). A “nucleoside” refers to a compound containing a sugar molecule (e.g., a pentose or ribose) or a derivative thereof in combination with an organic base (e.g., a purine or pyrimidine) or a derivative thereof (also referred to herein as “nucleobase”). A “nucleotide” refers to a nucleoside, including a phosphate group. Modified nucleotides may be synthesized by any useful method, such as, for example, chemically, enzymatically, or recombinantly, to include one or more modified or non-natural nucleosides.

[0797] Nucleic acids can comprise a region or regions of linked nucleosides. Such regions may have variable backbone linkages. The linkages can be standard phosphodiester linkages, in which case the nucleic acids would comprise regions of nucleotides.

[0798] In some embodiments, modified nucleobases in nucleic acids (e.g., RNA nucleic acids, such as mRNA nucleic acids) comprise 1-methyl-pseudouridine (m1ψ), 1-ethyl-pseudouridine (e1ψ), 5-methoxy-uridine (mo5U), 5-methyl-cytidine (m5C), and/or pseudouridine (ψ). In some embodiments, modified nucleobases in nucleic acids (e.g., RNA nucleic acids, such as mRNA nucleic acids) comprise 5-methoxymethyl uridine, 5-methylthio uridine, 1-methoxymethyl pseudouridine, 5-methyl cytidine, and/or 5-methoxy cytidine. In some embodiments, the polynucleotide includes a combination of at least two (e.g., 2, 3, 4 or more) of any of the aforementioned modified nucleobases, including but not limited to chemical modifications.

[0799] In some embodiments, a mRNA comprises 1-methyl-pseudouridine (m1ψ) substitutions at one or more or all uridine positions of the nucleic acid.

[0800] In some embodiments, a mRNA comprises 1-methyl-pseudouridine (m1ψ) substitutions at one or more or all uridine positions of the nucleic acid and 5-methyl cytidine substitutions at one or more or all cytidine positions of the nucleic acid.

[0801] In some embodiments, a mRNA comprises pseudouridine (ψ) substitutions at one or more or all uridine positions of the nucleic acid.

[0802] In some embodiments, a mRNA pseudouridine (ψ) substitutions at one or more or all uridine positions of the nucleic acid and 5-methyl cytidine substitutions at one or more or all cytidine positions of the nucleic acid.

[0803] In some embodiments, a mRNA comprises uridine at one or more or all uridine positions of the nucleic acid.

[0804] In some embodiments, mRNAs are uniformly modified (e.g., fully modified, modified throughout the entire sequence) for a particular modification. For example, a nucleic acid can be uniformly modified with 1-methyl-pseudouridine, meaning that all uridine residues in the mRNA sequence are replaced with 1-methyl-pseudouridine. Similarly, a nucleic acid can be uniformly modified for any type of nucleoside residue present in the sequence by replacement with a modified residue such as those set forth above.

[0805] The nucleic acids may be partially or fully modified along the entire length of the molecule. For example, one or more or all or a given type of nucleotide (e.g., purine or pyrimidine, or any one or more or all of A, G, U, C) may be uniformly modified in a nucleic acid, or in a predeter-

mined sequence region thereof (e.g., in the mRNA including or excluding the poly(A) tail). In some embodiments, all nucleotides X in a nucleic acid (or in a sequence region thereof) are modified nucleotides, wherein X may be any one of nucleotides A, G, U, C, or any one of the combinations A+G, A+U, A+C, G+U, G+C, U+C, A+G+U, A+G+C, G+U+C or A+G+C.

[0806] The mRNAs may comprise one or more regions or parts which act or function as an untranslated region. Where mRNAs are designed to encode at least one polypeptide of interest, the nucleic acid may comprise one or more of these untranslated regions (UTRs). Wild-type untranslated regions of a nucleic acid are transcribed but not translated. In mRNA, the 5' UTR starts at the transcription start site and continues to the start codon but does not include the start codon; whereas the 3' UTR starts immediately following the stop codon and continues until the transcriptional termination signal. The regulatory features of a UTR can be incorporated into the polynucleotides to, among other things, enhance the stability of the molecule. The specific features can also be incorporated to ensure controlled down-regulation of the transcript in case they are misdirected to undesired organs sites. A variety of 5'UTR and 3'UTR sequences are known and available in the art.

Pharmaceutical Compositions

[0807] The lipid nanoparticles described herein may be used for treating or preventing diseases, disorders, or conditions. For example, the lipid nanoparticles may be used for treating or preventing a symptom of diseases characterized by missing or aberrant protein activity, by replacing the missing protein activity or overcoming the aberrant protein activity.

[0808] In some embodiments, a subject to which a composition comprising a nucleic acid, a lipid, and/or a compound of Formula I or Formula II is administered is a subject that suffers from or is at risk of suffering from a disease, disorder, or condition, including a communicable or non-communicable disease, disorder or condition. As used herein, "treating" a subject can include either therapeutic use or prophylactic use relating to a disease, disorder or condition, and may be used to describe uses for the alleviation of symptoms of a disease, disorder or condition, uses for vaccination against a disease, disorder or condition, and uses for decreasing the contagiousness of a disease, disorder or condition, among other uses.

[0809] In some embodiments the nucleic acid is an mRNA vaccine designed to achieve particular biologic effects. Exemplary vaccines feature mRNAs encoding a particular antigen of interest (or an mRNA or mRNAs encoding antigens of interest). In exemplary aspects, the vaccines feature an mRNA or mRNAs encoding antigen(s) derived from infectious diseases or cancers.

[0810] In some embodiments, microbial growth within a composition disclosed herein is inhibited. In some embodiments, microbial growth is inhibited by the compound (e.g., compound of Formula I or Formula II). In some embodiments, a composition disclosed herein does not comprise a pharmaceutical preservative. Non-limiting examples of pharmaceutical preservatives include methyl paraben, ethyl paraben, propyl paraben, butyl paraben, benzyl alcohol, chlorobutanol, phenol, meta cresol (m-cresol), chloro cresol, benzoic acid, sorbic acid, thiomersal, phenylmercuric nitrate, bronopol, propylene glycol, benzylkonium chloride,

and benzethonium chloride. In some embodiments, a composition disclosed herein does not comprise phenol, m-cresol, or benzyl alcohol. Compositions in which microbial growth is inhibited may be useful in the preparation of injectable formulations, including those intended for dispensing from multi-dose vials. Multi-dose vials refer to containers of pharmaceutical compositions from which multiple doses can be taken repeatedly from the same container.

[0811] Compositions intended for dispensing from multi-dose vials typically must meet USP requirements for antimicrobial effectiveness. In some embodiments, a composition disclosed herein comprising a compound (e.g. a compound of Formula I, or of Formula II) has antimicrobial effectiveness, and may be dispensed from a multi-dose vial.

[0812] In some embodiments, "administering" or "administration" means providing a material to a subject in a manner that is pharmacologically useful. In some embodiments, a composition disclosed herein is administered to a subject enterally. In some embodiments, an enteral administration of the composition is oral. In some embodiments, a composition disclosed herein is administered to the subject parenterally. In some embodiments, a composition disclosed herein is administered to a subject subcutaneously, intraocularly, intravitreally, subretinally, intravenously (IV), intracerebro-ventricularly, intramuscularly, intrathecally (IT), intracisternally, intraperitoneally, via inhalation, topically, or by direct injection to one or more cells, tissues, or organs.

[0813] To "treat" a disease as the term is used herein, means to reduce the frequency or severity of at least one sign or symptom of a disease, disorder or condition experienced by a subject. The compositions provided herein are typically administered to a subject in an effective amount, that is, an amount capable of producing a desirable result. The desirable result will depend upon the active agent being administered. For example, an effective amount of a composition comprising a nucleic acid, a lipid, and a compound of Formula I may be an amount of the composition that is capable of increasing expression of a protein in the subject. A therapeutically acceptable amount may be an amount that is capable of treating a disease or condition, e.g., a disease or condition that that can be relieved by increasing expression of a protein in a subject. As is well known in the medical and veterinary arts, dosage for any one subject depends on many factors, including the subject's size, body surface area, age, the particular composition to be administered, the active ingredient(s) in the composition, the intended outcome of the administration, time and route of administration, general health, and other drugs being administered concurrently.

[0814] In some embodiments, a subject is administered a composition comprising a nucleic acid, a lipid, and/or a compound of Formula I in an amount sufficient to increase expression of a protein in the subject.

[0815] Compositions formed via the methods provided herein may be particularly useful for administering an agent to a subject in need thereof. In some embodiments, the compositions are used to deliver a pharmaceutically active agent. In some instances, the compositions are used to deliver a prophylactic agent. The compositions may be administered in any way known in the art of drug delivery, for example, orally, parenterally, intravenously, intramuscularly, subcutaneously, intradermally, transdermally, intrathecally, submucosally, sublingually, rectally, vaginally, etc.

[0816] Once the compositions have been prepared, they may be combined with pharmaceutically acceptable excipients to form a pharmaceutical composition. As would be appreciated by one of skill in this art, the excipients may be chosen based on the route of administration as described below, the agent being delivered, and the time course of delivery of the agent.

[0817] Pharmaceutical compositions provided herein and for use in accordance with the embodiments provided herein may include a pharmaceutically acceptable excipient. As used herein, the term “pharmaceutically acceptable excipient” means a non-toxic, inert solid, semi-solid or liquid filler, diluent, encapsulating material or formulation auxiliary of any type. Some examples of materials which can serve as pharmaceutically acceptable excipients are sugars such as lactose, glucose, and sucrose; starches such as corn starch and potato starch; cellulose and its derivatives such as sodium carboxymethyl cellulose, methylcellulose, hydroxypropylmethylcellulose, ethyl cellulose, and cellulose acetate; powdered tragacanth; malt; gelatin; talc; excipients such as cocoa butter and suppository waxes; oils such as peanut oil, cottonseed oil; safflower oil; sesame oil; olive oil; corn oil and soybean oil; glycols such as propylene glycol; esters such as ethyl oleate and ethyl laurate; agar; detergents such as Tween 80; buffering agents such as magnesium hydroxide and aluminum hydroxide; alginic acid; pyrogen free water; isotonic saline; citric acid, acetate salts, Ringer’s solution; ethyl alcohol; and phosphate buffer solutions, as well as other non-toxic compatible lubricants such as sodium lauryl sulfate and magnesium stearate, as well as coloring agents, releasing agents, coating agents, sweetening, flavoring and perfuming agents, preservatives and antioxidants can also be present in the composition, according to the judgment of the formulator. The pharmaceutical compositions can be administered to humans and/or to animals, orally, rectally, parenterally, intracisternally, intravaginally, intranasally, intraperitoneally, topically (as by powders, creams, ointments, or drops), buccally, or as an oral or nasal spray.

[0818] Liquid dosage forms for oral administration include pharmaceutically acceptable emulsions, microemulsions, solutions, suspensions, syrups, and elixirs. In addition to the active ingredients (i.e., the particles), the liquid dosage forms may contain inert diluents commonly used in the art such as, for example, water or other solvents, solubilizing agents and emulsifiers such as ethyl alcohol, isopropyl alcohol, ethyl carbonate, ethyl acetate, benzyl alcohol, benzyl benzoate, propylene glycol, 1,3 butylene glycol, dimethylformamide, oils (in particular, cottonseed, groundnut, corn, germ, olive, castor, and sesame oils), glycerol, tetrahydrofurfuryl alcohol, polyethylene glycols and fatty acid esters of sorbitan, and mixtures thereof. Besides inert diluents, the oral compositions can also include adjuvants such as wetting agents, emulsifying and suspending agents, sweetening, flavoring, and perfuming agents.

[0819] Injectable preparations, for example, sterile injectable aqueous or oleaginous suspensions may be formulated according to the known art using suitable dispersing or wetting agents and suspending agents. The sterile injectable preparation may also be a sterile injectable solution, suspension, or emulsion in a nontoxic parenterally acceptable diluent or solvent, for example, as a solution in 1,3-butanediol. Among the acceptable vehicles and solvents that may be employed are water, Ringer’s solution, ethanol, U.S.P.,

and isotonic sodium chloride solution. In addition, sterile, fixed oils are conventionally employed as a solvent or suspending medium. For this purpose, any bland fixed oil can be employed including synthetic mono or diglycerides. In addition, fatty acids such as oleic acid are used in the preparation of injectables.

[0820] The injectable formulations can be sterilized, for example, by filtration through a bacteria retaining filter, or by incorporating sterilizing agents in the form of sterile solid compositions which can be dissolved or dispersed in sterile water or other sterile injectable medium prior to use.

[0821] Compositions for rectal or vaginal administration may be suppositories which can be prepared by mixing the particles with suitable non-irritating excipients or carriers such as cocoa butter, polyethylene glycol, or a suppository wax which are solid at ambient temperature but liquid at body temperature and therefore melt in the rectum or vaginal cavity and release the particles.

[0822] Solid dosage forms for oral administration include capsules, tablets, pills, powders, and granules. In such solid dosage forms, the particles are mixed with at least one inert, pharmaceutically acceptable excipient or carrier such as sodium citrate or dicalcium phosphate and/or a) fillers or extenders such as starches, lactose, sucrose, glucose, mannitol, and silicic acid, b) binders such as, for example, carboxymethylcellulose, alginates, gelatin, polyvinylpyrrolidone, sucrose, and acacia, c) humectants such as glycerol, d) disintegrating agents such as agar, calcium carbonate, potato or tapioca starch, alginic acid, certain silicates, and sodium carbonate, e) solution retarding agents such as paraffin, f) absorption accelerators such as quaternary ammonium compounds, g) wetting agents such as, for example, cetyl alcohol and glycerol monostearate, h) absorbents such as kaolin and bentonite clay, and i) lubricants such as talc, calcium stearate, magnesium stearate, solid polyethylene glycols, sodium lauryl sulfate, and mixtures thereof. In the case of capsules, tablets, and pills, the dosage form may also comprise buffering agents.

[0823] Solid compositions of a similar type may also be employed as fillers in soft and hard filled gelatin capsules using such excipients as lactose or milk sugar as well as high molecular weight polyethylene glycols and the like.

[0824] The solid dosage forms of tablets, dragées, capsules, pills, and granules can be prepared with coatings and shells such as enteric coatings and other coatings well known in the pharmaceutical formulating art. They may optionally contain opacifying agents and can also be of a composition that they release the active ingredient(s) only, or preferentially, in a certain part of the intestinal tract, optionally, in a delayed manner. Examples of embedding compositions which can be used include polymeric substances and waxes.

[0825] Dosage forms for topical or transdermal administration of a pharmaceutical composition include ointments, pastes, creams, lotions, gels, powders, solutions, sprays, inhalants, or patches. The particles are admixed under sterile conditions with a pharmaceutically acceptable carrier and any needed preservatives or buffers as may be required. Ophthalmic formulation, ear drops, and eye drops are also possible.

[0826] The ointments, pastes, creams, and gels may contain, in addition to the compositions provided herein, excipients such as animal and vegetable fats, oils, waxes, paraffins,

starch, tragacanth, cellulose derivatives, polyethylene glycols, silicones, bentonites, silicic acid, talc, and zinc oxide, or mixtures thereof.

[0827] Powders and sprays can contain, in addition to the compositions provided herein, excipients such as lactose, talc, silicic acid, aluminum hydroxide, calcium silicates, and polyamide powder, or mixtures of these substances. Sprays can additionally contain customary propellants such as chlorofluorohydrocarbons.

[0828] Transdermal patches have the added advantage of providing controlled delivery of a compound to the body. Such dosage forms can be made by dissolving or dispensing the compositions in a proper medium. Absorption enhancers can also be used to increase the flux of the compound across the skin. The rate can be controlled by either providing a rate controlling membrane or by dispersing the compositions in a polymer matrix or gel.

[0829] In other embodiments, the stabilized compositions are loaded and stored in prefilled syringes and cartridges for patient-friendly autoinjector and infusion pump devices.

[0830] Kits for use in preparing or administering the compositions are also provided. A kit for forming compositions may include any solvents, solutions, buffer agents, acids, bases, salts, targeting agents, etc. needed in the composition formation process. Different kits may be available for different targeting agents. In certain embodiments, the kit includes materials or reagents for purifying, sizing, and/or characterizing the resulting compositions. The kit may also include instructions on how to use the materials in the kit. The one or more agents (e.g., pharmaceutically active agent) to be contained within the composition are typically provided by the user of the kit.

[0831] Kits are also provided for using or administering the compositions. The compositions may be provided in convenient dosage units for administration to a subject. The kit may include multiple dosage units. For example, the kit may include 1-100 dosage units. In certain embodiments, the kit includes a week supply of dosage units, or a month supply of dosage units. In certain embodiments, the kit includes an even longer supply of dosage units. The kits may also include devices for administering the compositions. Exemplary devices include syringes, spoons, measuring devices, etc. The kit may optionally include instructions for administering the compositions (e.g., prescribing information).

[0832] The term “pharmaceutically acceptable salt” refers to those salts which are, within the scope of sound medical judgment, suitable for use in contact with the tissues of humans and lower animals without undue toxicity, irritation, allergic response, and the like, and are commensurate with a reasonable benefit/risk ratio. Pharmaceutically acceptable salts are well known in the art. For example, Berge et al. describe pharmaceutically acceptable salts in detail in *J. Pharmaceutical Sciences*, 1977, 66, 1-19, incorporated herein by reference.

[0833] Pharmaceutically acceptable salts of the compounds provided herein include those derived from suitable inorganic and organic acids and bases. Examples of pharmaceutically acceptable, nontoxic acid addition salts are salts of an amino group formed with inorganic acids, such as hydrochloric acid, hydrobromic acid, phosphoric acid, sulfuric acid, and perchloric acid or with organic acids, such as acetic acid, oxalic acid, maleic acid, tartaric acid, citric acid, succinic acid, or malonic acid or by using other methods known in the art such as ion exchange. Other pharmaceutically acceptable salts include adipate, alginate, ascorbate, aspartate, benzenesulfonate, benzoate, bisulfate, borate, butyrate, camphorate, camphorsulfonate, citrate, cyclopentanepropionate, digluconate, dodecylsulfate, ethane-

sulfonate, formate, fumarate, glucoheptonate, glycerophosphate, gluconate, hemisulfate, heptanoate, hexanoate, hydroiodide, 2-hydroxy-ethanesulfonate, lactobionate, lactate, laurate, lauryl sulfate, malate, maleate, malonate, methanesulfonate, 2-naphthalenesulfonate, nicotinate, nitrate, oleate, oxalate, palmitate, pamoate, pectinate, persulfate, 3-phenylpropionate, phosphate, picrate, pivalate, propionate, stearate, succinate, sulfate, tartrate, thiocyanate, p-toluenesulfonate, undecanoate, valerate salts, and the like. Salts derived from appropriate bases include alkali metal, alkaline earth metal, ammonium, and $N^+(C_{1-4} \text{ alkyl})_4^-$ salts. Representative alkali or alkaline earth metal salts include sodium, lithium, potassium, calcium, magnesium, and the like. Further pharmaceutically acceptable salts include, when appropriate, nontoxic ammonium, quaternary ammonium, and amine cations formed using counterions such as halide, hydroxide, carboxylate, sulfate, phosphate, nitrate, lower alkyl sulfonate, and aryl sulfonate.

[0834] The terms “composition” and “formulation” are used interchangeably.

[0835] As used herein, the term “intercalating small molecule” or “small molecule nucleic acid intercalating agent” refers to a compound containing aromatic or heteroaromatic ring systems that can insert between adjacent base pairs of double stranded DNA or folded or double stranded regions of mRNA. Intercalating agents typically but not necessarily, contain planar polyaromatic rings and cationic substituents. Intercalation between adjacent base pairs may be full or partial. A typical small molecule intercalating agent contains three or four fused rings that absorb light in the UV-visible region of the electromagnetic spectrum.

[0836] The following examples are intended to illustrate certain non-limiting embodiments. Without further elaboration, it is believed that one skilled in the art can, based on the above description, utilize the disclosed compositions and methods to the fullest extent. The following specific embodiments are, therefore, to be construed as merely illustrative, and not limitative in any way whatsoever.

EXAMPLES

Example 1—PEG-DMG Modulation Study

[0837] This example describes the impact of increasing the amount of PEG-DMG added during the mixing and excipient addition steps.

[0838] Four separate nanoprecipitation operations were completed, each with different levels of % PEG-DMG present during the Mixing Unit Operation (vortex mixing of lipids). The batches were split into multiple aliquots and spiked with differing amounts of PEG-DMG to produce up to 10 different sublots for each batch (Table 1). Multiple methods of PEG-DMG addition could be used to achieve the same theoretical levels of Total PEG-DMG in the DP. All LNPs were formed via a nanoprecipitation reaction using a V-mixer. The post addition procedure comprised (i) a nanoprecipitation reaction between the lipids dissolved in ethanol and the mRNA in aqueous solution followed by pH adjustment, (ii) tangential flow filtration (TFF), (iii) exposure of the filtered particles to a solution comprising a certain weight percentage of PEG-DMG, and (iv) a final filtration step. The mol % of PEG-DMG used in the nanoprecipitation reaction varied depending on the amount of PEG-DMG used in the post-filtration exposure step. The table below provides the total amount of PEG-DMG in the composition resulting from varying amounts of PEG-DMG added during the nanoprecipitation reaction (y axis) and during the post-addition (PA) step (x axis).

TABLE 1

Concentrations of PEG-DMG in each DP Lot											
Batch	mol % PEG-DMG present during V-mixing	mol % PEG-DMG PA spike									
		0	0.5	0.75	1	1.25	1.5	1.75	2	2.25	2.50
1	0.25	.25	.75	1	1.25	1.5	1.75	2	2.25	2.5	2.75
2	0.5	.5	1	1.25	1.5*	1.75	2	2.25	2.5	2.75	3
3	0.75	.75	1.25	1.5	1.75	2	2.25	2.5	2.75	3	3.25
4	1.0	1	1.5	1.75	2.0	2.25	2.5	2.75	3	3.25	3.5

[0839] The size of the LNPs was measured using dynamic light scattering (DLS) after the v-mixing, pH adjustment, TFF harvest and just before excipient addition (pre-spike) 1 day after TFF harvest. As shown in FIG. 1, increased PEG-DMG content present during v-mixing better stabilized the LNP, leading to less particle size growth over TFF. Considerable size growth was observed for LNPs with 0.5 lipid-mol % PEG-DMG or less present during the v-mixing unit operation.

[0840] The amount of PEG-DMG present impacted particle size growth of the pH adjusted VMP (FIG. 2). VMP with 0.25 mol % present grew at the highest rate over time. The other lots showed similar behavior, except at the last time point where Batch 2 pH adj TMP exhibited accelerated growth.

[0841] FIG. 3 shows the size stability of the filtered VMP of each lot. Filtered VMP intermediates began to increase in size at the last time point observed for Batch 2 and Batch 3 (post 400 minutes after mixing), while no increase in size was observed for Batch 1 and Batch 4. Trending was similar to the pH-adjusted VMP.

[0842] Excipient post addition of PEG-DMG was also examined. It was found to be necessary to prevent particle size growth from occurring during multiple freeze/thaw (F/T) cycles. The DP average diameter after (5 F/T)/(TO DP size) (Z_{avg}/Z_0) was tabulated as a function of mol % PEG-DMG present during the V-mixing (y-axis) and mol % PEG-DMG added during excipient addition (x-axis) in Table 2 below. The addition of 1 mol % PEG-DMG or greater was found to prevent particle growth over time independent of the PEG-DMG level in the core of the particle.

(250 ng mRNA or 1000 ng mRNA). Eighteen to 24 hours later, the cells were fixed and processed using an ELISA and mean fluorescence intensity (MFI) was measured and plotted against mol % total PEG-DMG (FIGS. 4A-4B).

[0844] Dose-dependent protein expression was observed. Additionally, the amount of PEG-DMG present impacted in vitro expression (FIG. 4A). It was found that, when the content of PEG-DMG is very low (<1 mol % total), expression is reduced. When the total PEG-DMG level is at least 1 mol %, there is no significant difference (i.e., greater than 2-fold differences) observed in the percentage of cells expressing the protein at both doses. This result is independent of the total PEG-DMG content (provided it is <1 mol %), and the percent in the core and in the post addition.

[0845] At the high dose of 1000 ng, a similar pattern was observed as at the lower dose (FIG. 4B). Additionally, at levels above 2.5 mol % PEG-DMG, the material with 1 mol % PEG-DMG present in V-mixing (Batch 4) had lower mean fluorescent intensity than the batches with less PEG-DMG present during mixing.

Example 3—In Vivo Testing

[0846] As indicated by the cells bolded in Table 1, a selection of the batches were tested in animals to examine the effects of PEG-DMG levels on immunogenicity. A clear relationship was observed between the amount of PEG-DMG present during V-mixing and the titer levels (FIG. 5). Mice injected with DP that had less PEG-DMG present during V-mixing had higher protein-specific IgG titer levels than mice given DP with greater amounts of PEG-DMG

TABLE 2

Average DP Diameter after 5 Freeze/Thaw Cycles										
Batch	Mol % PEG-DMG in Vmix	Mol % PEG-DMG PA Spike								
		0	0.5	0.75	1	1.25	1.5	1.75	2	2.25
1	0.25	3.54	1.58	1.04	1.03	0.96	0.97	0.97	1.00	1.01
2	0.5	2.09	1.57	1.29	1.12	1.01	0.98	0.99	1.02	
3	0.75	2.54	1.55	1.45	1.20	1.04	1.00	1.02	1.03	1.04
4	1	1.55	1.08	1.04	1.03	1.01	0.99	0.98	0.98	0.97

Example 2—In Vitro Testing

[0843] The DPs were tested in vitro in Hep3B cells to determine protein expression via flow cytometry. In this study, 20,000 Hep3B cells were seeded in 100 μ l per well in a 96 well plate. Twenty to 24 hours later, the cells were transfected in triplicate with DP diluted in formulation buffer

present during V-mixing. At the level of PEG-DMG of 0.5 mol % present during mixing, no decrease in titer levels was observed as more PEG-DMG was added during the excipient addition step. However, a titer level drop was observed in mice given Batch 3, which had 0.75 mol % PEG-DMG present during V-mixing when PEG-DMG was added during the excipient addition step.

[0847] Therefore, a significant difference in the in vivo immunogenicity of DP with 0.5 mol % PEG-DMG present during v-mixing compared to DP with 0.75 mol % and 1.0 mol % PEG-DMG present during v-mixing was demonstrated. At the 0.5 mol % PEG-DMG level, at least 1.25 mol % PEG-DMG added during the excipient addition step was needed to provide adequate freeze/thaw stability. Titer levels in mice given Batch 2 (0.5 mol % present in lipid stock) were not dependent on the amount of PEG-DMG added during excipient addition, suggesting that more PEG-DMG could be added without a negative impact on immunogenicity.

[0848] This study was designed such that at 1.5 mol % PEG-DMG, there would be 50 mol % SM102, 38.5 mol % cholesterol, and 10% DSPC in each lot. Sublots at other levels of total mol % PEG-DMG necessarily had differing amounts of other lipids.

Example 4—Total PEG-DMG Mole Fraction Variation Studies

[0849] In this Example, in vitro expression of mRNA encoding an antigen, formulated in LNPs having different total PEG-DMG mole fractions was measured in Hep3B cells. Briefly, 20,000 Hep3B cells were seeded in 100 μ l per well in a 96 well plate. Twenty to 24 hours later, the cells were transfected in triplicate with DP diluted in formulation buffer (62.5 ng, 250 ng, or 1000 ng mRNA). Eighteen to 24 hours later, the cells were fixed and processed using an ELISA. The batches were manufactured using identical parameters, except that the lipid composition was varied to modulate the final PEG-DMG levels. Similar expression was observed at all lipid composition compared to control, regardless of PEG-DMG level (FIG. 6). The mRNA and lipid compositions tested are shown in Table 3 below.

TABLE 3

mRNA and Lipid Composition						
Lot	Average Mole % Ionizable Amino Lipid	Average Mole % Cholesterol	Average Mole % DSPC	Average Mole % PEG-DMG	mRNA concentration	Lipid to mRNA ratio
1	45.4	40.5	12.7	1.4	0.33	24.1
2	47.5	38.9	11.8	1.8	0.34	27.4
3	46.8	39.6	11.4	2.2	0.34	25.2
4	45.6	40.9	10.4	3.1	0.34	25.4

Example 5—Long-Term Effects of PEG-DMG and mRNA Concentration

[0850] This example examines the encapsulation efficiency and bulk average diameter of lipid nanoparticles comprising mRNA encoding an antigen after refrigerated or room temperature storage for six months. Briefly, an engineering lot of the mRNA, with an initial concentration of 0.77 mg/mL and initial PEG-DMG concentration of 1.5% mol lipid was diluted to varying mRNA and PEG-DMG concentrations. The mRNA was varied from 0.01-0.39 mg/mL and the PEG-DMG concentration varied from 1.5-3.5 mol %. Bulk average diameter was measured using dynamic light scattering and mRNA encapsulation efficiency was determined using methylene blue hypochromic shift. A central composite designed experiment with three

center points was undertaken to study the effects of changes in mRNA concentration in increases in PEG-DMG concentration. In total, 13 runs were used to estimate the five model terms and the overall average of the responses.

[0851] According to the models, the predicted diameter is minimized when mRNA concentration is low and PEG-DMG concentration is equal to approximately 3.1 mol % after six months of refrigerated storage. Similarly, the predicted diameter was found to be low with decreasing mRNA concentration after six months of room temperature storage. The encapsulation efficiency following six months of refrigerated storage was found to be maximized when the PEG-DMG concentration is approximate 2.3 mol %. At room temperature, the encapsulation efficiency was minimized when both mRNA concentration and PEG-DMG were low. The predicted diameter and percent encapsulation efficiency (EE) of four combinations of mRNA concentration and PEG-DMG mol % concentration after six months of refrigerated storage (5° C.) are shown in Table 4.

TABLE 4

Table of Predictions (6 months at 5° C.)			
[mRNA] (mg/mL)	% mol PEG	Diameter (nm)	% EE
0.1	1.5	157	96.9
0.1	2.5	117	97.4
0.2	1.5	161	96.9
0.2	2.5	121	97.4

EQUIVALENTS

[0852] While several embodiments have been described and illustrated herein, those of ordinary skill in the art will readily envision a variety of other means and/or structures for performing the function and/or obtaining the results and/or one or more of the advantages described herein, and each of such variations and/or modifications is deemed to be within the scope of the embodiments described herein. More generally, those skilled in the art will readily appreciate that all parameters, dimensions, materials, and configurations described herein are meant to be exemplary and that the actual parameters, dimensions, materials, and/or configurations will depend upon the specific application or applications for which the teachings is/are used. Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments described herein. It is, therefore, to be understood that the foregoing embodiments are presented by way of example only and that, within the scope of the appended claims and equivalents thereto, embodiments may be practiced otherwise than as specifically described and claimed. Embodiments are provided that are directed to each individual feature, system, article, material, kit, and/or method described herein. In addition, any combination of two or more such features, systems, articles, materials, kits, and/or methods, if such features, systems, articles, materials, kits, and/or methods are not mutually inconsistent, is included within the scope of the disclosure.

[0853] All definitions, as defined and used herein, should be understood to control over dictionary definitions, definitions in documents incorporated by reference, and/or ordinary meanings of the defined terms.

[0854] All references, patents and patent applications disclosed herein are incorporated by reference with respect to the subject matter for which each is cited, which in some cases may encompass the entirety of the document.

[0855] The indefinite articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.”

[0856] The phrase “and/or,” as used herein in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B,” when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc. As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e. “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of.” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.

[0857] As used herein in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally

including more than one, B (and optionally including other elements); etc. Each possibility represents a separate embodiment.

[0858] It should also be understood that, unless clearly indicated to the contrary, in any methods claimed herein that include more than one step or act, the order of the steps or acts of the method is not necessarily limited to the order in which the steps or acts of the method are recited. In the claims, as well as in the specification above, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” “composed of,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of” shall be closed or semi-closed transitional phrases, respectively, as set forth in the United States Patent Office Manual of Patent Examining Procedures, Section 2111.03.

1. A method of producing a lipid nanoparticle composition, the method comprising:

- a) mixing a lipid solution comprising 0.25 mol % to 0.75 mol % of a first PEG-lipid, an ionizable lipid, a phospholipid, and a structural lipid with an aqueous buffer to thereby form a precursor lipid nanoparticle; and
- b) adding a lipid nanoparticle modifier comprising 1.5 mol % to 2.5 mol % of a second PEG-lipid to the precursor lipid nanoparticle thereby forming a modified lipid nanoparticle.

2. The method of claim 1, further comprising mixing an mRNA with the lipid solution and the aqueous buffer to thereby form the precursor lipid nanoparticle, which is a precursor nucleic acid lipid nanoparticle.

3. The method of claim 1, further comprising mixing an mRNA with the precursor lipid nanoparticle to thereby form an mRNA-lipid nanoparticle.

4. The method of claim 1, wherein the lipid nanoparticle modifier is added in two separate stages and the total amount of the second PEG-lipid added is 1.5 mol % to 2.5 mol %.

5. The method of claim 1, wherein the lipid nanoparticle modifier is added in three, four, or five separate stages and the total amount of the second PEG-lipid added is 1.5 mol % to 2.5 mol %.

6. The method of claim 1, wherein the lipid solution comprises 0.5 mol % of the first PEG-lipid.

7. The method of claim 6, wherein the lipid nanoparticle modifier comprises 2.0 mol % of the second PEG-lipid.

8. The method of claim 1, wherein the modified lipid nanoparticle comprises a total of 2.5 mol % of the first PEG-lipid and the second PEG-lipid.

9. (canceled)

10. The method of claim 1, wherein the first PEG-lipid is PEG-DMG or 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetradetracontaoxatetradetracontahexyl stearate.

11. The method of claim 10, wherein the second PEG-lipid is PEG-DMG or 134-hydroxy-3,6,9,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93,96,99,102,105,108,111,114,117,120,123,126,129,132-tetradetracontaoxatetradetracontahexyl stearate.

12. The method of claim 1, wherein the ionizable lipid is compound I ((heptadecan-9-yl 8 ((2 hydroxyethyl)(6 oxo 6-(undecyloxy) hexyl)amino) octanoate)), compound II (heptadecan-9-yl 8-((2-hydroxyethyl)(8-(nonylloxy)-8-

oxooctyl)amino)octanoate), compound III (heptadecan-9-yl 8-((3-((2-(methylamino)-3,4-dioxocyclobut-1-en-1-yl) amino)propyl)(8-oxo-8-(undecan-3-yloxy)octyl)amino)octanoate), or compound IV (3-butylheptyl 8-((8-(heptadecan-9-yloxy)-8-oxooctyl) (2-hydroxyethyl)amino)octanoate).

13. The method of claim **12**, wherein the phospholipid is distearoylphosphatidylcholine (DSPC).

14. The method of claim **13**, wherein the structural lipid is a sterol.

15. (canceled)

16. The method of claim **1**, wherein the lipid solution comprises ethanol.

17. The method of claim **1**, wherein the mixing in (a) comprises turbulent mixing ("T-mix"), vortex mixing ("V-mix"), microfluidic mixing, or a combination thereof.

18. (canceled)

19. A composition comprising the mRNA-lipid nanoparticle of claim **3**.

20-21. (canceled)

22. The composition of claim **19**, wherein the composition is stable for at least six months at a temperature of 2° C. to about 6° C.

23. The composition of claim **19**, wherein the composition is stable for at least six months at room temperature.

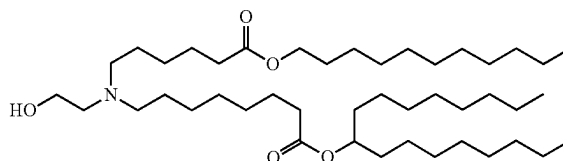
24. The composition of claim **19**, wherein the mRNA-lipid nanoparticle comprises about 2-3 mol % PEG-modified

lipid, 5-25 mol % neutral lipid, 25-55 mol % sterol, and 20-60% ionizable amino lipid.

25.-40. (canceled)

41. A composition comprising a lipid nanoparticle (LNP) and a nucleic acid, wherein the LNP comprises 2.5 mol % 1,2-dimyristoyl-sn-glycerol methoxypolyethylene glycol (PEG-DMG), 11 mol % 1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC), 38.5 mol % cholesterol, and 48 mol % Compound I:

(Compound I)



(heptadecan-9-yl 8-((2 hydroxyethyl)(6 oxo 6-(undecyloxy) hexyl)amino) octanoate), and

wherein 0.25 to 0.5 mol % of the PEG-DMG is in the core of the LNP.

42.-44. (canceled)

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